

Part V

New Paradigm of Central Bank Policy

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Chapter 13

Monetary Policy and Foreign Capital Flows

13.1. Introduction

As explained in Chapter 2, the GFC precipitated three fundamental changes in the monetary policy of the central bank. *First*, in addition to the mandate of price stability, the central bank must also play a role in maintaining financial system stability and, therefore, monetary policy must be integrated with macroprudential policy. *Second*, foreign capital flows inundated various countries, especially EMEs, with high volatility and, therefore, monetary policy optimization was required combined with exchange rate policy and foreign capital flow management (CFM). *Third*, communication as an important instrument of monetary policy and, therefore, the need to institute reforms to increase institutional credibility, independence, and transparency at the central bank. This chapter explores the second aspect, while the first aspect will be discussed in Chapter 14 and the third aspect was elaborated in Part III of this book.

Globalization has created massive cross-border foreign capital flows, which even became more pronounced after the GFC in line with abundant liquidity and low global interest rates due to monetary policy easing in advanced countries to stimulate the economic recovery. On one hand, the foreign capital flows helped to fund financing and, therefore, economic growth in emerging market economies (EMEs). The magnitude of foreign capital flows was influenced by the economic factors specific to each respective EME, particularly economic fundamentals, the degree of openness and the exchange rate system applied, as well as the macroeconomic policies instituted. Sounder economic conditions in a specific EME led to larger foreign capital inflows to that country. On the other hand, however, the deluge of foreign capital flows also increased economic fragility to external shocks in the global economy (global spillovers). Monetary policy and financial system stability may have been disrupted. Consequently, understanding foreign capital flows is crucial in terms of formulating policy, including monetary policy, to maximize the benefits and minimize the risks of foreign capital flows to the national economy.

Foreign capital flows exacerbate the complexity of monetary policy in an open economy. The ability of monetary policy to achieve the domestic economic goals, such as inflation and economic growth, will be determined by exchange rate volatility and foreign capital flows from external factors, as explained by the impossible trinity. According to that theory, if the central bank genuinely desires more

effective monetary policy to achieve the domestic economic goals, a fixed exchange rate regime and foreign exchange (FX) controls are one option. Nevertheless, that choice opposes the advance of globalization, which offers numerous international trade and investment benefits for the domestic economy. Consequently, an optimal alternative is required encompassing monetary policy, exchange rate policy, and foreign CFM so that the economic benefits of an open economy can be realized and the domestic economic goals can be met. In other words, an optimal solution is required to address the policy trilemma in an open economy.

This chapter presents an in-depth discussion of this issue in six sections. After the introduction, the theory and driving forces will be discussed along with what attracts the foreign capital flows. The third section presents the empirical studies concerning the effects of foreign capital flows on economic growth, as well as the invaluable lessons learned, in order to optimally exploit the foreign capital flows. The fourth section details several empirical studies regarding the effect of foreign capital flows on monetary stability and the implications on the policy trilemma and the policy mix response required. Before the closing remarks, the fifth part discusses the principles, targets and instruments of foreign CFM and its application in several EMEs as part of the policy mix to address the impossible trinity of monetary policy.

13.2. Theoretical Dimension and Foreign Capital Flow Developments

Foreign capital flow developments are recorded in the capital and financial account of the balance of payments (BOP). The capital account records capital transfers, for instance from written-off foreign debt and the acquisition of non-financial assets, such as copyrights and goodwill. Meanwhile, the financial account consists of three components, namely foreign direct investment (FDI), portfolio investment (PI), and other investment (OI). Per the International Monetary Fund (IMF), FDI is defined as “a category of cross-border investment associated with a resident in one economy having control or a significant degree of influence on the management of an enterprise that is resident in another economy,” which differs from PI, which is defined as “cross-border transactions and positions involving debt or equity securities, other than those included in direct investment or reserve assets” (IMF, 2009). FDI can be in the form of goods or funds (capital and debt) as an investment in a company that is resident in another country. PI involves non-resident purchases of stocks and debt securities (bonds and money market instruments) in the financial markets. Meanwhile, OI includes trade credits, foreign loans, term deposits, and other receivables that are typically transacted through the banking industry and other financial institutions.

13.2.1. Neoclassical Theory and PI Theory

Neoclassical theory positions its rationale on the advantages or benefits of cross-border investment for the originating economy and beneficiary economy. Consequently, liberalization of the BOP and capital and financial account is a

worthy policy in the era of globalization. Simply put, foreign capital flows can be explained by the following:

$$F = CA = Y(K, L) - C(r, Y) - I(r, Y) \quad (1)$$

$$S = Y(K, L) - C(r, Y) \quad (2)$$

$$F = CA = S - I \quad (3)$$

where F = foreign capital flows or capital account in the BOP, CA = current account in the BOP, Y = domestic GDP, which is determined by capital (K), labor (L), and productivity, C = domestic consumption, which is influenced by interest rates (r) and income (Y), I = Investment, which is influenced by interest rates (r) and the economic outlook (Y), and S = domestic savings deposits. Fundamentally, the equations show that, in developing countries, which generally maintain a current account deficit ($CA < 0$), foreign capital inflows ($F > 0$) are useful to bridge the gap between savings deposits and investment ($S - Y < 0$) to stimulate economic growth. Meanwhile, in advanced countries that typically enjoy a current account surplus ($CA > 0$), foreign capital outflows to capitalize on excessive savings deposits ($S - I > 0$) to invest in another country would also stimulate economic growth.

Based on the neoclassical theory, where countries face a constant return to scale, homogenous capital and labor factors, coupled with truly efficient and open financial markets, capital flows from rich countries to poor based on the law of diminishing returns so that the marginal product of capital is higher in poorer countries. For instance, the production function follows Cobb-Douglas: $y = Ak^\beta$, where y is income per worker and k is capital per worker, then the marginal product of capital is: $r = A\beta k^{\beta-1}$ or equivalent to $r = \beta A^{1/\beta} k^{(\beta-1)/\beta}$ if expressed as production unit per worker. Therefore, the influx of foreign capital from advanced countries, such as the United States, to EMEs, such as Indonesia, shows that the return on capital investment is higher in Indonesia than in the US, which is mutually beneficial to both economies.

Several assumptions underlie neoclassical theory to better understand the behavior, benefits and risks of foreign capital flows. *First*, foreign capital flows are required for investment in the real sector, supported by high productivity, thus catalyzing national economic growth. This implies that foreign capital flows play a complementary (rather than substitutionary) role in terms of domestic savings for economic financing. It is also possible that a country can still capitalize on foreign capital inflows, while simultaneously investing a portion of savings offshore as long as productivity is increased, thus driving real economic growth. In short, the propitious impact of foreign capital flows on investment and economic growth will be enjoyed if supported by productivity, a favorable investment climate and other factors of competition in the receiving country.

Second, the financial sector in the receiving country has developed efficiency to transmit the foreign capital flows into real sector economic financing. In the case of FDI, the investment climate must draw foreign capital flows and channel them

to productive sectors in the economy. In terms of PI and external debt, domestic financial institutions and financial markets are sound conduits in the form of credit instruments, bonds or stocks for economic financing. Regarding government bonds, financing the fiscal deficit would not crowd-out the financial markets or investments in the real sector. More efficient transmission leads to greater benefits from foreign capital flows, implying that most of the foreign capital can be channeled to the real sector and only a little, as excess liquidity, will circulate and be traded in the financial sector. In brief, financial market deepening in terms of the instruments, setting interest rates, liquidity, and infrastructure are crucial to channel foreign capital flows to productive economic financing.

Third, the foreign exchange system implemented in the receiving country could facilitate foreign capital inflows (perfect capital mobility), especially useful for productive economic financing with low volatility. In other words, current account liberalization (CAL) must be carefully designed in terms of the order, scope, and timing to achieve both goals. This is not a simple matter because the preliminary conditions in countries are very diverse. Countries still applying foreign exchange controls enjoy more freedom to design the order, scope, and timing of CAL policy. This is especially true of low-income developing countries. Nevertheless, many EMEs have already implemented a freer foreign exchange system and, therefore, are no longer in a position to optimally design CAL policy. For such countries, the best option would be CFM to optimize the benefits for economic financing and mitigate the adverse effects of the volatility that may appear.

It is more appropriate to apply neoclassical theory to the analysis of foreign capital flows that successfully boost productivity and are generally longer term, such as FDI and long-term external debt. Meanwhile, short-term foreign capital flows are typically determined by the return and risks of a particular investment. Such analyses can be applied to flows of PI on equity, securities, bonds and other financial securities. Theoretically, foreign PI is based on the International Capital Asset Pricing Model (IAPM), which optimizes the portfolio of domestic and global financial assets in line with the return profile and level of risk:

$$E(R_p) = R_F + \beta_w R_w + \sum_{i=1}^k \delta_i S_i \quad (4)$$

where $E(R_p)$ = expected return of the investor's portfolio, R_F = return on risk-free assets (such as the interest rate on US T-Bills), R_w = global benchmark PI return and S_i = magnitude of change in the exchange rate. Forming a global portfolio is based on optimization per Markowitz theory that produces an Efficient Frontier as follows:

$$R_w = \sum_{i=1}^k \omega_i R_i \quad \text{therefore:} \quad \sigma_w^2 = \omega_i^2 \sigma_j^2 + 2 \sum_{i=1}^k \sum_{j=1}^l \omega_i \omega_j \sigma_i \sigma_j \sigma_{ij} \quad (5)$$

where ω_i = share of asset i in the global portfolio, R_i = return on financial assets in country i , and σ^2 = risk associated with financial asset i .

From the theory, we can see that international PI could prompt greater diversification beyond the financial assets of the home country. Therefore, to optimize

the portfolio return, investors will also include financial assets from EMEs, which offer a return and contain risks that do not correlate with the financial assets in the investor's home country (advanced country). Due to the cross-border nature of the portfolio, investors will pay due regard to exchange rate fluctuations, in addition to the returns on the financial assets. How far the investor considers the return and the exchange rate would depend on the motive of the investor. Long-term investors (real money) will prioritize the yield and exchange rate stability, while short-term investors (carry trade) will maximize profit from exchange rate fluctuations. Furthermore, due to asymmetric information and other factors, there is a tendency for investors to perpetuate a home bias rather than a foreign bias.

13.2.2. Determinants of Foreign Capital Flows: Push and Pull Factors

From the two theories mentioned previously, international financial integration, or financial globalization, can be measured by the magnitude of cross-border capital flows. By this measure, the scale of global foreign capital flows has continued to increase since the end of World War II, accelerating in the 1990s and then really taking off in the final decade prior to the GFC. Measured as a percentage of GDP, foreign capital flows have increased from 5.7% in 1980–1989 to 6.2% in 1990–1999 before spiking at 13.3% in 2000–2007 and then falling dramatically back to 6.2% in 2008–2012 (Table 13.1). In terms of composition, PI and bank funds have exploded, while FDI has accelerated more gradually. From Table 13.1, flows of PI and bank funds have increased from 2.3% and 1.9% of GDP, respectively, in 1990–1999 to 4.2% and 5.0% in 2000–2007, before dropping back to 1.4% and 0.4% of GDP in 2008–2012. Meanwhile, after accelerating robustly in the previous two periods, FDI flows have remained relatively stable at 2.9% of GDP in 2000–2007 and in 2008–2012.

Congruent with the global economic developments, foreign capital flows to EMEs have also posted significant gains since 2004. After plummeting in 2008 due to the GFC, foreign capital inflows subsequently skyrocketed until 2013 before fading due to the Fed's Taper Tantrum. In fact, net foreign capital flows to EMEs halved from USD 1,074 billion in 2014 to just USD 548 billion in 2015 (Institute

Table 13.1: Global Foreign Capital Flows (% of GDP).

	Annual Average			
	1980–1989	1990–1999	2000–2007	2008–2012
FDI	1.0	1.5	2.9	2.9
Portfolio investment	1.2	2.3	4.2	1.4
Other investment	2.7	1.9	5.0	0.4
Reserve assets	0.8	0.5	1.2	1.5
Total	5.7	6.2	13.3	6.2

Source: James, McLoughlin, and Rankin (2014).

of International Finance, 2015). As a percentage of GDP, foreign capital inflows accounted for just 2%, down from the peak of 8% in 2007. Sharp declines were recorded in terms of PI and OI, particularly the flow of bank funds, while FDI has remained comparatively stable. Furthermore, foreign capital outflows from EMEs have also increased because of global financial market turbulence caused by US monetary policy uncertainty. Such conditions exacerbated pressures on reserve assets, exchange rates, and asset prices in EMEs. Consequently, in 2015, foreign capital flows to EMEs were negative for the first time since 1988, recording a net outflow totaling USD 540 billion.

The developments described above raise an important question: which factors influence foreign capital flows to EMEs? The literature review revealed a number of global push factors and domestic pull factors (Koepeke, 2015). The push factors include global economic growth, international interest rates, and the risk premium in global financial markets, while the pull factors include domestic economic growth, investment returns (such as interest rates, bond coupons, and stock dividends), exchange rates, and domestic risk indicators. In general, the relationship between foreign capital flows and the push and pull factors can be expressed as follows;

$$F_t = \alpha + \beta X_t + \delta Y_t + \phi Z_t \quad (6)$$

where F_t = foreign capital flows (FDI, PI, and OI), X_t = push factor vector, Y_t = pull factor vector, and Z_t = other factors, such as the foreign exchange system, the depth of the financial markets and institutional conditions (legal, political, and so on).

Numerous empirical studies have been performed to show which factors were more dominant in terms of influencing foreign capital flows to EMEs, with a different focus and findings. After foreign capital flows to EMEs accelerated in the wake of the crisis in México in the 1980s, for instance, Ghosh and Ostry (1993) proved that fundamental factors were more dominant in relation to the inflation stabilization program, privatization and capital market liberalization undertaken in EMEs. Meanwhile, Calvo, Leiderman, and Reinhart (1992) focused on the cyclical decline in the United States as the most dominant factor. Nevertheless, the continuation of foreign capital inflows to EMEs during the US economic recovery in the 1990s evidenced structural factors as the most dominant. The World Bank (1997) referred to this structural change in terms of the proliferation of institutional investors in advanced countries, financial market deregulation in EMEs and the impact of information technology. Thereafter, in the 2000s, Baek (2006) showed that the interest rate and risk premium were the most dominant push factors in terms of explaining PI trends. Meanwhile, De Vita and Kyaw (2008) demonstrated that pull factors, particularly productivity and competitiveness, were the dominant factors of FDI inflows to EMEs, especially Emerging Asia.

Several empirical studies have also explored the relative effect of push and pull factors for each type of foreign capital flow to EMEs. Of the three types of foreign capital flow, FDI was least affected by global economic fluctuations. Of the pull factors, domestic economic growth was the biggest draw for FDI. Meanwhile, asset yield indicators on financial markets (such as interest rates and dividends) were not an important factor because FDI is typically long term. In fact, aspects

of governance, including the investment climate, legal assurance, and political stability were significant FDI pull factors. Likewise, exchange rate stability was another important pull factor. Concerning the push factors, several empirical studies have shown that global economic growth was not always a significant factor. The positive impact of world economic growth was shown to be significant for vertical FDI, namely investment as part of the global or regional production chains, but not significant for horizontal FDI, namely to satiate the domestic markets. In terms of international interest rates, most empirical studies found an insignificant effect. Meanwhile, the impact of global risk indicators on FDI inflows was inconclusive, with some studies showing a positive effect of the Volatility Index (VIX) and others evidencing the adverse impact of country risk indicators.¹

In terms of PI to EMEs, the push factors were more dominant. All studies showed that PI to stocks and bonds in EMEs was significantly, adversely, and immediately affected by changes in the perception of global risk, as measured by the VIX or CDS spread.² Similarly, international interest rates (usually based on US interest rates) had an adverse and more dominant impact on bonds than stocks. In relation to the effect of economic growth in advanced countries, not many studies have demonstrated a significant direct impact, except improving the probability of PI inflows to EMEs (Forbes & Warnock, 2012). Of the pull factors, nearly all studies have shown domestic economic growth performance as an important pull factor, but the elasticity declines as the data frequency increases. Likewise, yield indicators were a critical pull factor, especially stock and bond yields. The monetary policy rate has not always been shown as an important factor because the effect has already been detected through bond yields and stock prices, except when the monetary authority implements a change of policy stance (Ahmed & Zlate, 2014), while real exchange rate volatility was shown to have a negative impact (Baek, 2006; World Bank, 1997). Indicators of external vulnerability also had an inverse impact, reflected by the external debt to GDP ratio (World Bank, 1997) and sovereign credit rating (Kim & Wu, 2008).

Foreign capital flows through the banking system (banking flows) were shown to respond negatively to the perception of global risks, along with interest rates in advanced countries, as the push factors. The impact was subsequently strengthened, however, by the risk-taking channel, where risk perception could exacerbate the impact of interest rates during periods of high global financial market volatility. Meanwhile, the influence of economic growth in advanced countries on banking flows proved inconclusive. Of the pull factors, strong evidence was provided for the effect of

¹VIX is an abbreviation for the Chicago Board of Exchange (CBOE) Volatility Index as a measure of volatility implied by Standard & Poor (S&P) index options. Colloquially referred to as the “fear index,” VIX measures market expectations of stock market volatility for the upcoming 30 days.

²CDS or credit default swap is a swap agreement where the CDS seller will compensate the buyer (in general, a creditor or bondholder) in the event of default by the borrower or bond issuer. Non-resident investors that invest in the bonds of EME can purchase CDS to hedge against default risk.

domestic economic growth, yields and risk indicators on banking flows to EMEs. The positive impact of yields was shown through various indicators, such as stock yields, exchange rate appreciation and, specifically, banking sector stock performance. Risk indicators, such as a decline in the external debt to GDP ratio or upgraded sovereign rating were also shown to significantly increase banking flows to EMEs.

In summary, the effect of push and pull factors on foreign capital flows to EMEs is presented in [Table 13.2](#).

In addition to the studies mentioned above, a number of empirical studies have also explored the effect of international investors' behavior on foreign capital flows to EMEs. Calvo and Reinhart (1996), for instance, showed that contagion could occur due to declines in domestic vulnerability indicators combined with the flight-to-safety phenomenon on global financial markets. The effect of contagion originates from changes in the behavior of investment managers due to changes in investment perception by institutional investors. Another study examined the impact of changes in investor behavior in terms of adjusting financial asset portfolio allocation due to yield indicator performance on domestic financial markets, otherwise known as rebalancing versus return chasing (Curcuro, Thomas, Warnock, & Wongswan, 2011). Meanwhile, the effect of asymmetric information and transaction costs were not detected in the analysis of push and pull factors (Portes & Rey, 2005).

The empirical studies have provided important inputs in terms of understanding the determinants of various types of foreign capital flows to EMEs. In general, FDI is more influenced by domestic economic fundamentals, while global factors have a greater impact on foreign capital flows in the form of PI and banking flows. Nevertheless, the relative effect of push and pull factors can change due to short-term developments in the global economy. Fratzscher (2012), for instance, showed that push factors were more dominant in terms of PI during the GFC, while PI was pushed more by macroeconomic fundamentals, institutional

Table 13.2: Determinants of Foreign Capital Flows: Push Versus Pull Factors.

		Portfolio Equity	Portfolio Debt	Banking Flows	FDI
PUSH	Global risk perception	Negative strong	Negative strong	Negative strong	Inconclusive
	Interest rates in advanced countries	Negative strong	Negative strong	Negative	Inconclusive
	GDP in advanced countries	Positive	Positive	Inconclusive	Inconclusive
PULL	Domestic GDP	Positive	Positive	Positive strong	Positive strong
	Domestic yields	Positive	Positive	Positive strong	Inconclusive
	Sovereign credit rating	Negative	Negative	Negative strong	Negative

factors and EME policy in the period following the GFC. Such conditions confirm the importance of maintaining economic fundamentals as well as instituting an appropriate and prudent macroeconomic policy response to maximize the benefits and minimize the risks of foreign capital flows.

13.3. Foreign Capital Flows and Economic Performance

There is a general belief, as explained by neoclassical theory, that foreign capital will flow from rich countries to poor and stimulate economic growth in both. Nonetheless, the empirical evidence does not always concur. In fact, during the GFC, foreign capital flows, which had previously flooded EMEs, reversed back to advanced countries. Despite recording gains since the GFC, foreign capital flows to EMEs and developing countries experienced a net outflow in 2015. There was also disparity among the EMEs and developing countries, namely most foreign capital flowed to Emerging Asia (predominantly China) and only a small portion went to Africa, which contradicts the neoclassical prediction that foreign capital flows would flow to countries with additional productivity from higher levels of capital per worker, in this case Africa, followed by Asia and then advanced countries.

13.3.1. Empirical Phenomenon: “Lucas Paradox”

Critical observations of cross-border foreign capital flows that contradict the neoclassical theory were raised by Lucas (1990), which subsequently became known as the Lucas Paradox. Namely, that globalization and cross-border financial international do not always benefit economic performance in developing countries and EMEs or the global economy. This issue has been more enthusiastically debated internationally since the GFC in relation to the global economic rebalancing agenda between advanced and developing countries. Lucas (1990) showed that average income (relative to the US) in net exporters of capital has declined well below that of net capital importers. This implies that capital indeed flows from poor countries to rich, not “downwards” but “upwards.” The combination of a US current account deficit and current account surplus in China is an important cause of why capital flowed upwards. Nonetheless, many advanced countries run a deficit and, similarly, many developing countries maintain a current account surplus. This is not a new phenomenon because similar trends were observed in the 1980s.

As explained by neoclassical theory, capital should flow to developing countries and EMEs that achieve robust growth and, therefore, are attractive draws for investment. Prasad, Rajan, and Subramanian (2006) analyzed foreign capital flows and economic growth trends in 59 developing countries for the period from 1970 to 2004.³ During this period, many advanced countries were experiencing a

³Growth trends were grouped from low to high and the 59 developing countries were divided into three categories with a balanced population, with China and India the only exceptions. After including the current account deficits, foreign capital flows of USD were calculated in real terms using CPI.

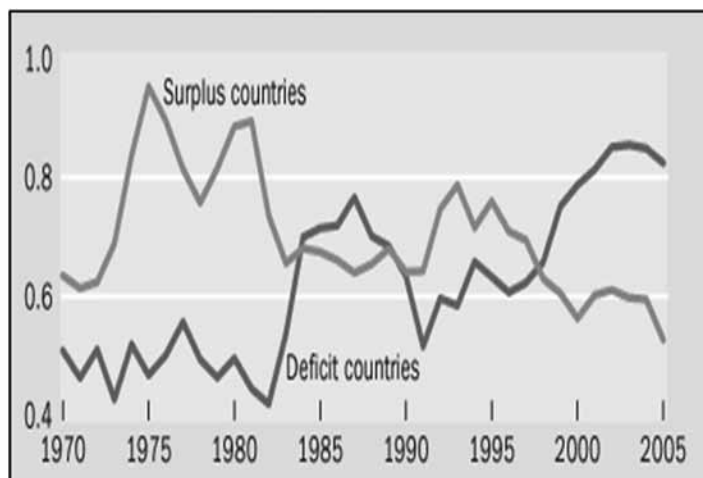


Fig. 13.1: Aggregate GDP per Capita of Capital Exporters (Weighted Current Account). *Source:* Prasad et al. (2006).

current account deficit, while many developing countries and EMEs were enjoying a current account surplus. As illustrated by Fig. 13.1, advanced countries (deficit countries) actually posted increases in income per capita, while the developing countries and EMEs (surplus countries) suffered declining income per capita. This implies, contradicting neoclassical predictions, that developing countries and EMEs, which were more dependent on foreign capital flows, did not achieve stronger economic growth in the long term.

Furthermore, among the developing countries and EMEs, as illustrated in Fig. 13.2 during the period from 1970 to 2004 net foreign capital flows to the group of countries with the highest economic growth was actually lower than to the low- and middle-income growth countries. The high-growth group of EMEs, averaging of 4.1% (yoy), received foreign capital flows totaling around USD400 million, while the medium- and low-growth countries, averaging 0.3% (yoy) and 1.6% (yoy), respectively, received around USD 900 billion and USD 600 billion in foreign capital flows. For a shorter observation period, the results were even more stark, with China and India as well as high and medium-growth EMEs all exporting capital to advanced countries, while the low-growth EMEs received significant foreign capital flows. Therefore, the flow of foreign capital contradicted the growth trend predicted by the neoclassical theory.

13.3.2. Important Takeaways from the Lucas Paradox

The Lucas paradox might be less controversial if analyzed in more depth. There were various issues discovered in developing countries and EMEs that meant the risk-adjusted return was not always the same as the measures of marginal productivity of capital and labor. In general, there are three explanations for the Lucas paradox. *First*, fundamental factors linked to different production structures

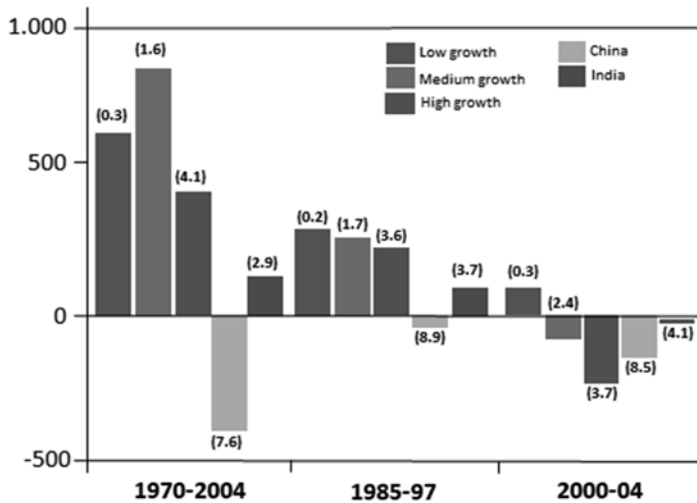


Fig. 13.2: Foreign Capital Flows to EMEs and Developing Countries (Annual Average in Billions of USD).

in the respective economies of each country. *Second*, domestic financial market deepening in terms of mobilizing savings and financial intermediation for productive financing in-country. *Third*, imperfect global financial markets, including asymmetric information and home-bias investors as well as disparate sovereign ratings and credit default risks.

13.3.2.1. Economic Fundamentals. The production and institutional structures in a particular country could create disparity between investment returns and the measures of marginal productivity of capital and labor. Several factors play an important role in the production structure, including technology, infrastructure, and other production factors, as well as government policy, such as tariffs, taxes, FX controls, and other impediments to investment, which affect total productivity but are not reflected in the measures of capital and labor productivity. Similarly, the institutional structure, including legal assurance and political stability, also influences the returns on investment from foreign capital flows.

Other than capital and labor, another important production factor is human capital as a factor of education, technology as well as research and development. Lucas (1990) included human capital in the production function, expressed as: $y = Ak^\beta h^\gamma$, where y is income per effective worker and h is human capital per worker. Basically, human capital was interpreted as a labor productivity multiplier at all skill levels, such as its function as a multiplier of total factor productivity. By including human capital, Lucas (1990) estimated the ratio of investment returns in India and the US. The results showed that the ratio of returns, which had previously been 58 times higher, was nearly the same if human capital was taken into consideration. Despite acknowledging that measuring human capital and the linkages with productivity was not easy, the analysis showed that one cause of the paradox of foreign capital flows and economic growth was a production factor that omitted land, human capital, and so on.

The institutional structure of each respective country also affected the desired returns (*ex ante*) and the actual returns (*ex post*). Although a number of technologies are available to various countries, implementation constraints, or the inefficient utilization of such technology have a significant impact of the rate of return of capital investment. Under such circumstances, similar yields of capital productivity per worker, measured using $A_{it}f^1(k_{it}) = r = A_{jt}f^1(k_{jt})$, do not mean that the capital will flow from rich countries to poor. One important contributing factor of total factor productivity in a country is institutional quality. The study by Alfaro, Kalemli-Ozcan, and Volosovych (2011) provided empirical evidence that institutional quality was an important factor to explain the Lucas paradox, after omitting other factors such as education, income per capita and constraints to capital flows. Likewise, Ju and Wei (2006) demonstrated the importance of financial system quality and the protection of copyright laws in terms of explaining the Lucas paradox. Such rationale and empirical evidence proves that policy to strengthen political stability, improve bureaucratic quality, ensure copyright protection, eradicate corruption, and strengthen law enforcement are critical factors to increase investment and productivity from foreign capital flows in terms of stimulating economic growth.

13.3.2.2. Domestic Financial Sector Deepening. The domestic financial sector plays a vital role in mobilizing savings and stimulating foreign capital flow intermediation efficiently for productive financing of economic growth. One neoclassical assumption is the close relationship between savings and investment, in terms of the total and productivity, to stimulate production gains and economic growth. The function of foreign capital flows should be complementary (not a substitution) to the savings and investment gap. If the domestic financial sector works efficiently, public funds will be mobilized optimally with an interest rate (after calculating the normal intermediation cost) that is relatively the same as the return on investment. Such conditions draw foreign capital flows to the country involved.

This rationale underlies the theoretical and empirical study performed by Gourinchas and Jeanne (2013), who explained the Lucas paradox due to a misallocation in the relationship between savings, investment and economic growth. Two causes could occur, namely an investment wedge or a saving wedge. The investment wedge encompasses the various distortions that cause non-resident investors to receive a lower return than the return on investment in the country involved due to the cost of bureaucracy, corruption, and barriers to the flow of foreign exchange. Meanwhile, the saving wedge encompasses various factors that distort the public's propensity to save in order to maximize their current and future levels of consumption, such as banking interest rate regulations and limited financial products for investment. The theoretical model developed was subsequently calibrated using data from 68 developing countries and EMEs for the period from 1980 to 2000 to analyze the relationship between net foreign capital flows, productivity, the investment wedge, and the saving wedge.

The study produced a number of interesting findings. *First*, consistent with the Lucas paradox, countries with a net inflow of foreign capital tended to have lower

total factor productivity.⁴ *Second*, countries with a high investment wedge tended to enjoy a productivity catch-up to the low average level of productivity of all 68 countries. In other words, the investment wedge tended to exacerbate the misallocation of foreign capital flows in terms of productivity catch-up, as demonstrated by the Lucas paradox. *Third*, countries with a high level of productivity catch-up were those that could mobilize domestic savings, while those countries that could not mobilize savings would remain left behind in terms of productivity. Therefore, the ability to optimally and efficiently mobilize domestic savings was shown to be a leading determinant of cross-border foreign capital flow misallocation and the ability of developing countries to boost investment in order to improve productivity and stimulate economic growth.

The study emphasized the importance of overcoming the various distortions in the relationship between savings and investment, especially in the domestic financial sector. A shallow financial sector could limit gains in terms of domestic savings and investment to stimulate economic growth. Therefore, the public's propensity to save for future consumption would also be restricted. Similarly, inefficient financial intermediation would make investment less responsive to productivity and economic growth. Misallocation could occur if the banks guaranteed a company with low productivity. A shallow financial system could encourage the public to save excessively for precautionary motives rather than for productive investment purposes.

13.3.2.3. Imperfect International Finance. Imperfect international finance creates deviation between the actual and desired capital returns, thus engendering distortions between foreign capital flows, investment, and economic growth. Obstfeld and Rogoff (2010), for instance, showed that the Lucas paradox occurred because of the relationship between international trade and finance. Consequently, $A_{it}f^1(k_{it})(1-\tau_{it}) = r = A_{jt}f^1(k_{jt})(1-\tau_{jt})$, where t reflects tax, foreign exchange controls, tariffs, transportation costs, and other constraints. Likewise, Razin, Rubenstein, and Sadka (2003) explained the large start-up cost as a cause of distortions in terms of foreign capital flows. That cost means investment in small firms becomes unfeasible despite achieving marginal productivity that could generate an attractive return for foreign capital flows.

Asymmetric information could also impede foreign capital flows despite high returns on investment, therefore investors preferred investing in their home country and in assets or countries that offer sufficient investment alternatives, a phenomenon known as home bias. Ahearn, Grier, and Warnock (2004) conducted a study on investor behavior in the United States in terms of offshore equity investments and proved that countries in which the corporate sector would fail to meet US regulations tended to receive a smaller portfolio allocation (underweight). Concerning FDI, a study by Levis, Muradoglu, and Vasileva (2007) showed that

⁴South Korea, for example, which achieved average TFP growth of 4.1% per annum and average rate of investment of 34% per annum, was not a recipient of large foreign capital flows, while Madagascar, with average TFP growth of 1.5% and average investment of less than 3% received foreign capital flows to the tune of around 7% of GDP.

corporate investors favored neighboring countries (home bias) or countries in the same economic union or with the same legal system. In other words, investors are more confident and upbeat about investing in countries that they already know despite the chance of higher returns in other countries.

Default risk is another factor that distorts foreign capital flows from investment congruent with the level of productivity. Reinhart and Rogoff (2004) explained default risk as the main reason why not much foreign capital will flow from advanced countries to poor countries. In effect, countries that experience default, generally the poorer countries, will find it difficult to borrow offshore. Foreign capital flows in the form of FDI and PI would also be low due to the negative reputation of the country after defaulting on a loan. This evidence is robust in terms of explaining the Lucas paradox. In other words, the problem is not so much the flow of foreign capital from rich countries to poor, but too much foreign capital flowing to countries with high default risk.

13.4. Foreign Capital Flows and Monetary Stability

Capital flow volatility, especially PI and short-term external debt, may induce risks to monetary, financial system, and macroeconomic stability in general. The risks are transmitted through various channels. Foreign capital flows directly influence supply and demand on the foreign exchange market and, therefore, capital flow volatility would directly affect exchange rate volatility. Capital flows as PI also influence the development of domestic financial asset prices, including stock prices and bond yields. Similarly, the impact of liquidity from capital flow volatility would exacerbate liquidity risk and undermine banking industry capacity to extend credit for the economy. Furthermore, capital flow volatility and the risk-taking behavior of international investors could intensify systemic risk in the domestic financial system.

13.4.1. Empirical Studies on the Effect of Capital Flow Volatility

Various empirical studies have shown the adverse effect of capital flow volatility on monetary, financial system and macroeconomic stability.⁵ Fuertes, Phylaktis, and Yan (2014), for instance, showed that capital flow volatility to EMEs was primarily attributable to flows of PI and short-term foreign loans, otherwise known as hot money. The study utilized monthly PI flow data (stocks and bonds) from the US to 18 EMEs (nine in Asia and nine in Latin America) for the period from 1988:1 to 2012:12, as well as quarterly data on banking flows for the same period. A Kalman filter was also used to separate temporary and short-term capital flows from permanent and long-term capital flows.

⁵Box 13.1 also contains several empirical studies that confirmed an increase in capital flow volatility due to global spillovers in the wake of the global financial crisis in 2008/2009, including the adverse impact of monetary policy uncertainty from the US Federal Reserve.

The empirical results of the study showed that flows of bank loans and PI from the US to EMEs in the 1990s were more permanent and long term, later usurped by temporary and short-term flows. The empirical findings were made even more robust with the inclusion of various push and pull factors, combined with other temporary and permanent decomposition techniques. The increasing domination of hot money contributed to the Asian Financial Crisis of 1997/1998 and served as the main conduit of global spillovers from the US subprime mortgage debacle that subsequently unfolded into the GFC, of which the adverse effects quickly spread to EMEs. Consequently, various macroeconomic and macroprudential policies were required, along with foreign CFM, to mitigate the harmful impact of short-term PI and bank loans on macroeconomic and financial system stability.

In a different study, Caporalea, Ali, Spagnolo, and Spagnolo (2015) provided empirical findings for the effect of equity portfolio flows (stocks and bonds) on exchange rate volatility in EMEs. The study was based on bilateral monthly data between the US and EMEs as well as developing countries in Asia (India, Indonesia, South Korea, Pakistan, Hong Kong, Thailand, the Philippines and Taiwan) for the period from 1993:01 to 2012:11. The results showed that such portfolio flows exacerbated exchange rate volatility. Moreover, using a Markov-switching model, the study also showed that a net stock (bond) portfolio flow precipitated high (low) exchange rate volatility. Specifically, bond portfolio flows raised the probability of low volatility in Pakistan, Thailand, and the Philippines, while in Indonesia the level of volatility remained high. Meanwhile, stock portfolio flows were shown to increase the probability of high volatility in India, Indonesia, South Korea, Hong Kong, and Taiwan. The empirical evidence provided by that study reinforced the return-chasing hypothesis for international investors and, consequently, foreign CFM of PI has become an effective instrument to stabilize the exchange rate.

Ananchotikul and Zhang (2014) also conducted a study regarding the effect of portfolio flows and global risks on asset prices in 17 EMEs using weekly portfolio flow data from the Emerging Portfolio Fund Research (EPFR) for the period from the beginning of 2013 until February 2014. Global risk perception was measured using the VIX, while asset prices included stock yields, bond yields and exchange rates. Three salient findings emerged from the study. *First*, PI flows had a significant impact on asset prices in the domestic financial markets, which increased during the GFC. The impact increased five- to tenfold during the period of global distress due to the simultaneous effect of a foreign capital reversal, tight market liquidity and herding behavior among investors. *Second*, the empirical results also demonstrated that global risk perception had a significant effect on all three components of asset prices in EMEs, with the magnitude depending on the specific characteristics of each country. The impact of VIX on stock price volatility was shown to increase as financial openness in the country increased, measured using total international financial obligations to GDP. As a country opened itself up to international capital flows, the impact of volatility from global risk spillovers to the domestic stock market also increased. Nevertheless, a similar pattern was not found for bond investments. The sensitivity of bond yield volatility to VIX was shown to correlate more closely with domestic economic fundamentals in terms of macroeconomic stability, particularly inflation and the current account, with a stronger influence as the duration of bond tenors increased.

In terms of exchange rate volatility, the impact of spillovers from perceived global risks depended on the exchange rate system applied, with the most significant impact affecting flexible exchange rate systems and a relatively more moderate effect in managed exchange rate systems. The empirical evidence showed the importance of exchange rate stabilization policy to mitigate to adverse impacts of global risk perception on exchange rate volatility. Intervention policy on the foreign exchange market could mitigate the impact of such global spillovers on exchange rate volatility, but must be supported by stabilization policy on the bond and stock markets. More importantly, solid macroeconomic fundamentals play a significant role in terms of strengthening resilience to international shocks. Specifically, maintaining low inflation and preventing a large current account deficit would significantly reduce domestic financial asset price sensitivity to global spillovers.

13.4.2. Monetary Policy Trilemma in an Open Economy

The various studies mentioned above showed that foreign capital flow volatility created additional complexity for the central bank, known as the monetary policy trilemma or impossible trinity. Chapter 4 explained a theoretical review of the impossible trinity, using the Mundell-Flemming model, Donbusch's overshooting model and the portfolio balance model. Basically, with perfect capital mobility, the exchange rate system will determine the ability of the central bank's monetary policy to achieve the domestic goals. The central bank must consider its preferences in terms of three policy goals that are not aligned, namely monetary autonomy for the domestic goals of price stability and economic growth, exchange rate stability or foreign capital flow mobility (Fig. 13.3). The simple view of the impossible trinity shows that a country embracing a free foreign exchange system but with a fixed exchange rate regime will lose its monetary policy autonomy to achieve the domestic economic goals. A more moderate view considers the trilemma as a policy trade-off and, therefore, the central bank must optimize monetary policy to control inflation with policy to stabilize the exchange rate and apply foreign CFM.

Obstfeld, Shambaugh, and Taylor (2005) studied the monetary policy trilemma in 103 countries covering a period of 130 years. Monetary policy autonomy was measured by comparing domestic interest rates with interest rates in other countries. Using a *de facto* and *de jure* approach as explained in Chapter 4, the exchange

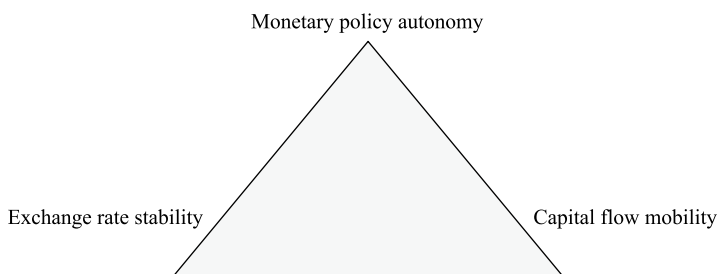


Fig. 13.3: Monetary Policy Trilemma.

rate system was categorized as a pegged system or non-pegged system, while the foreign exchange system was divided into capital mobility and capital control. Furthermore, the study was divided into three periods, namely the Gold Standard (1870–1914), Bretton-Woods (1959–1970), and post-Bretton-Woods (1973–2000). Based on the international theory of uncovered interest rate parity (UIP), the policy trilemma was proven in the study using the following equation:

$$\Delta R_{it} = \alpha + \beta \Delta R_{it}^* + \varepsilon_{it} \quad (7)$$

where R = domestic interest rate and R^* = comparable interest rate. Peer countries using a pegged system included: UK for the Gold Standard period, US for the Bretton-Woods period, and various countries in the post-Bretton-Woods era.

In general, the study confirmed that the policy trilemma was indeed valid as a monetary policy framework guideline in an open economy. *First*, the pegged exchange rate system was shown to create a closer link between the domestic interest rate and interest rate in a peer country rather than a non-pegged exchange rate regime. Interest rates in a pegged country tended to react more and have a stronger long-term correlation to the interest rate in the peer country. *Second*, by easing controls on foreign capital flows, pegged countries experienced a decline of monetary policy autonomy. Simultaneously, non-pegged countries enjoyed a greater degree of autonomy, particularly in the post-Bretton-Woods period, even under conditions of perfect capital mobility. *Third*, pegged countries, in reality, did not always actually peg their exchange rate due to devaluation or maintaining a range. On the other hand, non-pegged countries were also guilty of not always maintaining a floating exchange rate and often followed the peer country when setting the domestic interest rate.

Several other studies have shown how the central bank overcomes the policy trilemma. Calvo and Reinhart (2002), for instance, revealed a phenomenon they called a fear of floating in the exchange rate policy pursued by various countries. As explained in Chapter 4, classifying the exchange rate systems adopted in various countries showed that more and more countries are embracing an intermediate system rather than a hard peg or purely floating. The fear of floating is a way to overcome the policy trilemma by stabilizing the exchange rate. In fact, in countries that implemented a flexible exchange rate system, the central bank also intervened on the foreign exchange market to stabilize the exchange rate.

Several empirical studies have explored the extent to which monetary policy considered the exchange rate when determining the policy rate under the Inflation Targeting Framework (ITF). The methodology included the real exchange rate when determining the policy rate à la Taylor rule (Batini, Harrison, & Millard, 2003; Taylor, 2001) with the following model specification:

$$i_t = i_t^* + \rho i_{t-1} + (1 - \rho)[\lambda_1(\pi_t - \pi_t^*) + \lambda_2(y_t - y_t^*) + \lambda_3(q_t - \lambda_4 q_{t-1})] \quad (8)$$

where q_t = real exchange rate deviation in period t from the equilibrium. The specification enabled the researchers to examine the effect of volatility on determining the policy rate: (1) if $\lambda_4 = 0$, the central bank reacted symmetrically to deviations in the real exchange rate from equilibrium; (2) if $\lambda_4 = 1$, the central bank reacted

to fluctuations in the real exchange rate consistently by limiting exchange rate volatility; and (3) if $0 < \lambda_4 < 1$, the central bank gave a certain weight to rapid changes in the exchange rate and limited exchange rate misalignment.

Aizenman, Hutchison, and Noy (2011), for instance, studied a modified Taylor rule in the determination of the monetary policy rate using panel data for 16 EMEs during the period from 1989:Q1 to 2006:Q4. The results showed that ITF EMEs applied a strategy where the central bank would respond to inflation and the real exchange rate when setting the policy rate. A study by Mohanty and Klau (2004) confirmed similar results using a modified Taylor rule for 13 EMEs, where the central bank reacted to actual inflation, the output gap and exchange rate fluctuations. In general, the results proved that central banks reacted to real exchange rate volatility. All countries, with Chile the only exception, were shown to lean against the wind by raising the interest rate in the event of exchange rate depreciation. Furthermore, there was a high degree of persistence in the monetary policy response to exchange rates in South Korea, India, México, Peru, Thailand, and South Africa. The monetary policy response to the exchange rate was, in fact, larger than that to inflation and the output gap, further proof of the fear of floating.

Foreign exchange market intervention is a tool generally employed by the central bank for stabilization purposes to ensure exchange rate flexibility remains in line with economic fundamentals and supports the achievement of price stability. Theoretically, the effectiveness of foreign exchange intervention can be analyzed using the portfolio balance model, as discussed in Chapter 4. Most economists agree that sterilized intervention is effective in terms of influencing the exchange rate but the effectiveness of non-sterilized intervention is considered inconclusive from a theoretical and empirical standpoint. In this case, the effectiveness of sterilized intervention on the exchange rate works through the signal channel because intervention is seen by the markets as a change in the intention of the central bank's monetary policy as well as the portfolio balance channel because intervention affects the asset composition of investors. The latest study by Blanchard, Adler, and Filho (2015) explored the effectiveness of foreign exchange intervention, particularly sterilized foreign exchange intervention, as an instrument to mitigate exchange rate pressures originating from foreign capital flows. Consistent with the portfolio balance channel, the empirical findings of the study showed that relatively large intervention created comparatively lower exchange rate appreciation in response to foreign capital inflows. The magnitude of effect proved that foreign exchange intervention was a valid instrument of macroeconomic management.

The importance of exchange rate stabilization persuaded a number of economists, including Ostry, Ghosh, and Chamon (2012a), to opine that under certain conditions, dual targeting – namely exchange rate stabilization to support achievement of the inflation target – could strengthen monetary policy credibility at the central bank. The conditions are, based on ITF, that exchange rate targeting is used to mitigate the undesired impacts of capital flow shocks on inflation expectations, either directly through exchange rate pass-through or indirectly through domestic demand. With the effect of exchange rate fluctuations included in the inflation assessment and projections of the central bank, direct exchange rate targeting was more effective than an interest rate response to the impact of

the exchange rate on inflation. Ostry, Ghosh, and Chamon (2012b) recommended that central banks in EMEs pursued dual targets. Moreover, central banks in EMEs have access to a second instrument, namely foreign exchange intervention. In other words, dual target monetary policy, targeting inflation, and exchange rate stabilization, using two instruments, namely interest rates and foreign exchange intervention, would increase, not decrease, central bank credibility.

Several subsequent studies demonstrated the importance of foreign CFM to circumvent the policy trilemma. Klein and Shambaugh (2013) examined how far foreign CFM policy, such as directed and temporary capital controls or limited exchange rate flexibility, could realize full monetary policy autonomy. The results showed that foreign CFM under a flexible exchange rate regime could increase monetary policy autonomy. Nevertheless, foreign CFM under a fixed exchange rate system tended not to increase monetary policy autonomy compared to perfect capital mobility. As is widely practiced, extensive and long-term capital controls tend to break the relationship between domestic and international interest rates under a fixed exchange rate system. In addition, inflation is typically more controllable under a fixed exchange rate system and, therefore, the increase to monetary policy autonomy is not significant.

Rey (2013) also extolled the importance of foreign CFM to overcome the monetary policy trilemma. This was based on the increasing comovement of capital flows, asset prices and economic growth across borders in the global financial cycle. Consequently, the central bank was recommended to convert the policy trilemma into a dilemma, namely to strengthen monetary policy autonomy and exchange rate stabilization through direct capital controls or macro-prudential policy. The study revealed the dominant determinants of the global financial cycle, namely monetary policy in advanced countries, which influenced capital flow leverage, as well as credit growth in the international financial system. The impact was transmitted to monetary conditions in many countries, especially EMEs. In fact, interest rate policy and exchange rate flexibility were unable to protect a country's economy from the global financial cycle.

Obstfeld (2015) went further, stating that the impact of financial globalization has not only created a monetary policy trilemma but also a financial trilemma. The study evaluated the ability of monetary policy in EMEs to dampen the effect of global financial and monetary fluctuations. Consistent with the monetary policy trilemma, the study confirmed that EMEs with a flexible exchange rate were in a better position in terms of monetary policy autonomy. Nonetheless, changes in the exchange rate itself were unable to protect the domestic economy from global financial and monetary shocks. The important channels of international monetary and financial transmission operate through long-term interest rates and risk premiums, while financial globalization was shown not only to spur capital flows but also propagate the risks across jurisdictions. Although it would be possible to mitigate global spillovers through exchange rates, monetary policy cannot mitigate the exacerbating impact of capital flows and global financial risks on risks in the domestic financial system. The monetary policy trilemma has expanded into a financial trilemma, namely policy autonomy with global financial integration now a strong influence on maintaining domestic financial system stability.

13.4.3. Foreign Capital Flows and Monetary Policy Dynamics in Indonesia

In the case of Indonesia, as a small open economy, the deluge of foreign capital flows⁶ heightened the complexity of monetary control challenges. *First*, exchange rate fluctuations became more responsive to foreign capital flows rather than current account developments, which intensified domestic financial market volatility and, in turn, strengthened the transmission of further shocks (shock amplifier). *Second*, persistent foreign capital inflows reduced the potential effectiveness of monetary controls, considering the measures taken by Bank Indonesia to control liquidity in the economy were ultimately offset by the significant surge of foreign capital inflows.

The two salient challenges outlined above had fundamental implications for ITF-based monetary policy implementation in the context of a small open economy, like Indonesia, considering, on one hand, the assumed role of the exchange rate as a shock absorber was reversed, while, on the other hand, there was a tendency to orient monetary policy, directly and indirectly, toward managing exchange rate fluctuations within a certain range, in line with economic fundamentals. With the influx of foreign capital, a policy orientation toward managing external balances could be counterproductive to policy to manage the internal balances. Therefore, it is important to monitor the degree of monetary policy autonomy enjoyed by Bank Indonesia in terms of managing rupiah stability against external influences.

As is widely accepted, in line with changes in the global environment over the past few years due to behavioral changes in the financial system or the impact of financial crises in the last two decades, international monetary policy management in numerous EMEs has not been fully anchored to the impossible trinity, with the rationale starting to shift toward the need for a possible trinity.⁷ In Indonesia's case, the policy trilemma index estimated by Aizenman, Chinn, and Ito (2008) showed that during the past two decades, against a backdrop of increasing financial market integration (foreign capital flow dynamics), although depreciating on the period prior to the Asian Financial Crisis of 1997/1998 and monetary policy autonomy, the rupiah exchange rate tended to be more stable (Juhro, 2015). Such developments revealed misalignment between the theoretical perspective and empirical facts in Indonesia (Table 13.3).

Observations of foreign capital flow behavior (Net Foreign Assets – NFA), exchange rates and the interest rate differential also revealed the start of a shift in the monetary policy paradigm of Indonesia, moving away from the impossible

⁶Over the past few years, particularly after the GFC, the economy of Indonesia, as with other EME, faced the problem of excessive foreign capital flows. The period of widespread quantitative easing policy to exit from the crisis in various advanced countries, including the US, EU, and Japan, triggered an influx of foreign capital to Indonesia's economy. In contrast, however, subsequent indications of monetary policy normalization reduced foreign capital inflows and heightened the risk of a sudden capital reversal.

⁷The possible trinity implies that sub-optimal monetary policy (not independent) could be formulated by restricting overly flexible exchange rate fluctuations, coupled with measures to restrict the mobility of foreign capital flows.

Table 13.3: Monetary Policy Trilemma Index in Indonesia.

Policy Trilemma Index	1997–2000 Asian Crisis	2001–2005 ITF Transition	2006–2008 Pre-crisis ITF	2009–2013 Global Financial Crisis
Exchange rate stability	0.11	0.27	0.25	0.28
Monetary policy autonomy	0.45	0.30	0.50	0.57
Financial market integration	0.74	0.69	0.69	0.71

Sources: Juhro (2015, spring); Juhro & Goeltom (2015, February)

Table 13.4: A Decomposition of the Dynamic Correlation between Exchange Rates Versus Foreign Capital Inflows and Interest Rates.

Decomposition of the Correlation with Exchange Rates	1997–2000 Asian Crisis	2001–2005 ITF Transition	2006–2008 Pre-crisis ITF	2009–2013 Global Financial Crisis
Foreign capital inflows (NFA)	0.86	0.74	0.56	0.53
Interest differential	0.14	0.26	0.44	0.47

Source: Juhro & Goeltom (2015, February)

trinity. Rolling standard deviation data was used to show the trend of comovement between the three variables, which have evolved over time since the introduction of a floating exchange rate system in August 1997. This implies a decline in the degree of monetary policy autonomy in Indonesia due, among others, to monetary policy orientation, which not only strived to control inflation per se, but also to manage foreign capital flows and maintain rupiah exchange rate stability through active intervention. In other words, exchange rate dynamics were not completely removed or influenced by market forces, but strongly influenced by domestic monetary policy. Fundamentally, this phenomenon was also reported in numerous EMEs.

Quantitatively, a decomposition of comovement between the dynamics of exchange rate fluctuations and foreign capital flows experienced a significant decline from 86% to 54%, while comovement between the dynamics of exchange rate fluctuations and interest rate differential experienced a significant increase from 14% to 46% (Table 13.4).

The decline in the degree of monetary policy autonomy also represents another portrait of this unique phenomenon, where a surge of foreign capital flows could undermine the effectiveness of monetary controls. This could occur when the central bank intervenes with foreign capital inflows and subsequently increases liquidity in the domestic economy, which must be reabsorbed (sterilized) in order to mitigate the inflationary pressures from total money supply growth. Nonetheless, measures to control liquidity will cause the interest rate to rise

(and interest rate differential), thus drawing more foreign capital inflows to offset the decline of total domestic liquidity.

For Indonesia's case, Juhro (2010) examined the magnitude of offset and sterilization for the period from 2000 to 2010 through the reaction function of the central bank's monetary policy, referring to Kouri-Porter (1974) and Cumby and Obstfeld (1982). An offset coefficient of 0.7 showed that the high degree of economic openness in Indonesia allowed external factors to subvert monetary policy implementation by Bank Indonesia because around 70% of the liquidity absorbed by Bank Indonesia (monetary controls) was offset by more foreign capital inflows. On the other hand, however, a comparatively low sterilization coefficient of 0.5 implied that around 50% of rupiah liquidity expansion, stemming from foreign capital flow intervention policy, was reabsorbed (neutralized) by Bank Indonesia. The relatively high offset coefficient compared to the sterilization coefficient was symptomatic of a low degree of monetary policy autonomy in Indonesia in terms of controlling foreign capital flow dynamics.

There are at least three fundamental policy implications. The *first* relates to how to formulate an optimal possible trinity against a backdrop of ubiquitous uncertainty blighting the international environment. To that end, calculations on the use of monetary policy instruments, exchange rate developments and reserve asset management cannot merely be based on macroeconomic considerations but also integrated with the microstructure of the financial markets. The *second* relates to efforts to formulate an optimal strategy of monetary control instruments within the ITF despite the relatively limited role played by the interest rate. With the deluge of foreign capital inflows and excess liquidity on the financial markets, non-interest rate instruments (such as reserve requirements and other macroprudential instruments) may be used as the main or supporting instruments to control liquidity in the economy. The *third* relates to the need for coordinated liquidity control measures on the domestic financial markets by managing potential foreign capital inflows. Coordination is critical, not only in terms of efforts to unwind external and internal imbalances but also to manage the impact of monetary policy optimally, without exacerbating or negating conditions.

The need for Bank Indonesia's monetary policy response to symmetrically react to deviation in the real exchange rate from the equilibrium value was supported by the data. Juhro and Mochtar (2009) performed a simple analysis of Bank Indonesia's monetary policy response for the period from 2000 to 2009, with and without including real exchange rate dynamics in the Taylor rule. The specification of the simple standard Taylor rule was expressed as follows:

$$i_t = \delta + \rho i_{t-1} + (1 - \rho) [\gamma_\pi \pi_t + \gamma_y y_t] \quad (9)$$

where $\delta = (1 - \rho)\alpha$

Then, by including the ad hoc role of the exchange rate in the Taylor rule, the hypothetical reaction function of the enhanced Taylor rule, with the real exchange rate, was formulated as follows:

$$i_t = \delta + \rho i_{t-1} + (1 - \rho) \left[\gamma_\pi \pi_t + \gamma_y y_t + \sum_{k=0}^{k=K} \gamma_{ek} e_{t-k} \right] \quad (10)$$

where e_t is the expression of exchange rate deviation from the equilibrium value and represented by the real exchange rate. If the monetary policy regime adopted is pure ITF, hypothetically $\gamma_{e0} = \dots = \gamma_{eK} = 0$. The adjusted rule represents an interesting format because it takes into consideration the flexibility of the exchange rate's role (enhanced rule). Despite some uncertainty concerning the application of that format, which is considered inconsistent with the basic rationale of ITF, empirical observations of economies with large exchange rate pass-through as well as relatively high and unstable inflation demonstrated the feasibility of the enhanced rule (Edwards, 2006). Congruently, Bask (2006) concluded that, technically, for small open economies, the addition of an exchange rate variable in the design of the Taylor rule could facilitate system stability, as long as the data used in the rule was contemporaneous. In the case of lagged data, such an addition would be unnecessary.

Solving the two aforementioned equations produced three salient conclusions. *First*, despite different coefficients of determination (*R*-squared), namely 90% for the simple rule and 98% for the enhanced rule, it was generally concluded that the behavior of both rules was adequately explained by the data. The independent variables were in line with the theoretical predictions and statistically significant. All parameters in the two equations, namely inflation deviation, output gap, (effective) real exchange rate, and interest rate lag, were statistically significant at 5%. *Second*, inflation deviation was two to three times higher than output growth, therefore, ITF implementation in Indonesia was consistent with the fundamental rule of ITF, that controlling inflation was the overarching priority of monetary policy implementation. Furthermore, the large autoregressive interest rate parameter (0.7–0.8) indicated a conservative monetary policy response through interest rate smoothing (gradualism). *Third*, the short-term impact of real exchange rates was estimated from the design of the enhanced rule at 0.3 for the case of Indonesia, compared to zero in Chile but 0.8 in México. In general, the results provided sufficient justification to include the behavior of exchange rates in the implementation of monetary policy based on the ITF, which was supported by the data. In fact, during certain periods, the policy response in the design of the enhanced rule allowed a more optimal countercyclical policy response compared to the simple rule.

13.5. Foreign Capital Flow Management: Theories and Practices

A theoretical and empirical review of the various countries mentioned in previous sections showed that foreign capital flows, in addition to stimulating economic growth, could also trigger macroeconomic and financial system stability risks. Consequently, a mix of macroeconomic policies (fiscal and monetary), financial system stability and structural reforms are critical at the national level. The combination of the policy mix will depend on local conditions in the country involved. Furthermore, structural reform policy is crucial to amplify the benefits

of foreign capital flows in terms of productivity and economic growth. Macroeconomic policy must also be adjusted, especially during periods of pressure on macroeconomic stability, such as high inflation or a wide current account deficit. Consequently, prudent and consistent monetary and fiscal policies play an important role as the first line of defense against foreign capital flows. Thereafter, adequate FX reserves, coupled with bilateral and multilateral swap arrangements, also play an important role as the second line of defense. In addition, strengthening and deepening the financial sector, as well as improving institutional capacity have been shown to improve a country's ability to face foreign capital flows.

13.5.1. Principles, Targets, and Instruments

In the context of the central bank, a monetary policy mix, exchange rate stabilization, and foreign CFM are required to achieve an optimal balance in terms of addressing the policy trilemma described above. Experience since the global financial crisis of 2008/2009 has shown that interest rates alone, as the main policy instrument, cannot respond to the volatility and magnitude of foreign capital flows to EMEs, even when complemented with foreign exchange intervention as a secondary instrument. In the face of an influx of capital flows, for instance, foreign exchange intervention would also increase the accumulation of reserve assets. Despite reinforcing the external resilience of EMEs, the cost of accumulating FX reserves is large because the central bank is, therefore, required to sterilize the resultant liquidity. Under such conditions, foreign CFM must be considered by the central bank. Foreign CFM is also required in the event of large foreign capital outflows because the interest rate response is not always effective, while foreign exchange intervention could also restrict the scope of the reserve assets owned by the central bank.

13.5.1.1. Formulation and Implementation Principles. In general, the following conditions serve as a reference for when foreign CFM is applicable (IMF, 2012). *First*, foreign CFM is required when the space to make further macroeconomic policy adjustments becomes more limited, for instance after tightening monetary or fiscal policy, but foreign capital inflows remain beyond control. Such conditions are typically found during an episode of economic overheating or when signs appear of asset price bubbles, accumulating external debt, overvalued exchange rates as well as excessive and expensive reserve assets.

Second, foreign CFM is required if there is a lag in the macroeconomic policy adjustment process or in terms of the economic impact. For example, a change in the budget as part of fiscal policy generally requires time for parliamentary approval. Monetary policy effectiveness could also be delayed by the transmission mechanism in terms of influencing inflation expectations and domestic demand.

Third, foreign CFM is also required if the surge in foreign capital inflows amplifies the risk of financial system instability. Macroprudential policy would be more appropriate for overcoming systemic risk that does not originate from foreign capital flows. Nevertheless, an influx of foreign capital could also prompt excessive lending due to expansive liquidity conditions as well as increasing risk-taking behavior. Under such conditions, restricting foreign capital flows through

foreign CFM supports macroeconomic, monetary, and macroprudential policy effectiveness in terms of maintaining macroeconomic stability, monetary stability, and financial system stability.

Foreign CFM implementation must take into consideration aspects of effectiveness and efficiency (IMF, 2013a). Likewise, the design and implementation of foreign CFM must be transparent, targeted, temporary and, wherever possible, must not discriminate between residents and non-residents.

- *Transparent and targeted:* clear communication on the targets and instruments of foreign CFM is critical to avoid market distortions and misinformed public expectations. Foreign CFM, targeting certain aspects that pose the most risks, such as PI flows as well as short-term external debt and speculation, is considered more effective. Striking a balance between the scope, effectiveness and side-effects of foreign CFM requires an assessment specific to the country involved. Targeted foreign CFM, therefore, is easier to monitor and avoids triggering the undesired side-effects.
- *Temporary:* As applied, foreign CFM could be tightened if foreign capital inflows increase and loosened as the flows ebb or even reverse. Foreign CFM can be maintained as required to support financial system stability, but not the BOP.
- *Non-discriminatory:* In general, foreign CFM that does not discriminate between residents and non-residents is preferred. Nevertheless, if such efforts undermine the effectiveness of foreign CFM, instruments that discriminate against non-residents (otherwise known as capital controls) may be applied. The preference for non-discriminatory foreign CFM is based on the fair and impartial treatment expected of all IMF members.

13.5.1.2. Targets: Macroeconomic Stability and/or Financial System Stability.

Foreign CFM requires a clear target, whether it be macroeconomic stability, financial system stability or both. The answer will depend upon the types and volatility of the capital flows, combined with the economic fundamentals and financial sector dynamics in the respective country. As mentioned previously, volatile PI flows impact exchange rate, stock price, and bond yield volatility. If a country is not suffering from inflation and current account issues, sound monetary policy may be implemented through the interest rate or foreign exchange market intervention to mitigate asset price volatility. In this case, foreign CFM to achieve macroeconomic stability would support monetary policy, especially if inflation is relatively low and the position of FX reserves is inadequate to stabilize the exchange rates. Nonetheless, if inflation and the current account deficit are high, foreign CFM is required to complement the macroeconomic policies required, through monetary policy, fiscal policy or policies in the real sector.

Under other circumstances, however, capital flow volatility may not only threaten macroeconomic stability but also spur pressures on financial system stability. Excessive asset price volatility would exacerbate market risk. Similarly, pressures on financial system stability may also appear from expansive liquidity and bank lending due to excessive foreign capital inflows. Furthermore, external debt in the banking industry would not only impact macroeconomic stability

but also financial system stability. The extent of the impact would depend upon financial sector resilience as well as the effectiveness of the macroprudential and microprudential policies instituted. In this regard, foreign CFM to control capital flows that targeted macroeconomic stability would support macroprudential policy effectiveness in terms of maintaining financial system stability.

Occasionally, there is the perception that foreign CFM encompasses macroprudential policy, even though they both have different targets. Foreign CFM aims to restrict capital flows, while macroprudential policy aims to mitigate systemic risk and maintain financial system stability. Besides, foreign CFM is oriented toward controlling the total and composition of foreign capital flows, while macroprudential policy aims to mitigate any accumulation of systemic risk, irrespective of whether the risks originated externally or domestically. For example, taxes imposed on certain types of foreign capital flows constitute foreign CFM and only indirectly affect financial system stability. On the other hand, the additional capital surcharge and countercyclical buffer imposed on systemically important financial institutions are macroprudential policies and only indirectly affect capital flows.

Nevertheless, there are several situations where foreign CFM and macroprudential policy become mutually complementary (IMF, 2015a). If the capital flows trigger systemic risk in the financial sector, both foreign CFM and macroprudential policy may be used. For example, capital flows to the banking sector in the form of external debt or PI could prompt a credit boom and domestic asset price bubbles. Regulations to limit external debt in the banking industry, through reserve requirements, debt-to-capital ratio, net open position (NOP) or even applying different risk weights to capital requirements, could restrict capital flows and dampen exchange rate volatility, which would simultaneously control excess liquidity in the banking industry as well as excess credit growth and help prevent domestic asset price bubbles from forming. Under such conditions, the regulations instituted to control capital flows and mitigate systemic risk in the financial sector may be considered as foreign CFM and macroprudential policy.

Another important consideration is how and when to unwind or exit from foreign CFM and macroprudential policy. When the capital flows have ebbed and the associated volatility has subsided, foreign CFM could become an undesirable economic expense or even become ineffective and must, therefore, be removed. Nonetheless, certain macroprudential policies should be maintained to mitigate systemic risk in the financial sector that may persist after the influx of capital flows and period of high volatility. It is important to formulate macroprudential policy that targets financial system stability and not to control capital flows, a topic that will be discussed in Chapter 14. Additionally, in the case of persistent or permanent foreign capital flows, further reform policies would be required to deepen the financial sector, boost investment, and productivity in the real sector and strengthen the institutional aspects.

13.5.1.3. Instrument Selection: Prudential Regulations or Capital Controls.

Which instruments are available to mitigate the risks associated with foreign capital flows? In general, the policy options depend on the conditions specific to the country involved. Several instruments are available to control capital flows in the

form of administrative instruments, including capital controls and prudential regulations. Administrative instruments directly target specific types of capital flows, while prudential regulations aim to improve the ability of the financial or corporate sector to mitigate the associated risks.

The administrative instruments of foreign CFM are instituted through certain policies targeting the capital transactions of residents or non-residents in the form of specific taxes imposed on the capital flows of non-residents, additional reserve requirements without interest remuneration, special licensing policies, and even restrictions or outright bans. Such regulations could be applied to all capital flows, or specifically target flows based on type or period (debt, equity, FDI; short term vs long term). Furthermore, the regulations could target the economy as a whole or specific sectors (normally the financial sector) or specific industries (e.g., strategic industries). In general, the terminology of foreign CFM implies non-discriminatory instruments, meaning the requirements should apply to residents and non-residents alike. Nevertheless, if there are differences between residents and non-residents, the term capital controls is used. For example, a higher reserve requirement applied to non-residents than residents would be a form of capital control but differentiating the reserve requirement based on currency, applicable to residents and non-residents alike, would constitute foreign CFM.

In general, prudential foreign CFM regulations target foreign exchange transactions based on currency and not the residency of the parties engaged in transactions. The regulations are imposed on domestic financial institutions, specifically the banking industry. The instruments include restrictions on the NOP based on capital or restrictions on banking sector investment in foreign currency assets. The banks may also be limited in terms of disbursing foreign currency loans, particularly to customers that do not hedge, or face different reserve requirements on obligations in the domestic currency and a foreign currency. In addition to financial institutions, prudential foreign CFM regulations may also be applied to the corporate sector, primarily to mitigate the risks associated with external debt. The corporate sector may have to meet hedging regulations to mitigate currency risk and liquidity regulations to ensure repayment capacity as well as a maximum debt-to-equity ratio or minimum credit rating to mitigate corporate default risk.

The various foreign CFM instruments are typically applied to control foreign capital inflows. There are, however, a few regulations that can be used to mitigate the adverse impact of foreign capital outflows on macroeconomic stability (IMF, 2015b). For example, taxes or reserve requirements may be loosened or removed altogether to overcome the risk of a capital reversal, depending on the intensity of the problems faced. Several other foreign CFM instruments, particularly the prudential regulations, such as the NOP and restrictions on foreign loans, may be maintained to strengthen risk management for capital flow volatility. In addition, other foreign CFM instruments are also available to mitigate foreign capital outflows, including: (1) restrictions on resident investments in offshore financial instruments (Iceland); (2) a minimum holding period for non-residents to sell their investments (Chile, 1990s); a minimum holding period before transferring the proceeds of an investment sale (Ukraine); a tax on the transfer of the proceeds of an investment sale (Malaysia); (3) restrictions on converting and

transferring assets in the domestic currency (Iceland) as well as restrictions on withdrawing term deposits (Argentina, Greece). Restricting non-resident access to the local currency would also make speculation more difficult.

Similar to the case of foreign capital inflows, the effectiveness of foreign CFM instruments to control foreign capital outflows would be improved by integrating the measures into a comprehensive policy package, supported institutionally and by a healthy legal system. In other words, foreign capital outflow management is not a replacement for sound macroeconomic policy, particularly if the foreign capital outflows are due to domestic economic fragilities. Specifically, foreign capital outflow management would be more effective if formulated appropriately and implemented consistently. To be effective, comprehensive foreign CFM is necessary, tailored to the specific problems faced.

13.5.2. Foreign CFM Practices

The previous section dealt with various foreign CFM instruments together with the macroprudential policies available to the central bank in order to mitigate the adverse impacts of capital flows on monetary and financial system stability. How could the various instruments be applied to EMEs and how effective would they be? The answer depends on the specific conditions of the country involved, particularly the complexity of the problems faced and the ability of the relevant authority to respond.

In general, there are a number of key takeaways. *First*, foreign CFM instruments are typically not applied across the board but are targeted and adjusted over time (either loosened or tightened) depending on the issues faced. *Second*, instrument choice depends on the institutional arrangements and jurisdiction of the authorities in each respective country. In general, the instruments under the preserve of the central bank are simpler and faster to implement or adjust as required. *Third*, the effectiveness of foreign CFM instruments is influenced by the ability of the relevant authorities to detect regulatory loopholes exploited by the banking sector and market players early and, therefore, the authorities must be ready to refine the regulations as necessary. *Fourth*, foreign CFM instruments and macroprudential policy are complementary and, therefore, oftentimes implemented in tandem to enhance the effectiveness of controlling capital flows. And, most importantly, prudent macroprudential policy, on the monetary and fiscal sides, along with structural reforms to increase economic and financial sector capacity is the most optimal response in the face of foreign capital flows.

13.5.2.1. Foreign CFM Effectiveness: Brazil and Colombia. Brazil and Colombia apply foreign CFM to mitigate the undesired impact on competitiveness and monetary policy autonomy (Ostry et al., 2011). In 2007, Colombia imposed unremunerated reserve requirements (URR) of 40% at six months on external borrowing as well as portfolio inflows, stocks and bonds. The URR was increased to 50% in May 2008 and the scope expanded to inflows of FDI to remain in the country for a minimum of two years in order to overcome strong exchange rate appreciation. Brazil, in March 2008, imposed a 1.5% tax on portfolio inflows to stocks and

bonds, which was subsequently removed due the onset of the global financial crisis in 2008. In October 2009, however, the tax was reintroduced at 2% on portfolio inflows to stocks and bonds but the rate was lowered to 1.5% if the non-resident investors converted American Deposit Receipts (ADR) into domestic stocks.

Although the instruments used were different, there were a number of similarities between foreign CFM implementation in Brazil and Colombia. Despite initially only targeting foreign loans using URR in Colombia, both countries subsequently expanded the scope to include portfolio flows to stocks and bonds in order to prevent non-resident investors from entering the stock market and then converting to bonds. From the beginning, Colombia exempted stock issuances through ADR, while Brazil initially imposed the same regulation but then changed to a tax on ADR issuances to close the regulatory loophole by converting ADR to local stocks.

Onshore and offshore derivatives markets began to develop in both countries to circumvent the regulations. For example, because the URR and tax were imposed on a mandatory margin on a derivative transaction, foreign investors tended to seek profit by selling dollar forwards and speculating on exchange rate fluctuations (carry trade). Likewise, non-resident investors could cooperate with the domestic banks through branch offices operating internationally because such transactions were not subject to the URR or tax requirements. Foreign investor loans to international branches could subsequently be exchanged into the domestic currency and then used to repay the loan to the foreign investor. Colombia closed this loophole by restricting the banks' position on derivative instruments, a good example of prudential regulations to strengthen the URR instrument. Meanwhile, the tax instrument used in Brazil was unable to close the loophole through such a derivative instrument. In other words, foreign CFM effectiveness was also influenced by the ability of the authorities to detect and close loopholes exploited by the markets through other foreign CFM instruments or with prudential regulations.

13.5.2.2. Macprudential Regulations on Capital Flows: South Korea. In South Korea, exporters – particularly shipbuilding companies with scheduled dollar receipts – were required to hedge against future export proceeds. Such exporters wished to sell their dollars through forward contracts to receive Korean won to pay for their operating activities. When strong won appreciation was expected, however, exporters tended to over-hedge in order to profit from a level of appreciation that exceeded the interest rate differential. Domestic banks (typically foreign bank branches) as the counterparty, purchased dollar forwards and sold Korean won forwards, which were paid for by loans from banks abroad, including subsidiary banks, exchanged into Korean won and invested in domestic assets to fulfill the forward to the exporter. In this case, domestic banks were fully protected from foreign currency risk and sought profit from carry trade on exchange rate fluctuations.

To mitigate the rapid increase of short-term foreign loans and its impact on currency risk and financial system stability, in June 2010, the Bank of Korea promulgated a comprehensive package of prudential regulations to prevent over-hedging and sudden stop risk from the short-term loans. The package strengthened regulations issued in November 2009, requiring banks

to increase their share of long-term foreign loans from 80% to 90% and then to 100%. In addition, the 2010 package also contained restrictions on the banks' position of foreign exchange derivatives, tight restrictions on provisions for foreign currency loans and a more stringent liquidity ratio imposed on domestic banks.

The various regulations outlined above successfully restricted foreign capital flows in the form of short-term foreign loans and bank exposure to foreign exchange. The regulations were imposed on domestic banks; therefore, the exporters were still able to meet their hedging requirements through offshore banks. To that end, the offshore banks would generally cover their short position on Korean won from non-deliverable forwards (NDF) through investments in government bond portfolios. In other words, although the regulations successfully limited foreign loans in the domestic banking sector, foreign capital inflows to South Korea were maintained in other forms, including government bonds.

13.5.2.3. Prudential Regulations on Foreign Loans: Croatia. Foreign banks have dominated the Croatian banking system since the year 2000 and have played an important role in terms of foreign capital flows to the country. A surge of capital flows prompted issues with macroeconomic and financial system stability, particularly the large share of household consumer loans. The problems became more complex because the central bank of Croatia (Croatian National Bank – CNB), as a member of the Eurosystem, had limited access to monetary instruments, while fiscal policy was constrained by structural budget problems. Consequently, the CNB had to rely on prudential regulations imposed on the financial sector, including foreign CFM instruments and capital controls.

In 2003, the CNB imposed limits on credit growth disbursed by the banking industry and required a minimum level of retained earnings to be maintained if the limits were exceeded. Nonetheless, the banks were able to bypass the regulations by selling a portion of their credit portfolio to an affiliated leasing company, thereby transferring the risk to the balance sheet of the parent bank abroad. In July 2004, the CNB changed the regulations into capital controls on domestic banks by introducing URR in the form of a marginal reserve requirement (MRR) to be deposited at the CNB if the position of foreign currency liabilities exceeded the upper bound as of June 2004. In the subsequent period, the CNB tightened the URR regulation and expanded its scope to close the loophole. In addition, the capital controls were also tightened. Consequently, the banks were required to meet a minimum Capital Adequacy Ratio (CAR) of 12% if credit growth was maintained below 12%, with CAR progressively increasing if credit growth accelerated.

The experiences of Croatia show how difficult it can be to control foreign capital flows if the central bank has limited space to maneuver as part of a monetary union and fiscal policy is also constrained. This implies that prudent macroeconomic policy, as the first line of defense, would be unavailable. Consequently, capital flows expose the domestic economic and financial cycles to the global financial cycle. In fact, economic advancement in Croatia created complex problems as foreign capital flows increased. As a member of the Eurosystem, Croatia was also not in a position to use exchange rate policy to control the foreign capital flows. Prudential regulations and capital controls successfully lowered credit growth in the banking industry and temporarily eased the deluge of capital flows.

Nevertheless, credit growth remained high because domestic loans were replaced by direct foreign loans, thus changing the structure of foreign capital flows that went beyond the reach of prudential regulations and capital controls.

13.6. Concluding Remarks

This chapter has discussed various aspects relating to the surge and dynamism of foreign capital flows in line with increasing global economic and financial integration. The benefits and risks of capital flows depended of the conditions specific to each respective country. Macroeconomic resilience, particularly low inflation, a sound current account, controlled fiscal deficit, and healthy financial sector, especially the banking sector, are critical. An appropriate macroeconomic policy response, from the monetary or fiscal side, is necessary to ensure that domestic economic fundamentals are maintained. Equally important is structural reform policy, encompassing the investment climate, increasing productivity, and competitiveness, including technology and human capital, as well as institutional strengthening of the government sector and corporate sector. Such measures would redirect the foreign capital flows to long-term and productive investments, hence supporting economic growth. Financial market deepening is also important to mobilize domestic savings deposits and stimulate more efficient intermediation of foreign capital flows to productive economic financing.

From the central bank's perspective, a mix of interest rate policy, exchange rate stabilization, and foreign CFM would produce the most optimal results in terms of monetary and financial system stability. Interest rate policy must still be oriented toward achieving price stability by paying due consideration to general economic and financial developments. Furthermore, exchange rate flexibility could help as a domestic economic shock absorber to external shocks. Notwithstanding, if exchange rate fluctuations and volatility deviate from their fundamentals, foreign exchange intervention would be necessary to achieve price stability and, therefore, dampen the impact on economic growth. Foreign CFM and macroprudential policy, individually or combined, could be implemented to mitigate the impact of capital flows on monetary and financial system stability.

This chapter has also demonstrated the close relationship between monetary stability and financial system stability. Global financial integration has tightened the linkages between the domestic economic and financial cycles with the global economic and financial cycles. Foreign capital flows not only affect exchange rates and other financial asset prices, but also impact liquidity conditions and credit growth in the banking sector. Furthermore, global risks also influence capital flow volatility as well as the behavior of the banking industry and domestic financial markets. Monetary stability not only correlates closely with financial system stability, but both are heavily influenced by global financial and economic developments. Consequently, monetary and macroprudential policy integration has become more important in terms of strengthening monetary and financial system resilience in the face of greater global financial integration. Chapters 14 and 15 will provide a detailed and comprehensive discussion on financial system stability, the macroprudential policy response required as well as integration with monetary policy in a central bank policy mix.

Box 13.1: The Impact of Global Spillovers from US Monetary Policy

The global financial crisis of 2008/2009 precipitated extraordinary foreign capital flows to emerging market economies in terms of their scale and volatility. Several studies have examined the impact of global spillovers from the monetary policy of the US Federal Reserve. Two periods, in particular, have received special attention, namely the period of unconventional monetary policy (UMP) after the global financial crisis to stimulate the US recovery, and the period of monetary policy normalisation from May 2013 until the Fed's proposal to hike the Federal Funds Rate (FFR).

In general, the impact of global spillovers from the Fed's UMP on foreign capital flows, asset prices and economic activity in emerging market economies was transmitted through various channels (Lavigne, Sarker, & Vasishta, 2014). *First*, the portfolio balance channel, namely the purchase of long-term assets, such as government bonds and mortgage-backed securities (MBS) by the Fed, which reduced supply and the risk premium of such assets on the market. *Second*, the signaling channel, namely Fed commitment to loosen monetary policy after the global financial crisis, which lowered long-term bond yields. A combination of those two factors pushed investors to adjust their portfolio composition through the purchase of assets in emerging market economies with a higher risk-adjusted return, otherwise known as rebalancing versus return chasing. *Third*, the exchange rate channel, namely expansionary monetary policy by the Fed, which led to USD depreciation. In addition to the deluge of foreign capital flows to emerging market economies, such developments decreased US imports and EME exports to the United States through the trade channel.

Several studies have examined the impact of the Fed's expansionary monetary policy on foreign capital flows to emerging market economies. Ahmed and Zlate (2014), for instance, studied the determinants of foreign capital flows to EMEs by including the effect of the Fed's UMP, using net portfolio investment flow data to 12 emerging market economies from Asia (India, Indonesia, South Korea, Malaysia, the Philippines, Taiwan, and Thailand) and Latin America (Argentina, Brazil, Chile, Colombia, and México). The sample period, from 2002:Q1 to 2013:Q2, covered the Fed's UMP through to the Taper Tantrum in the middle of 2013. The specification of the empirical model using panel data is as follows:

$$(F/Y)_{i,t} = \beta_0 + \beta_1(g_{i,t} - g_t^{AE}) + \beta_2(R_{i,t} - R_t^{US}) + \beta_3LSAP_t + \beta_4RA_t + \varepsilon_t \quad (11)$$

where F/Y = net portfolio investment to GDP ratio, $(g_{i,t} - g_t^{AE})$ = economic growth differential with advanced countries, $(R_{i,t} - R_t^{US})$ = interest rate differential with US, $LSAP_t$ = the Fed's UMP through large-scale asset purchases (LSAP), and RA_t = global risk perception. The research produced several interesting findings. *First*, as predicted, the Fed's expansionary

monetary policy through LSAP increased portfolio investment to emerging market economies. *Second*, the economic growth differential and interest rate differential as well as global risk perception had a significant impact of foreign capital flows to EMEs. *Third*, there was a significant change in the dynamics of portfolio investment before and after the global financial crisis of 2008/2009, particularly in terms of becoming more sensitive to the interest rate differential. *Fourth*, the capital controls introduced by a number of emerging market economies since the global financial crisis reduced portfolio investment flows to EMEs.

Anaya, Hachula, and Offermanns (2015) examined how global spillovers from the Fed's UMP affected macroeconomic conditions and the policy response in EMEs. A Global VAR (GVAR) empirical model was used, thus facilitating interactions between push factors from the US, including the Fed's UMP, and pull factors in emerging market economies. Such interaction would be indicative of a contagion effect from the global spillovers. Monthly data was used from 2008 to 2014 for 20 EMEs: Argentina, Brazil, Chile, Colombia, Hungary, India, Indonesia, Israel, Korea, Mexico, Malaysia, Peru, the Philippines, Poland, Russia, Singapore, South Africa, Thailand, and Turkey. The empirical findings confirmed that the Fed's UMP significantly increased portfolio flows to emerging market economies. The surge of portfolio flows elevated real economic growth, stock returns, and exchange rate appreciation, while lowering lending rates in emerging market economies. Furthermore, EMEs reacted by easing monetary policy in response to US expansionary monetary policy in order to address the monetary policy trilemma.

Meanwhile, several studies have examined the impact of the Fed's plan to normalize monetary policy since May 2013 on foreign capital flows to emerging market economies. Gauvin, McLoughlin, and Reinhardt (2014), for instance, investigated the extent to which macroeconomic policy uncertainty in advanced countries triggered global spillovers to EMEs through portfolio flows to stocks and bonds. Monthly portfolio data was used covering 20 EMEs for the period from 2004 to 2011. The model was expressed as follows:

$$F_{i,t} = \sum \alpha_n F_{i,t-n} + \beta_0 + \beta_1 \Delta PU_t + \beta_2 X_{i,t} + \delta_i + \varepsilon_{i,t} \quad (12)$$

where $F_{i,t}$ = portfolio flows to bonds and stocks in EMEs, ΔPU = policy uncertainty vector in the US and EU, $X_{i,t}$ = vector of the control variable, and δ_i = country-specific fixed effects. Policy uncertainty was measured using the difference between economists' predictions and a Big-Data search of newspaper articles. The control variable $X_{i,t}$, covered the push factors, including risk perception (VIX), stock returns, global liquidity (M2 growth in the US, EU, and Japan), interest rates and oil prices, as well as the pull factors, including CDS spreads, stock returns, and domestic interest rates.

The empirical findings showed that the effect of policy uncertainty on portfolio flows to EMEs was very different depending on the origin, namely from the US or EU. Policy uncertainty in the US reduced portfolio flows to stocks and bonds in EMEs. Conversely, policy uncertainty in the EU reduced portfolio flows to bonds but increased portfolio flows to stocks. The level and direction of the global spillovers was not linear, namely that the impact would increase during periods of high global risk. Regarding portfolio flows to stocks, the level of risk in each respective country was also significant, namely that EMEs with a low level of risk would experience an increase of such flows despite high global risks.

Koepke (2014) looked at the effect on portfolio investment flows to EMEs of market expectations concerning FFR hikes by emphasising the element of surprise in US monetary policy. The empirical model used in this study included two additional dummy variables, namely $D_1 = 1$ for months when the FFR was expected to rise and $D_2 = 1$ for months when the FFR was expected to fall. Market expectations were measured using FFR futures contracts.

$$F_t = \alpha_0 + \alpha_1 F_{t-1} + \beta_1 D_1 MP_t + \beta_2 D_2 MP_t + \sum \lambda_i PULL_{i,t} + \sum \delta_i PUSH_{i,t} + \varepsilon_t \quad (13)$$

The results confirmed that changes in market expectations for US monetary policy were a significant determinant of portfolio flows to emerging market economies, particularly bonds. Furthermore, the effect was asymmetric, with expectations of monetary policy tightening having a larger adverse impact on portfolio flows to bonds (and relatively lower on stocks) compared to the positive impact of expectations of monetary policy easing. In other words, the emphasis on the surprise view in this study showed that changes in market expectations had a greater impact on foreign portfolio flows to EMEs than the actual impact of the Federal Reserve hiking the Federal Funds Rate (FFR).

Rai and Suchanek (2014) examined the impact of four Federal Reserve tapering announcements in 2013 on foreign portfolio flows and the financial markets in 19 emerging market economies. Using an event-study approach, the empirical model was estimated as follows:

$$R_{it} = \beta_1 Z_{it} + \beta_2 Z_{it} S_{it} + \beta_3 S_{it} + \beta_4 X_t + \varepsilon_{it} \quad (14)$$

where R = vector of exchange rate fluctuations, stock prices, bond yields, and net portfolio flows to EMEs; Z = vector of the dummy variable for each announcement date (May 22, 2013, June 19, 2013, September 18, 2013, and December 18, 2013); S = vector of specific EME variables, such as GDP growth, inflation and the current account balance; and X = vector of global variables, namely global stock and bond yields. The empirical results showed that the Fed's announcements in May and June 2013 had the adverse effect of precipitating foreign capital outflows, exchange rate

depreciation, higher bond yields, and lower stock prices in numerous emerging market economies. In contrast, the very same indicators reversed after the Fed announced its plans to delay tapering in September 2013. Reaction to the tapering announcement in December 2013 was more muted, however, because the markets had already anticipated the move and the FOMC's forward guidance had improved. In addition, more severe impacts were observed in more vulnerable EMEs, namely those with lower economic growth, a large current account deficit, high external debt and a low position of reserve assets. The impact manifested in larger capital outflows (capital flight), exchange rate depreciation and elevated bond yields. This research confirmed that solid economic fundamentals and an appropriate macroeconomic policy response could help to mitigate the global spillovers stemming from the Fed's plan to normalize its monetary policy stance.

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Chapter 14

Macprudential Policy and Financial System Stability

14.1. Introduction

In addition to the influence it exerted on foreign capital flow volatility as elucidated in Chapter 13, the global financial crisis (GFC) of 2008/2009 also raised awareness concerning the importance of maintaining financial system stability (FSS). A crisis can occur at anytime, anywhere, in advanced and developing countries. Differing from previous crisis episodes, however, the GFC that originated in the US stemmed from an accumulation of leverage in the housing sector, facilitated by securitized bank loans traded extensively in the financial sector. Subprime mortgage securities, coupled with structured products as well as excessive risk-taking behavior, created a build-up of leverage and mortgage delinquencies soared, bringing down the entire financial system. The leveraging process was also facilitated by financial institutions in Europe that exploited low interest rates stateside. Consequently, the crisis that began in the US quickly spread into a crisis in Europe and the GFC of 2008/2009. The adverse effect of the GFC was extraordinary, with an economic and financial depression felt around the world in advanced and developing countries, which took a protracted period to resolve. The crisis exposed the harmful side of the capitalist economy, which had spread worldwide through financial liberalization and globalization.

Maintaining FSS incorporates wide-ranging dimensions in the national interest and, therefore, cannot be the preserve of just one policy or institution. The sound of financial institutions is indeed important, both banks and nonbanks alike. The Lehman Brothers' crisis in the US showed that a crisis can occur due to fragilities at nonbank financial institutions. Nonetheless, sound financial institutions are not enough. A financial crisis can occur due to four salient factors: (1) financial sector fragilities; (2) imprudent macroeconomic policy; (3) deteriorating governance in the corporate sector and government; and (4) volatile capital flows (Eichengreen, 2004). Excessively expansive fiscal and monetary policy overheats the economy, prompts inflation to soar, and leads to runaway fiscal and current account deficits, which ultimately trigger a crisis. Similarly, foreign capital

flow volatility creates pressures on the balance of payments (BOP) and exchange rates, thus exacerbating the crisis. FSS must be able to anticipate systemic risk because of financial system and macroeconomic (macrofinancial) linkages, while facilitating prevention measures to avoid a crisis. Therefore, the early detection of FSS fragilities to the aforementioned factors is critical. Close policy coordination between the government, central bank, supervisory authority and deposit insurance corporation is, therefore, vital.

This chapter discusses various aspects of macroprudential policy in terms of supporting FSS. Following the introduction, the systematic discussion is divided into four sections. *First*, an explanation of FSS policy and the concept, including the central bank's role in supporting FSS with an emphasis in macrofinancial linkages. *Second*, a discussion on the theory and empirical evidence for financial procyclicality, which is considered an important trigger of financial crises, emphasizing the threat from asset price bubbles and excessive credit growth (credit boom). *Third*, an explanation of the theory and empirical evidence for interconnectedness and financial networks that oftentimes accelerate crisis propagation, emphasizing the interbank money market and systemic banking crises. *Fourth*, a summary of the macroprudential policy theory and practices implemented at various central banks, including the targets, approaches, and instruments used as well as the importance of coordination with the microprudential supervisory authority. The discussions in this chapter and in Chapter 13 form an integral part of the central bank's policy mix as discussed in Chapter 15.

14.2. Conceptual Dimension of FSS

The GFC of 2008/2009 again exposed the fragilities of the capitalist economy based on the trade of currency and capital through the financial system in order to finance investment activities. We witnessed the crisis in Latin America during the 1980s caused by excessive fiscal expansion and an accumulation of government debt. We felt the Asian Financial Crisis of 1997/1998, driven by an economic boom in the wake of financial liberalization during the 1980s, which fed through to excessive credit growth, property bubbles and a build-up of private external debt. And we were impacted by the GFC of 2008/2009 that originated in the US due to massive default on mortgage-backed securities (MBS) that imploded the financial system. The GFC of 2008/2009 and the previous crisis episodes have had severe consequences on the financial system, economy, fiscal burden and public prosperity.

That the roots of crises stem from an excessive build-up of debt, public and private, has been proven throughout the history of human civilization (Kindleberger, 1978; Reinhart & Rogoff, 2009). Likewise, that procyclical property bubbles and credit booms precede and trigger crises in various countries is also not a new idea (Claessens & Kose, 2013). And that crises are becoming more frequent and multidimensional, comprising of currency crises, debt crises and financial system crises, has also been proven in various jurisdictions (Bordo, Eichengreen, Klingebiel, & Peria, 2001). The problem, therefore, is why we have not learned the various causes of crisis in order to anticipate or prevent a crisis? And why have

we not focused on efforts to maintain FSS, protecting against macroeconomic instability, financial sector fragilities, weak corporate and government governance as well as risks from capital flow volatility in order to prevent the reoccurrence of crisis episodes?

14.2.1. The GFC of 2008 / 2009 and FSS

What is FSS and what is the central bank's role in terms of supporting FSS? Although academic references to financial instability are available from the thinking and ideas of Minsky (1982), the GFC of 2008/2009 brought such thinking to the fore, becoming the focus of serious attention by policymakers around the world. According to Minsky (1982), financial instability is a consequence of the capitalist economy where rising asset prices and the build-up of debt are potentially unchecked. During an expansionary economic period, supported by price (inflation) stability and low interest rates, such as during the two decades of Great Moderation in the US prior to the outbreak of the GFC in 2008/2009, the returns on investment and rising asset prices (financial and physical) outstripped lending rates or the cost of capital. The misplaced perception that returns would remain high during the economic boom period led to a proliferation of trade in debt and capital to finance the investment. The accumulation of debt, from bank loans and foreign capital, became increasingly uncontrollable. Speculative investors and Ponzi schemes began to dominate and outnumber prudent investors engaged in hedging activities. Ultimately, speculation and the Ponzi schemes incurred huge losses, triggered a credit crunch and prompted financial instability. Stability is destabilizing due to the boom-bust financial cycle, which eventually pulls the economy into a crisis.

Definitions of FSS are not always identical between policymakers and academics. Fundamentally, however, FSS refers to conditions where the financial system functions soundly in an economy and is resilient to the various shocks that could emerge (Allen & Wood, 2006; Mishkin, 1999b). Several academic definitions contrast FSS with conditions that could trigger a crisis. Borio and Drehmann (2009), for example, departed from the description of financial instability when defining FSS as follows:

financial instability is a set of conditions that is sufficient to result in the emergence of financial distress/crises in response to normal-sized shocks. These shocks could originate either in the real economy or the financial system itself. Financial stability is then defined as the converse of financial instability.

Meanwhile, Schinasi (2004) defined FSS in broader terms:

a financial system is in a range of stability whenever it is capable of facilitating (rather than impeding) the performance of an economy, and of dissipating financial imbalances that arise endogenously or as a result of significant and unanticipated events.

Central banks apply more practical definitions. For example, the European Central Bank (ECB) defines

financial stability as a condition in which the financial system – intermediaries, markets and market infrastructures – can withstand shocks without major disruptions in financial intermediation and in the general supply of financial services.

In the same vein, Act No 9 of 2016 concerning Financial System Crisis Prevention and Resolution (UU-PPKSK), defined “financial stability as a condition in which the financial system functions effectively and efficiently and is resilient to domestic and global shocks,” with reference to

the financial system as a system consisting of financial services institutions, financial markets and financial infrastructures, including the payment system, which interact to facilitate the accumulation of funds from the public and allocation to support national economic activities.

From the aforementioned definitions, at least five important aspects must be stressed in terms of maintaining FSS and preventing crises as follows. *First*, the soundness of individual financial institutions is necessary but not sufficient. FSS is linked to how the financial system functions to resist and survive shocks in the economy. The key, therefore, lies in the macrofinancial linkages of the financial system with the economic activities rather than the soundness of individual financial institutions. *Second*, history has shown four types of macrofinancial linkages that often occasion a crisis as follows: (financial and property) asset bubbles; credit booms; excessive leverage; and sudden-stop capital reversals (Claessens & Kose, 2013; Reinhart & Rogoff, 2009). Instability that emerges from those four macrofinancial linkages often leads to financial procyclicality, which accelerates economic growth during a boom period and, conversely, undermines economic growth and even exacerbates the economic cycle during periods of recession. The phenomenon of procyclicality, if left unchecked, will amplify the boom-bust cycle that can ultimately result in a crisis. Furthermore, financial procyclicality often foregoes financial crisis episodes (Claessens, Kose, & Terrones, 2011; Jorda, Schularick, & Taylor, 2011) and, therefore, must be anticipated and prevented early. We witnessed this phenomenon during the crises in Latin America, Asia, US, and Europe as well as the GFC of 2008/2009.

Third, policy to dampen shocks in the domestic economy and the ability to overcome external shocks are pivotal to support FSS. When domestic demand outstrips economic capacity, economic overheating is oftentimes the result, marked by high inflation and/or a current account deficit and/or a fiscal deficit. Such macroeconomic imbalances can either trigger or accompany macrofinancial imbalances as discussed previously. Furthermore, the risk of default, in terms of FSS, can originate from the spillover of global and/or regional economic shocks. The trigger may be in the form of a sudden crisis or change to policy in another

jurisdiction. Such an external shock could precipitate a sudden-stop capital reversal, thus prompting distress and a currency crisis, payment crisis, external debt crisis as well as financial and economic crises. Consequently, maintaining FSS requires macroeconomic policy (fiscal and monetary), structural reform in the real sector and foreign capital flow management.

Fourth, while a crisis can be triggered by default at one financial institution, the bursting of a bubble due to procyclical macrofinancial imbalances, or international or domestic economic shocks, the contagion process leading to a systemic crisis event throughout the entire financial system can progress with rapidity due to interconnectedness and financial networks in the markets and infrastructures, including the payment system (Allen, Babus, & Carletti, 2010; Acemoglu, Ozdaglar, & Tahbaz-Salehi, 2015; De Bandt & Hartmann, 2000). A currency crisis, for example, could be triggered by a foreign capital reversal or global or regional spillovers that propagate through contagion due to interconnectedness and collapse of the foreign exchange market (Calvo & Reinhart, 2000). In this case, investors are looking to buy but the supply of foreign exchange has evaporated, except from the central bank. Likewise, the default of an individual bank could prompt bank runs and spread to the banking system due to interconnectedness and insufficient liquidity on the interbank money market (Freixas, Parigi, & Rochet, 2000; Morris & Shin, 2004). Globalization has also exacerbated contagion worldwide due to interconnectedness and networks between international financial institutions, markets and infrastructures.

Fifth, a crisis typically peaks when contagion through interconnectedness and financial networks are accompanied by herding behavior and information contagion (Acharya & Yorulmazer, 2003; Bikhchandani & Sharma, 2001). Panic on the markets and among depositors, or even financial institutions, due to herding behavior and information contagion would further compound a financial crisis, resulting in fire sales, inadequate liquidity, and financial market collapse. The US subprime mortgage debacle demonstrated how contagion can propagate rapidly through fire sales on financial markets and tight credit in the banking system (Diamond & Rajan, 2010). Herding behavior and information contagion quickly turned the subprime mortgage fiasco into a full-blown and complex financial crisis, not only in the US but also in Europe and around the world.

The widespread and devastating impact of the crisis reinforced the need for joint measures and accountability. No one institution can, by itself, be accountable for the entire mandate of maintaining FSS. A mechanism is therefore required, or institutional policy coordination, to maintain FSS nationally, which has become common practice in many jurisdictions. The supervisory authority for banks and financial services is responsible for overseeing the soundness of individual financial institutions in order to protect the depositors, investors and consumers through microprudential regulation and supervision. The central bank, therefore, is responsible for macroprudential regulation and supervision to mitigate the risk of macrofinancial imbalances and systemic risk in the financial system, as well as lender of last resort (LOLR). A deposit insurance corporation also plays an important role in terms of mitigating information contagion from bank runs to protect depositors, as well as active involvement in the early resolution actions to

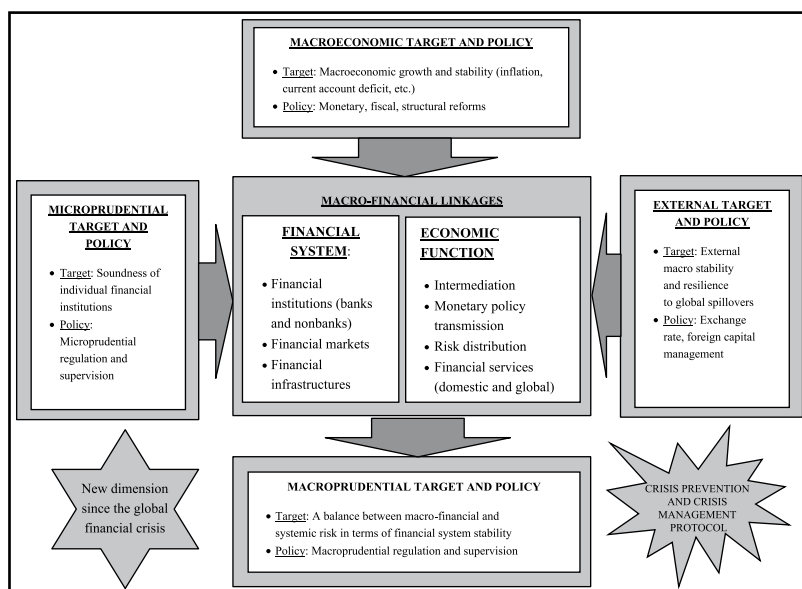


Fig. 14.1: Linkages between Macro–Micro Policy and FSS.

resolve a problem bank. Furthermore, the government, through the Ministry of Finance, needs to lead national FSS coordination efforts to prevent the onset of an economic crisis and the exorbitant fiscal and public costs incurred.

Fig. 14.1 illustrates how the financial system looks from four dimensions and the required policy responses, namely: microfinancial sector, macronational, macrointernational and macrofinancial linkages. The first three dimensions were already the subject of serious attention prior to the GFC in 2008/2009, while the fourth dimension has emerged in the wake of the catastrophic GFC. Before the GFC of 2008/2009, as presented in the first three parts of the schematic, attention in terms of economic management and crisis prevention were primarily focused on: (1) microprudential regulation and supervision to maintain the soundness of individual financial institutions; (2) macroeconomic (fiscal and monetary) policy for macroeconomic stability (inflation, exchange rates, fiscal deficit and current account deficit); and (3) foreign capital flow management to ensure resilience against contagion and international shocks. The new dimension, which has developed since the GFC of 2008/2009, as presented in the lower part of the schematic, is the need to include the financial system in the function of the economy, which could trigger macrofinancial instability. This must be overcome through macroprudential policy, namely the regulation and supervision of financial institutions from a macrofinancial and systemic risk perspective. The linkages between the various policies are required to support sustainable economic growth, while maintaining macroeconomic and FSS.

A coordinated crisis prevention and management protocol between institutions is, therefore, crucial to maintain FSS.

Placing FSS in the context of national macro–micro policy interconnectedness was seen as a correction to the reformulation of policy and institutional reforms of previous crises. After the Asian financial crisis of 1997/1998, for instance, the prevailing focus of many authorities was considered appropriate for that time. The Ministry of Finance was responsible for fiscal policy, while the central bank focused on its mandate of price (and exchange rate) stability through monetary policy. The regulation and supervision function of financial institutions has now been transferred away from the central bank, however, to a newly formed authority, such as what happened in South Korea and Indonesia. In fact, the independence of each respective authority was strengthened to ensure more effective policies at both. On one hand, this approach was motivated by the democratic movement that has increasingly become accepted as the global model. On the other hand, the change was driven by the academic community recommending the need to focus on one instrument for one goal, congruent with the Tinbergen rule. Nevertheless, the GFC exposed this approach to be inappropriate. The rule failed to acknowledge that various policies are mutually interconnected and, therefore, require a coordination mechanism to achieve robust economic growth and maintain stability. Furthermore, the GFC showed how important it is to consider the financial system from a macrofinancial perspective and that systemic risk must be mitigated through macroprudential policy. The central bank is the most suitable institution to issue macroprudential policy, on top of monetary and payment system policy as well as LOLR.

14.2.2. Central Bank's Role in FSS

The previous discussion showed that the central bank requires a mandate to maintain FSS, in addition to the mandate of price and exchange rate stability (Haldane, 2004; Padoa-Schioppa, 2003). The mandate of price (and exchange rate) stability through monetary policy (including foreign capital flow management) is already well known and accepted. The central bank's policy framework is already well established, supported by a credible Inflation Targeting Framework (ITF) as discussed in Chapters 8 and 9. That policy framework has already lowered long-term inflation as well as supported robust economic growth and lower interest rates in many jurisdictions (Berg, Hallsten, Queijo, Heideken, & Söderström, 2013). The salient features of the policy framework that supports monetary policy credibility include: a clear inflation target to be achieved, solid analytical models and macroeconomic projections, interest rate policy consistency to attain the inflation target, a formal and regular policy formulation process, as well as published inflation projections and other communications to anchor inflation expectations. The central bank also fundamentally applies the same policy framework to foreign capital flow management, as referenced in Chapter 13. Such a policy framework, applying macroeconomic analysis and projections, can easily be expanded to include macrofinancial linkages, particularly through the financial system, foreign capital flows and macroprudential policy by the central bank.

The central bank's mandate to support FSS is primarily achieved through macroprudential regulation and supervision of financial institutions from a macrofinancial perspective and a focus on systemic risk. As mentioned previously, macrofinancial linkages between financial institutions and the economy often create financial procyclicality, thus accelerating economic growth during an expansionary period but also exacerbating the downturn during a recession. Procyclicality, if not well managed, accelerates the boom-bust of the financial cycle, which could ultimately trigger a crisis. Macroprudential policy is, therefore, directed toward managing financial procyclicality, especially credit booms and asset bubbles, to avoid macrofinancial imbalances that often lead to financial crises. In addition, macroprudential policy is also directed toward managing interconnectedness and financial networks, in particular through the interbank money market and financial infrastructures, to mitigate systemic risk. While default at one financial institution could trigger a crisis, along with macrofinancial procyclicality, or international or domestic economic shocks, a systemic crisis event throughout the financial system could propagate with rapidity due to interconnectedness and financial networks.

In addition to macroprudential policy, the central bank also supports FSS through monetary and payment system policy as well as foreign capital flow management. Monetary policy supports FSS through avoidance of economic shocks as part of the third aspect of FSS mentioned previously, in addition to the overarching goal of price (and exchange rate) stability. This includes the regulation and supervision of money markets and the foreign exchange market as an integral part of monetary policy, which also supports FSS by mitigating systemic risk that could surface due to interconnectedness and financial networks as the fourth aspect of FSS. Similar considerations underlie central bank authority in terms of the payment system, in addition to the core purpose of maintaining a secure, efficient, and reliable payment system, which is inextricably linked to effective monetary policy. Meanwhile, the authority of the central bank to manage foreign capital flows supports FSS by avoiding economic shocks per the third aspect, on top of the main target to support exchange rate stability, while preventing sudden-stop and BOP crises.

The central bank is the most appropriate institution to implement macroprudential regulation and supervision. The central bank has the capacity for macroeconomic and macrofinancial surveillance, coupled with the required macroprudential policy instruments (Kawai & Morgan, 2012). Moreover, a Bank for International Settlement (BIS) (2011) study of 13 advanced and emerging market countries concluded that central banks must have direct involvement in FSS policy-making and implementation if the policy is to be effective based on three explanations. *First*, the monetary policy framework and task implementation focuses the central bank on macroeconomic analysis and projections, while complementing understanding of the financial markets, institutions, and infrastructure, all matters of great import in terms of macroprudential policy implementation. *Second*, financial system instability can be caused by and have a significant impact on economic performance, with fundamental consequences on

economic activities, price, and exchange rate stability, as well as monetary policy transmission effectiveness. Central banks are already familiar with evaluating financial system and economic interconnectedness, the macrofinancial aspects as the focus of macroprudential policy. *Third*, the central bank is a source of liquidity in the financial system and economy through monetary policy and the LOLR function, and the availability of liquidity is crucial when maintaining financial stability.

14.2.3. Central Bank Macroprudential Policy

Central bank urgency to strengthen the FSS framework requires solid financial infrastructures, including an adequate inspection and supervision function to support domestic financial market integration into an increasingly complex financial system. In this context, Borio (2003) stressed the need to strengthen the regulatory framework or macroprudential policy to limit risks if pressures or distress emerge on the financial markets for a protracted period, which could precipitate a real output slump in the economy.

Conceptually, macroprudential policy contains prudential regulatory instruments that aim to maintain FSS as a whole, not the soundness of individual financial institutions. Analogously, microprudential policy consists of prudential regulatory instruments that aim to maintain the soundness of individual financial institutions.

Macroprudential policy seeks to develop, oversee and deliver appropriate policy responses to the financial system as a whole. It aims to enhance financial system resilience and dampen systemic risks that spread through the financial system. (The G-30)

Therefore, macroprudential policy is used to prevent a boom-bust cycle in the supply of credit and liquidity that could trigger economic instability. Through its role of maintaining a stable supply of financial intermediation, macroprudential policy also supports the goals of monetary policy in terms of maintaining price stability and output.

There are two salient dimensions of macroprudential policy. *First* is the time-series dimension, namely that macroprudential policy is directed toward dampening the risk of excessive procyclicality in the financial system. To that end, macroprudential policy must be designed in such a way as to avoid or mitigate procyclicality. The principal is how to encourage financial institutions to prepare an adequate buffer during an expansionary economic phase, when imbalances tend to appear, and then how to utilize that buffer when the economic slumps.

The second is the cross-section dimension that shifts the focus from prudential regulation of individual financial institutions to the regulation of the financial system as a whole. History has shown that most financial crises around the world occur not because of problems at individual banks that subsequently propagate to the financial system. In contrast, large crises occur because of simultaneous

exposure to macrofinancial imbalances by most of the financial system. Therefore, a more holistic view of the financial system and linkages to the macroeconomy from various perspectives is desperately required.

In line with changes to the landscape of the financial sector, especially since the GFC of 2008/2009, many central banks have begun to apply broader macroprudential policy instruments. In this regard, several instruments previously thought of as monetary instruments (such as reserve requirements) have been used to prevent systemic risk and maintain FSS in the economic cycle, particularly for targeted economic sectors. The instruments are not focused, however, on efforts to mitigate risk at individual banks. Therefore, the policy instruments can be categorized from a broader macroprudential perspective. Examples of common macroprudential policy instruments are presented in [Table 14.1](#).¹

In addition to the policy goal (target) set and policy instruments available to achieve that target, another important aspect to be formulated in the macroprudential policy framework is the policy response. When formulating the macroprudential policy response, rule versus discretion is a prominent consideration. Like monetary policy, there is always a trade-off between using rules or discretion. Rules provide assurance to the markets and credibility to the central bank. Nonetheless, rules are too rigid, thus removing the flexibility to respond to structural changes or uncertainty that often befall financial markets. Considering the advantages and disadvantages of rules and discretion, decision-making models that combine the two are often preferred.²

Table 14.1: Examples of Macroprudential Policy Instruments.

Problem	Instrument
Leverage (potential procyclicality)	Adjust risk weighting in capital requirements Apply capital ratio to risk-weighted assets
Credit (debtor linkages and characteristics, pressure on macrostability – interconnectedness)	Apply countercyclical provisioning (on certain loans) Restrict the loan-to-value ratio for certain sectors (with potential bubbles) Limit lending to certain sectors (for instance property, credit cards) Adjust the reserve requirement, either across the board or targeted
Liquidity (potential risk on certain aspects)	Apply a buffer to reduce dependence on risky sources of funds Apply the loan-to-deposit ratio

Sources: Borio and Shim (2007), Hannoun (2010), and G-30 (2010).

¹A more detailed discussion on the various types and utilization of macroprudential policy instruments in various jurisdictions is presented in Section 14.5 of this chapter.

²More detailed discussion on this issue is presented in Box 14.1.

Box 14.1: Macprudential Policy Response: Rule Versus Discretion

When seeking the most optimal macroprudential policy response, it is imperative to have comprehensive understanding of business cycles and financial cycles. The main problem faced by central banks in the financial sector is identifying when the economic cycle enters an expansionary or contractionary phase and, thus, the policy response required. Departing from the existing monetary policy framework that has a firm foundation in academic thinking, theory and established practices, the macroprudential policy framework remains in the early stages of maturation, especially since the global financial crisis of 2008/2009. Consequently, prevailing thinking on designing the appropriate macroprudential policy response is an extremely interesting topic to discuss. In terms of policymaking, this helps to organize the various perspectives on certain macroprudential policies, while increasing policy transparency and reinforcing policy communication.

Referring to existing concepts, the macroprudential policy response can be categorized into two polar opposites: discretion and rule. As explained in Chapter 7, understanding discretion refers to a policy response based on certain judgments in accordance with current (past) developments, while rule refers to a policy response based on existing rules that have previously been implemented.

In a simple case, namely the application of countercyclical macroprudential policy (for instance a countercyclical capital buffer), the ultimate goal of policy is to prevent excessive credit imbalances or to smooth the financial cycles. In this case, the macroprudential policy response can be applied through a rule or based on discretion. Based on discretion, the central bank may require the banks to build capital buffers above the normal rate during an economic boom period. Based on pre-determined variables, the central bank will announce that the economic cycle has entered a boom period and decide if the boom has triggered excessive and dangerous risk-taking behavior, thus requiring the banks to provision additional capital as a buffer should the economy moderate in future periods. Conversely, when the economy contracts, based on central bank discretion, the banking industry shall be permitted to release or relax its capital buffers.

On one hand, countercyclical macroprudential policy eases procyclical-risk but, on the other hand, could increase the risk of default at individual banks. In fact, capital buffers are considered more appropriate for macroprudential purposes, namely by targeting risks and the wellbeing of the overall financial system, which is independent of the solvency situation of individual banks as the preserve of microprudential policy. In a broader context, discretion also suffers weaknesses in the form of adverse effects from the warnings and recommendations put forward by the central bank,

which could trigger self-fulfilling prophecies and, ultimately, significant political pressures on the central bank.

On the other hand, a rule-based response seeks to establish macroprudential policy per automatic mechanisms and triggers, thus avoiding time inconsistencies, even more so when the policy authority is forced to identify (subjectively) an optimal balance between the micro and macroobjectives in an integrated manner. Unfortunately, rules can be overly complex and difficult to apply, particularly for new policies and those not considered international best practices.

As a pragmatic measure, the use of simple (feedback) rules has been explored as stylized designs *à la* Taylor rule in monetary policy.

Table 14.4.1: Monetary and Macroprudential Policy Rules.

Risks	Rule Design
Inflation	Monetary Policy Rule: Policy Rate = $f(\text{inflation gap; output gap})$
<i>Cyclical risks</i>	<i>Macroprudential Policy Rule:</i>
Balance sheet composition	Instruments representing the structure of the balance sheet = $f(\text{target cyclical risk}^a)$ <i>For instance:</i> countercyclical capital buffer = $f(\text{credit to GDP ratio})$
Terms and conditions of financial transactions	Instruments representing the terms and conditions = $f(\text{target cyclical risk}^b)$ <i>For instance:</i> loan to income (LTI) ratio = $f(\text{property or house prices})$
Cross-sectional risks	<i>Macroprudential Policy Rule:</i>
Market structure	Instruments representing market structure = $(\text{target cross-sectional risk}^c)$

Source: Adapted from Fic, Tatiana (2012).

^a Instruments representing the structure of the balance sheet: countercyclical capital buffer, sectoral capital requirements, leverage ratio, liquidity buffer and so on. Targets: credit to GDP ratio, credit growth, credit gap, and so on.

^b Instruments representing the terms and conditions: LTV, LTI. Targets: property or house price growth, house price deviation from equilibrium price, and so on.

^c Instruments representing market structure: wholesale funding taxation. Targets: composite index of interconnectedness, information from surveys, and so on.

Referring to the various rule designs, future macroprudential research and policymaking will certainly need to be directed towards setting targets and instruments in line with the specific risk (instability) category, namely balance sheets, terms and conditions of financial conditions, and

structural risks. Several macroprudential policy instruments introduced by the Bank of England, such as the countercyclical capital buffer (CCyB), sectoral capital requirements, and leverage ratios, fundamentally only overcome (directly) the risks associated with balance sheet composition. If a quantitative policy target has been set, however, empirical verification is still required to show which instruments are the most effective in meeting the policy target. Further identification is required of the macroprudential policy transmission channels, which is not a simple task.

With the distinct advantages and disadvantages of each macroprudential policy response based on rule or discretion, prevailing thought suggests a combination of the two, as is the case with monetary policy. The Bank of England (2009), for instance, proposes a constrained discretion macroprudential regime, namely to apply a response that is largely discretionary to allow policymakers to adapt as they learn, but remains systematic, transparent and accountable, in line with the pre-determined policy goals and constraints. Such a regime requires the support of clear policymaking targets and frameworks as well as measures of accountability that are acceptable to the stakeholders.

A combined response could also place more weight on using rules rather than discretion. Such a policy response would use a simple rule as the main reference, as is the case with the countercyclical capital buffer. For example, several key financial sector variables, which might fall under the auspices of different financial sector policy authorities, must be announced clearly and transparently (*ex ante*). The relevant policy authorities have the option to deviate from the rule, allowing the banks to adjust (raise/lower) the buffer in line with their information pertaining to economic conditions. This, however, must be the exception, rarely used and clearly explained to the markets. Therefore, the general policy response will adhere to the rule in daily practice, while discretion shall be available when absolutely necessary (Quagliariello & Libertucci, 2010).

14.3. Theoretical Model and Empirical Evidence of Financial Procyclicality

Per the Financial Stability Board (FSB, 2009), the term “procyclicality” relates to:

the dynamic interactions (positive feedback mechanisms) between the financial and the real sectors of the economy. These mutually reinforcing interactions tend to amplify business cycle fluctuations and cause or exacerbate financial instability.

Therefore, procyclicality is a phenomenon where the financial cycle accelerates the economic cycle, as illustrated in Fig. 14.2, which shows a procyclical financial

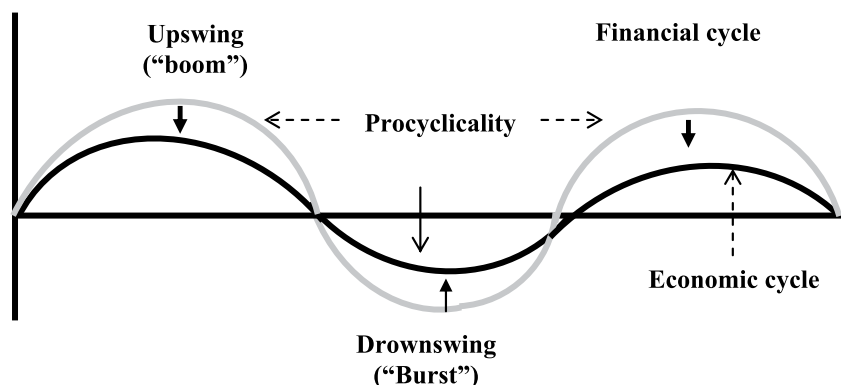


Fig. 14.2: Financial Cycle and Economic Cycle Procyclicality.

cycle accelerating the booms and busts of the economic cycle. During a period of economic expansion (boom), the financial cycle tends to outpace the economic cycle. Bank credit expands rapidly, while financial and property asset price bubbles appear, excessive debt accumulates and an influx of foreign capital floods domestic financial markets, all of which exceed what is actually required per economic fundamentals and, therefore, risks build and fragilities appear in the financial system. Oppositely, during a period of economic contraction (bust), the financial cycle exacerbates the downswing in the economic cycle. Consequently, a domestic or international shock, against a backdrop of financial system fragilities due to a build-up of risk, often leads to financial and economic crises.

Procyclicality is a consequence of dynamic interactions between the financial and the real sectors of the economy and macrofinancial linkages, caused by factors in the microfinancial sector and macroeconomy. Concerning the microfinancial sector, several factors lead to procyclical lending and other capital financing in the economy. *First*, asymmetric information between the lender and borrower causes credit rationing or the external finance premium. To overcome information asymmetry, the lender requests collateral on the loan or investment capital extended to the borrower, in the form of a business project or fixed asset, such as land or property. As the value of business projects and fixed assets fluctuate with the economic cycle, the loan or investment capital will also fluctuate. During an expansionary phase of the economic cycle, business prospects improve and corporate wealth increases, thus reducing credit rationing and lowering the risk premium. As investment also increases and accelerates economic growth, the acceleration becomes more pronounced due to (collateral) asset price bubbles during the boom period. In contrast, during an economic downswing, credit rationing, and the risk premium increase. Investment declines, along with economic growth, thus exacerbating the downturn. In the literature, this phenomenon is known as the financial accelerator.

Second, regulation and accounting methods can also compound procyclicality. In the banking industry, for instance, capital requirements are based on risk

contained on the bank balance sheet. Difficulties when evaluating the fair value of assets tend to underestimate risk during an expansionary phase and overestimate risk as the economy declines. Consequently, capital requirements based on risk assessments can aggravate procyclicality if the under/overestimates of risk are not adjusted. Fair value accounting can also create problems if the market value of the asset deviates far from the fundamental value. Such conditions can occur if asset price bubbles appear during an economic upswing. Price corrections and weak markets, as the economy moderates, heighten systemic risk that are often late to overcome.

Third, changing perceptions among economic players can also exacerbate asset price volatility in the financial and the economic cycles. During an economic boom, market confidence is high and asset prices are expected to keep rising. This occurs in the property sector and for financial assets, including stocks and bonds. As the markets increasingly tend to believe favorable economic news, interest grows in purchasing assets, the prices of which increase, leading to herding behavior. Consequently, asset price bubbles appear and continue to grow as economic optimism increases. Then, as the economy begins to moderate, a reversal of perception encourages investors to release assets in order to prevent further price declines. Once again, herding behavior, this time to release assets, exacerbates the downward price trend. Furthermore, investors may begin to panic, thus prompting fire sales to save themselves. Consequently, large price corrections occur and trigger systemic risk.

In terms of the macroeconomy, there are several determinants of the intensity of procyclicality in the financial system. In general, fiscal expansion accelerates during an economic upswing due to larger tax revenues, easier access to government debt and other factors. The impact of monetary policy on financial system procyclicality occurs through changes to interest rates, exchange rates, credit or risk-taking behavior in the financial sector. Stringent and lenient regulation and supervision by the authorities also affects financial sector behavior and activities in response to corporate and household demand for financing, as well as other financial services. Such conditions are typically linked to financial sector liberalization in a given jurisdiction. Likewise, foreign capital flows in various forms, including external debt, are normally relatively large during an economic upswing but vulnerable to a sudden reversal when the economy moderates. Macroeconomic developments, due to the various aforementioned factors, interact dynamically with the microfinancial sector factors and mutually reinforce financial system procyclicality in the economy.

Therefore, procyclicality is not merely the result of dynamic interactions between the business cycle and financial cycle, but also stems from the risk-taking cycle, namely behavior marked by excessive optimism during an economic upswing and excessive pessimism during a downswing (Nijathaworn, 2009). The business cycle is said to expand when the economy enters a growth phase and contract when the economy moderates. The financial cycle is characterized by more expansive banking behavior in the form of increased leverage in line with the expansionary phase of the business cycle. Conversely, the banks become more conservative through deleveraging as the business cycle

contracts. This interaction between the business cycle and financial cycle is determined by the behavior of economic agents to risk, which is also influenced by economic expectations, risk perception, regulations, and incentives. These changes to risk explain why investor behavior can change so drastically from extreme optimism when risk is low to extreme pessimism in order to avoid risk. Such changes to risk-taking behavior are based on sudden changes in financial system and economic activities.

Interaction between the three cycles, moving in the same direction and mutually reinforcing, create procyclicality in the financial sector (Table 14.2). The interactions are typically illustrated in the context of the boom-bust cycle. Initially, when the economy expands, characterized by macroeconomic stability and stronger economic growth, investor confidence boosts optimism in the economy, which encourages risk-taking behavior and triggers demand for credit and higher asset prices.

During a period of optimism, risk in the financial sector dissipates, interest rate spread narrows and loan loss provisions are reduced because banks are inclined to look at shorter horizons rather than the longer term. Higher asset prices raise the value of collateral, which boosts lending and, therefore, elevates market confidence and prompts risk-taking behavior, as reflected by increased leverage. As banks are more inclined to lend, the corporate sector undertakes more investment activities and households consume, which stimulates economic

Table 14.2: Interaction between the Business Cycle, Risk-taking Behavior, and the Financial Cycle.

	Business Cycle	Risk-taking Behavior	Financial Cycle
Expansionary Phase	• Macroeconomic stability • Stronger economic growth	• Greater confidence and optimism • More risk-taking behavior • Stronger demand for credit	• Lower risk assessment, narrower interest rate spread • Rising asset prices raise value of collateral • Leverage increases • Influx of foreign capital • More lending
Contractionary Phase	• Heightened macrovolatility • Less economic activity	• Lower market confidence • Risk averse • Less demand for credit	• Bank deleveraging • Larger loan loss provisions • Broader interest rate spread • Less lending • Decline of foreign capital inflow

Source: Modified from Nijathaworn (2010).

growth. In contrast, when confidence in the economy slumps, the behavior changes and investors become risk averse, thereby precipitating lower asset prices and the value of collateral declines. In response, the banks deleverage, shifting their portfolios away from risky loans to low-risk assets, such as Bank Indonesia Certificates (SBI) and government debt securities (SUN), in order to maintain adequate capital. The banks also reinforce loan loss provisions in anticipation of deteriorating credit quality. Such developments reduce lending and, in turn, exacerbate the economic downswing.

14.3.1. Financial Accelerator Theory

The literature shows that imperfections and asymmetric information in the financial sector accelerate the business cycle and economic cycle, a phenomenon known as the financial accelerator. The mechanism operates through capital availability or changes to the risk premium. In the banking sector, for example, the acclaimed work of Stiglitz and Weiss (1983) showed that credit rationing is a normal phenomenon. Information asymmetry between banks and borrowers concerning actual business conditions leads to adverse selection. Only borrowers with a viable business outlook can obtain loans from a bank and, therefore, imbalances appear on the credit market with the credit volume and lending rates below demand for loans.

Meanwhile, Williamson (1987) suggested that credit rationing occurs at different volumes and interest rates in accordance with an assessment of the borrower's business prospects. On the other hand, there is moral hazard among the borrowers which causes the banks to demand additional collateral beyond project feasibility (Bester & Hellwig, 1987). Consequently, the phenomenon of credit rationing in the banking sector could be caused due to limited supply from the banks, per Stiglitz and Weiss, or in line with project feasibility per Williamson, or even because the additional collateral is in the form of land, buildings or other fixed assets. Whatever the reason, the financial accelerator affects bank loans because the project value and additional capital fluctuate with the economic cycle due to fiscal and monetary policy as well as other macrofactors. Greenwald and Stiglitz (1993, 2003) analyzed the dynamic interactions of credit rationing with the macroeconomy.

Departing from the aforementioned factors of capital availability through credit rationing, Bernanke, Gertler, and Gilchrist (1999) analyzed the financial accelerator due to risk premium factors in terms of investment financing. Asymmetric information causes firms seeking external capital, such as bank loans, to incur the external risk premium. Therefore, the net worth of the project, based on the net-present value of profits obtained now and at a future date, is larger than the value of the loan received after calculating the interest. During an economic upswing, corporate net worth is generally higher due to higher financial expectations and asset market prices, which increase compared to the level at equilibrium. Such conditions create procyclicality. The procyclical net worth of borrowers, due to profit and asset price procyclicality, also leads to procyclical bank lending and lending rates.

The model developed by Kiyotaki and Moore (1997) explained the influence of fluctuating asset prices on project value and collateral in terms of financial system procyclicality. Lenders, including banks, can only enforce loan agreements through foreclosure on corporate net worth, including additional collateral in the form of land, buildings and other fixed assets. Therefore, the maximum loan value is determined by corporate net worth as collateral. Simultaneously, the value of collateral is influenced by total credit as an aggregate. In other words, there is a mutually reinforcing relationship between total credit, collateral value and economic growth. The financial accelerator and procyclicality occur. Fluctuations in the value of collateral affect credit volume, corporate performance and economic growth, asset prices, and so on, which exacerbates the impact of the financial cycle on the economy, thus triggering systemic risk.

Several important conclusions can be drawn from the financial accelerator phenomenon that creates procyclicality in the financial sector and economy. Early detection is crucial to measure how much procyclicality is excessive, among others, by comparing differences between credit and financing growth in the financial sector and the level considered normal in line with fundamentals. Likewise, rising financial asset and property prices demand vigilance to detect the presence of bubbles. Macroprudential policy can combat excessive procyclicality by, for instance, using the loan-to-value ratio (LTV) or more stringent capital requirements. Microprudential regulation and supervision are also vital for risk management and sound credit procedures. Furthermore, the availability of transparent information on corporate conditions must be improved, among others through a credit information system. Such measures are required to dampen the impact of procyclicality on systemic risk.

14.3.2. Empirical Evidence for Financial Procyclicality

Experience has shown that crises commonly occur with procyclicality in three key sectors: (1) financial asset prices, including property and securities; (2) bank credit and innovative structured products; and (3) foreign capital flows, particularly foreign debt and portfolio investment. The following studies provide empirical evidence for procyclicality in those three sectors. In essence, procyclical asset prices, lending and foreign capital flows cause an excessive accumulation of debt as the main root of financial crises (Reinhart & Rogoff, 2009).

14.3.2.1. Asset Price Bubbles. Hunter, Kaufman, and Pomerleano (2003) and Evanoff, Kaufman, and Malliaris (2012) conducted literature reviews covering various aspects of asset price bubbles and the implications on monetary policy, regulation, and international policy. Congruent with economic theory, asset price bubbles were said to occur if: “the market price of an asset exceeds its price determined by fundamental factors by a significant amount for a prolonged period.” In general, market price (P_t) shows the net-present value of return (R_t) on assets received now and at a future date and can be expressed as follows:

$$P_t = \sum_{i=0}^{\infty} E_t R_{t+i} + f(x) \quad (1)$$

where $P^* = \Sigma E_t R_{t+i}$ is the fundamental price of the asset and $f(x)$ is market price deviation from the fundamental value. The return on the asset is in the form of capital gain and the proceeds from investment in the asset. If the investment is in a financial asset, such as stocks, the return originates from higher share prices and dividends. If the investment is in property, the return is the increase in property price and any rental fees collected.

Asset price bubbles can be identified by studying market price deviation from the fundamental value, namely a function of $f(x)$ above. On an efficient market, prices generally fluctuate around the fundamental value. The function $f(x)$ is random with a normal distribution. Such conditions indicate an absence of asset price bubbles. Nonetheless, if market price deviation from the fundamental value increases, an asset price bubble is said to have occurred. In the economic literature, this phenomenon is known as growing bubbles. The spread of price deviation can occur at a constant speed or exponentially.

History has shown that price bubbles are frequently observed prior to, and are therefore an indication of, a financial crisis. Studies by Claessens et al. (2011) and Drehmann et al. (2010) showed how property price bubbles are an accurate indication of an impending crisis after the financial cycle has peaked. Similarly, from a study of crises in five advanced countries and 15 emerging market economies (EME), Kaminsky and Reinhart (1999) found that a crisis is preceded by excessive credit growth, accompanied by around a 40% jump in stock prices each year above normal levels. Property and other asset prices also bubble prior to a crisis. At some point, however, the bubbles burst, hence share and property prices collapse. Such developments are typically followed by bad credit (loss loans), default, and tight liquidity on the interbank money market. Furthermore, systemic risk in the banking industry, at financial institutions and on the financial markets also intensifies. Crises triggered by asset price bubbles, specifically property price bubbles, and excessive credit growth, were also found in many countries, including Japan, Latin America, and Europe (Hunter et al., 2003).

Such observations were clearly visible when the GFC of 2008/2009 began in the US, which was foreshadowed by a housing price bubble, accompanied by a price bubble of structured products at nonbank financial institutions, which subsequently triggered systemic risk and a financial crisis. Driven by strong demand, rising house prices were followed by a construction boom (Scherbina, 2013). Many new houses were built to supply the market at exorbitant prices, particularly in areas beset by rapidly rising prices. From 2004 to 2006, the Federal Reserve raised its policy rate 17 times from 1% to 5.25% to control inflation. Default on mortgage-backed loans became common. Then, when demand and prices began to decline, the inventory of unsold new homes prompted foreclosures by the banks, thus depressing prices further. What began as a housing crisis, credit crisis and structured products crisis, subsequently evolved into a financial crisis and economic crisis in the US (Brunnermeier, 2009).

Experience from the Asian financial crisis in 1997/1998 also revealed property price bubbles and a credit boom prior to the crisis (Collins & Senhadji, 2003). Property prices skyrocketed and the banking industry had the capacity to lend to the sector using rising property prices as collateral thanks to financial

liberalization in the 1980s. Nevertheless, this study showed that the impact of property price bubbles on the crisis event was also determined by the resilience of the banking industry in each respective jurisdiction. Weak banking sectors in Thailand, South Korea, and Indonesia were vulnerable to speculative attacks on the foreign exchange market that triggered a property crisis, external debt crisis, banking crisis, and economic crisis. Conversely, Hong Kong, Singapore, and Malaysia survived the speculative attacks due to stronger banking systems and sounder regulatory and supervisory systems. Consequently, when the property price bubble burst, those three countries avoided a deep economic crisis. Such experiences from the Asian Crisis demonstrated that sound banking regulation and supervision is a strong barrier to a crisis event when an asset price bubble bursts.

14.3.2.2. Credit Booms and Busts. The importance of credit as an indicator to understand the financial cycle and its impact on a crisis was also emphasized by Schularick and Taylor (2012). Using data covering 140 years from 14 advanced countries for the period of 1870–2008, this study showed that a shift in the role of money supply and credit has occurred since World War II. Financial liberalization has increased the importance of credit, while monetary policy using interest rates has proven incapable of controlling the credit cycle in the economy. During an expansionary economic period, credit growth was shown to boom, exceeding 7%, but then bust at -2% during an economic downswing in 14 advanced countries. Furthermore, the boom and bust periods of the credit cycle were shown to be inextricably linked to financial crisis episodes. The study recommended that central banks place more emphasis on assessing bank credit performance, in terms of pace of growth, application of lending standards, and risk accumulation. Consequently, an interest rate policy response would be insufficient, instead requiring control over bank credit, including macroprudential regulation and supervision.

Mendoza and Terrones (2008) conducted a study of 27 credit booms in advanced countries and 22 EMEs for the period of 1960–2006 using microcorporate and macroeconomic data. The results showed that credit booms correlate closely with economic expansion, elevated property and stock prices, real exchange rate appreciation and a broader current account deficit, followed by the reverse conditions when the economy slumps. The same dynamics were observed in corporate indicators of leverage, worth, and dependence on external financing, as well as banking indicators of asset quality, profitability, and credit activity. Moreover, credit booms have tended to synchronize internationally, peaking in the debt crisis in the 1980s, the ERM crisis in 1992 as well as external crises in EMEs. In addition to the various consistencies, several differences in the characteristics of credit booms between advanced countries and EME were also revealed as follows: (1) macroeconomic and microcorporate indicator fluctuations tended to be more pronounced, more persistent, and asymmetric in EMEs, with a strong correlation to the non-tradeable sector; (2) not all credit booms culminate in crisis, but many crises in EMEs were preceded by credit booms; and (3) credit booms in EMEs were often preceded by a deluge of foreign capital inflow rather than domestic financial reform or increased productivity, while credit booms in

advanced countries were generally prefaced by a bump in productivity or financial reform.

In the case of Asia, Craig, Davis, and Pascual (2006) conducted a study concerning the influence of financial system characteristics on accelerating procyclicality in five advanced countries (Australia, Hong Kong, Japan, New Zealand, and Singapore) and six EMEs (China, Indonesia, South Korea, Malaysia, the Philippines, and Thailand). The study evidenced strong but asymmetric procyclicality between property prices, credit growth, and GDP, with a stronger correlation during a cyclical downswing. Analysis also showed that asset prices and credit growth have deviated significantly from long-term trends, peaking during several episodes and, thus, triggering financial instability. Panel estimations using macro and individual data from nearly 300 banks found that the structural characteristics of financial systems in Asia were an important source of procyclicality, namely: excessive dependence upon credit guarantees, delayed recognition of loss loans (bad credit) and insufficient loan loss provisions, loose credit risk assessments that fed through to excessively small lending margins to cover the risk, lower quality credit assessments during a cyclical upswing, loan concentration at state-owned enterprises and credit groups, bank dependence upon foreign sources of funds, and financial liberalization.

14.3.2.3. Procyclical Capital Flows. Numerous studies have demonstrated procyclical capital flows. Kaminsky, Reinhart, and Vegh (2004), for instance, studied capital flow procyclicality and the interaction with fiscal and monetary policy. Using data from 104 countries for the period of 1960–2003, the study revealed several salient aspects of procyclical capital flows. *First*, capital flows were proven to be procyclical, namely that foreign loans would increase during an economic upswing and decrease during a downswing in most Organization for Economic Co-operation and Development (OECD) and developing countries. *Second*, fiscal policy is also procyclical, namely that government spending increases during an economic upturn and decreases during a downturn in the majority of developing countries. *Third*, in EMEs, monetary policy tends to be procyclical, such that interest rates are lowered as the economy improves and raised when the economy moderates. *Fourth*, in developing countries and, specifically, in EMEs, periods of foreign capital inflow correlate to expansive macroeconomic policy, while foreign capital outflow is associated with contractive macroeconomic policy. In other words, in EMEs and developing countries, when it rains it pours.

Meanwhile, Calderon and Kubota (2012) found that private capital flows could spur a credit boom. Based on quarterly data concerning foreign capital inflows, credit to the private sector and other macrofinancial indicators for 71 countries for the period of 1975–2010, the study revealed several interesting findings: (1) a surge of private capital flows is a relevant predictor of a credit boom; (2) the probability of a credit boom is more likely if the foreign capital flows are driven by other private investments and, with a smaller coefficient, by portfolio investment; (3) a deluge of inflow is also a relevant predictor of a bad credit boom, namely one that culminates in a financial crisis; (4) a bad credit boom is more commonly linked to capital inflows as other investments, but

mitigation is possible through an increase of foreign direct investment (FDI); (5) the predictive ability of other investment flows is primarily determined by external debt; and (6) an accumulation of leverage and overvaluation of real exchange rates strengthen the predictive capacity of credit booms that culminate in a systemic crisis. Empirical evidence that foreign capital inflows prompt a credit boom and might culminate in a financial crisis was also put forward by Sa (2006) and Amri et al. (2013).

Arena, Bouza, Dabla-Norris, Gerling, and Njie (2015) conducted a further study into foreign capital flows, credit booms, and macroeconomic dynamics in low and middle-income countries. Using data from 135 developing countries for the period of 1960–2011, the study provided several invaluable lessons. Foreign capital flows were shown to trigger a credit boom, with a similar frequency and level between the two groups of developing countries. Nevertheless, the impact on macroeconomic dynamics was more pronounced in middle-income countries, with larger fluctuations noted in terms of output, investment, private and public consumption, real exchange rates, money supply, and the current account balance. Furthermore, countries with a more rigid, or less flexible, exchange rate regime, experienced faster credit growth during a period of foreign capital inflow and, therefore, were more vulnerable to capital reversals. Experience has also shown that financial liberalization could prompt a credit boom if not accompanied by prudential banking regulations. The study reminded developing countries to carefully consider the implications of financial liberalization on the potential influx of foreign capital, which have been empirically shown to trigger a credit boom and create fragilities in terms of FSS.

14.4. Theoretical Model and Empirical Evidence for Interconnectedness and Financial Networks

A crisis can originate from bank default that propagates into bank runs. Furthermore, a crisis can also stem from a property bubble bursting, credit crunch, default on external debt payments, or sudden capital reversal. Nonetheless, the rapidity and propagation of bank default or macrofinancial imbalances into a systemic crisis is determined by interconnectedness and financial networks in financial markets and infrastructures, in addition to the spread of information. Experience showed the speed at which the speculative attack on the foreign exchange market in Thailand spread to South Korea and Indonesia, which quickly grew into the Asian Financial Crisis of 1997/1998. The public witnessed first-hand how defaults on housing loans and mortgaged-backed securities in the US culminated in the bankruptcy of Lehman Brothers in September 2008, which quickly evolved into the financial crisis that befell the US, Europe and the rest of the world.

14.4.1. Theory of Interconnectedness and Financial Networks

Financial interconnectedness is indicative of a network consisting of credit exposure, trade relationships, and other transactional linkages, as well as interdependence between financial players (Allen & Babus, 2007; DTCC, 2015). In general,

financial interconnectedness refers to the relationships between economic players created by through financial transactions and the supporting agreements. In the financial system, interconnectedness specifically occurs from linkages between: (1) financial institutions, banks and nonbanks; (2) providers of financial infrastructure services; and (3) supply and third-party linkages supporting the financial institutions. For example, banks that provide loans to and borrow from other banks become interconnected through credit exposure on the interbank money market. Similarly, contractual obligations between financial institutions, such as ownership, loans, derivatives and other contracts, also form mutual connections. Furthermore, when financial institutions invest in the same asset, they are interconnected through the same exposure to that asset.

Interconnectedness is extremely important because it facilitates contagion, which can exacerbate financial system instability and trigger a systemic crisis (Allen & Gale, 2000). The impact of default at a financial institution, a bank for instance, which is highly interconnected, will quickly propagate to the entire financial system. The strength of the interconnectedness through the financial network between financial institutions will determine how far the default will spur systemic risk in the financial system as a whole. In that context, financial networks can be categorized as complete, incomplete, or disconnected networks, as presented in Figs. 14.3–14.5, respectively, as follows:

In the case of a complete network, as shown in Fig. 14.3, all financial institutions are mutually interconnected to one another bilaterally or through another financial institution. Under such conditions, default at any of the financial

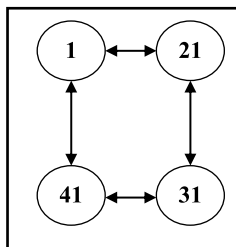


Fig. 14.3: Incomplete Networking.

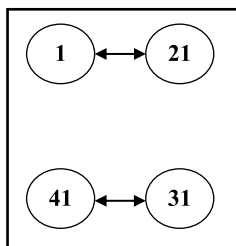


Fig. 14.4: Disconnected Network.

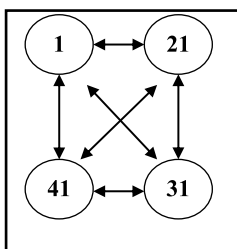


Fig. 14.5: Complete Network.

institutions would spread quickly to the others, thus creating systemic risk. Vigilance is also required in the case of incomplete networks, as presented in Fig. 14.4. To prevent a systemic impact from the default of one financial institution, for example, Financial Institution No. 3, prevention efforts must be directed toward Financial Institution No. 4 in order to prevent contagion at Nos. 1 and 2. Conditions are very different in the case of a disconnected network, as shown in Fig. 14.5. Default at a financial institution, for example, No. 3 could only propagate to No. 4 and, therefore, systemic risk is low.

Interconnectedness through financial networks can occur directly and indirectly (DTCC, 2015). Direct financial interconnectedness occurs through transactions, obligations, contracts and agreements as well as other financial linkages as follows. *First*, interbank credit exposure occurs through lending and borrowing activities on the interbank money market; swap, forward and other financial derivative transactions on the foreign exchange market; as well as certificates of deposit or securities issued by one bank and held by another. *Second*, direct interconnectedness through financial market infrastructure linkages, specifically payments, clearing, and transaction settlement through the payment system. *Third*, through supply and other third-party linkages, providing financial services to several banks, for instance information technology. *Fourth*, dependence on other financial services between banks, for example, the derivatives market in the case of hedging, custodian banks, and guarantee companies.

Meanwhile, indirect financial interconnectedness is possible through various channels, where problems at one financial institution can affect another despite no direct financial or transactional links between the two. *First*, exposure to securities or other financial assets jointly held by financial institutions, such as bonds, foreign exchange and other securities. Price fluctuations, including interest rates and exchange rate, as well as heightened risk on the securities, especially when the financial markets are distressed, create financial interconnectedness between the financial institutions. *Second*, mark-to-market losses that trigger fire sales on financial markets also create indirect financial interconnectedness. Fire sales occur when numerous financial assets are sold during a brief period due to capital shortfalls, sudden withdrawals (bank run) due to inadequate liquidity and so on. When a fire sale occurs, tumbling prices and tight liquidity persist. *Third*, margin calls and haircuts also create indirect financial

interconnectedness. Lower collateral prices erode liquidity at lending banks that must meet the margin calls.

Fourth, interconnectedness through shadow banking activities. Although most counterparties are still banks, financial product innovation has expanded the scope of counterparties to include nonbanks, such as mutual funds, pension funds, insurance and hedge funds. Housing loan securitization, as occurred in the US, for instance, began in the banking sector but trade in those securities, as well as various structured products, quickly spread to the shadow banking system. Furthermore, such transactions spread to the international financial markets, particularly in Europe. The problem is that nonbank regulation and supervision is not as rigid as for the banking industry in any one jurisdiction or across borders. Consequently, default on a financial security could occur at a nonbank financial institution and then quickly propagate to the conventional banking system and to the entire financial system in the country involved or even to other jurisdictions, thereby triggering a chain reaction of systemic risk from financial interconnectedness.

Fifth, financial interconnectedness through information spillovers. Difficulties or default at one bank or financial institution could be perceived by the markets as a negative signal on other banks or financial institutions. Such developments could spark a crisis of confidence on the markets and then trigger a contagious reaction by market players. Such information spillovers can occur quickly not only from one market player to another but also through the mass media, including the television, print media and online. Default at one bank could also trigger bank runs. For example, the default on a structured financial security from housing loans in the US triggered a contagious reaction on the financial markets that culminated in the failure of the entire financial system.

14.4.2. Empirical Evidence for Financial Interconnectedness

Realizing the importance of preventing systemic risk in the financial system, several approaches and analytical empirical studies of interconnectedness and financial networks were developed by central banks and academia (ECB, 2010). Central banks, including the Bank of England, Deutsche Bundesbank, ECB and Reserve Bank of India (RBI), used financial network models to analyze interconnectedness on interbank money markets for systemic risk assessment in the banking system as well as the impact on the financial system as a whole. In particular, central banks used financial network models to: (1) understand and visualize the linkages and interactions of interbank lending and borrowing that are not clearly explained by the numbers; (2) assess the risk of credit exposure, liquidity, bank behavior and probability of distress that could emerge from transactions on the interbank money market; and (3) simulate potential systemic risk from default at one bank on the other banks and financial system as a whole, including probability, contagion, and losses incurred.

Mapping interbank financial networks and understanding the topology are crucial to formulate regulations and design the restructuring required to mitigate systemic risk (Haldane, 2009). The data used for the analysis is relatively detailed

and varied, such as: (1) direct bilateral exposure between banks on the interbank money market; (2) various types of transaction on the financial markets, for instance unsecured and secured loans, foreign currency exposure, swaps, forwards, and securities; (3) counterparty data; and (4) data from the payment system. The analyses are conducted using the nominal value and frequency, namely daily, weekly, and monthly.

From the data, the first step is typically to map the financial networks. Each bank is represented as a node on the network in the interbank market, the size of which corresponds to the share of transactions at the respective bank. A larger market share equates to a larger node. Meanwhile, the interbank transactional linkages are illustrated as a link, the boldness of which corresponds to the transaction value. The links represent the direction of the transaction. In this case, the direction of the network shows the interconnectedness between banks in terms of assets and liabilities and, therefore, measures total transaction exposure (gross, not netting). The network links are weighted from the total transaction value on the interbank market, or are not weighted.

Mapping the financial network helps to analyze banks where potential systemic risk could appear. This is achieved through centrality analysis of the role of each respective bank in terms of transactions on the interbank money market. Fig. 14.6 shows that banks 1, 2, and 3 act as money centers on the interbank money market, while the other banks are interconnected through those three banks. The three banks exacerbate interconnectedness on the interbank money market. Consequently, banks 1, 2, and 3 could potentially trigger systemic risk in the event of default. For instance, if bank 1 experienced default, not only would banks 2 and 3 be at risk but also the other banks that transact with those three.

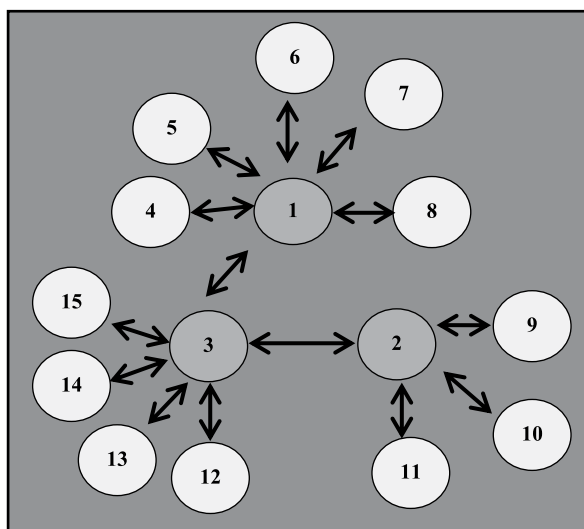


Fig. 14.6: Interconnectedness through Three Banks.

To deepen our understanding of interbank system dynamics, financial network analysis also explains the role of the bank on the interbank money market using indicators of cohesion, centrality and distance (Allen & Babus, 2007). Such indicators help investigate the statistical characteristics of the interbank system structure. The centrality indicator is useful to understand the distribution of banks on the financial network and determine the strength, effect and control of the respective bank. In Fig. 14.6, bank *N* is the centrality on the interbank market. The centrality may be in the form of central lender, central borrower, central counterparty or can also imply a balance between lending and borrowing. The cohesion indicator expresses the key linkages on the interbank market in terms of connectivity. Meanwhile, the distance indicator shows the extent of the financial network and how various information propagates through the interbank market.

Several models and empirical studies have revealed significant findings in the analysis of financial networks. In addition to the aforementioned study by Allen and Gale (2000), Sachs (2010) investigated the effect of credit exposure distribution between banks on the interbank money market. The study was motivated by observations of strong interconnectedness between large banks on a core network in the interbank money market with numerous small periphery banks that were connected to just one core but not the other banks. This kind of financial network became known as a money center. The study showed that FSS not only depends on the completeness and interconnectedness of the financial network, as revealed in the study by Allen and Gale (2000), but also on the distribution of credit exposure between banks in the interbank money market. Financial networks with a money center, as illustrated in Fig. 14.7, where the assets are concentrated at the core banks, tend to be more stable compared to a system where the banks are all of similar size in a random network.

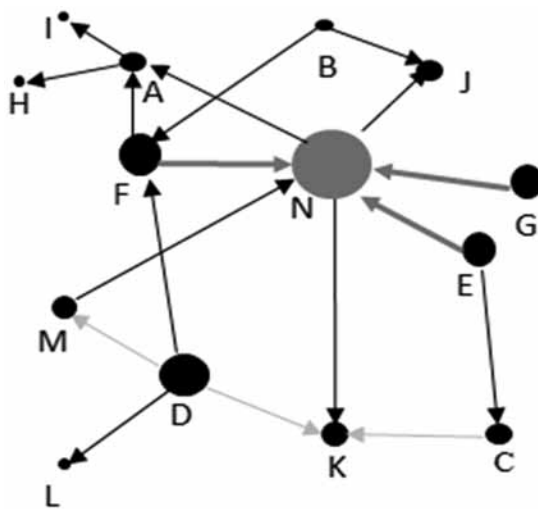


Fig. 14.7: Money Center Network.



Fig. 14.8: UK Banking Network.

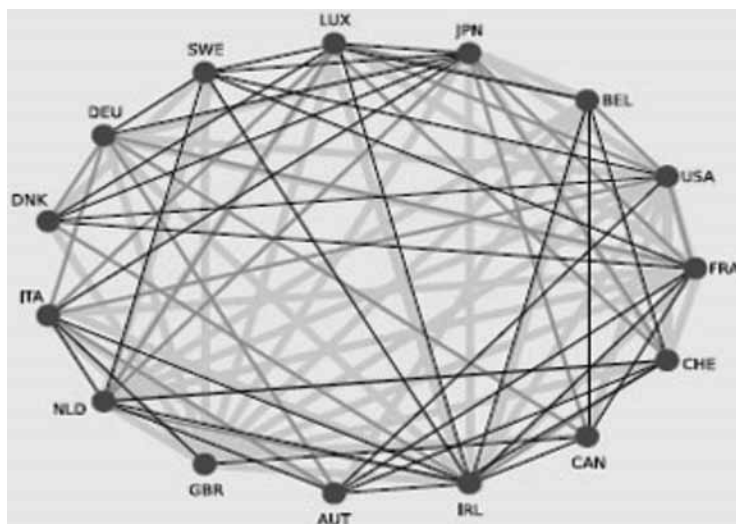


Fig. 14.9: Global Banking Network.

Several central banks use financial network analysis to assess systemic risk in the financial system. The Bank of England, for example, evaluates the network of large interbank exposures to assess systemic risk in the banking system in the UK, as presented in Fig. 14.8 using Q2/2009 data. Large exposures are defined as a value of liabilities that exceeds 10% of the capital required (Tier 1 and Tier 2). In that network, the nodes illustrate each bank in the UK, the size of which indicates total exposure of one bank to the other banks and nonbank financial

institutions in the banking system network. The boldness of the links is weighted proportionally from the exposure of a lending bank.

Minou and Reyes (2012) conducted an interesting study of the global banking network in 184 countries for the period of 1978–2010. In Fig. 14.8, the links show capital flows intermediated by banks between individual countries in 2007. The thin lines indicate a capital flow US\$ 1–5 billion, the medium lines indicate a capital flow of US\$ 5–10 billion and the bold lines indicate a capital flow exceeding US\$ 10 billion. The diagram shows the interconnectedness of banking activity in 15 countries, which explains the effect of spillovers and systemic risk from default at a bank in one country on banks in other countries. Moreover, the intensity of the global banking network also tends to be procyclical, implying that expansion and contraction follow the global capital flow cycle. Interconnectedness in the global network tends to increase prior to debt and banking crises and then declines dramatically post crisis.

Using financial network analysis, Haldan (2009) found several characteristics with important implications in the assessment of the impact on systemic risk. *First*, financial networks often appear robust yet fragile. To a certain degree, interconnectedness in a financial network acts as a shock absorber by diversifying and distributing the risks of various transactions among financial institutions. Nevertheless, in the event of default at a financial institution, financial networks also exacerbate conditions and accelerate propagation to the entire financial system. The US subprime mortgage debacle is a good example of this. The long-tailed distribution structure of the financial network was proposed as the cause. As presented in Fig. 14.5, default at a large bank, as a money center on the interbank money market, would trigger a systemic event throughout the banking system. In fact, default at a periphery bank could also have a systemic impact, known as the small-world phenomenon. This would occur if default at the periphery triggered problems at the respective systemic bank.

Second, financial network fragilities accelerate under market distress due to hoarding liabilities and fire sales of financial assets. Similar to contagion, reaction can be to hide or flight from the spillovers. The initial reaction of the banks when market conditions are under stress is to withdraw from transactions that create liabilities, holding on to liquid assets. The second reaction, however, is flight by selling assets to avoid deeper losses. In the financial system, the reaction to build liquidity, combined with fire sales, triggers a chain reaction through the financial network, which exacerbates and accelerates the propagation of bad assets or default at one bank into systemic failure.

Third, interconnectedness and transaction complexity in a financial network can create uncertainty and undermine price setting, risk assessment and counterparty transaction risk. In terms of credit default swaps (CDS), for instance, direct counterparty transaction risk assessment does not stop at the counterparty. A larger financial network implies that counterparty transactions could have many other counterparty transactions as well. Consequently, assessing the risk of default of counterparty transactions becomes more complex and less assured due to the potential effect on subsequent counterparties in the financial network chain.

Fourth, financial innovation also affects the complexity of the financial network chain. In the decade prior to the GFC of 2008/2009, for example, structured products exploded, thus breaking up the risk of financial assets and issuing new financial products, such as MBS and credit default orders (CDOs). Financial product innovation precipitated interconnectedness among increasingly complex claims contracts, thus blurring price setting and risk assessment. The problem is that innovative financial products are issued by an institution with a high rating, which are easy to sell due to guaranteed higher returns and risks that appear low. As more products are issued, however, risk assessment becomes less precise. Consequently, as a financial product turns bad, network complexity through financial product innovation exacerbates potential systemic risk.

Fifth, the benefits of financial product innovation diversification and phased risk management are lost due to replication and imitation on the financial markets. In the financial sector, diversification through financial product innovation, such as the securitization of housing loans and CDS, is desirable to increase returns and as a part of risk management. As such products begin to provide higher returns, however, other financial institutions imitate, leading to a proliferation of trade on the financial markets. Commercial banks begin to act as investment banks, for example, through housing loan securitization, selling such instruments on the financial markets as part of a new originate and distribute business model. Investment banks create financial products through fund managers, targeting maximum returns with risks that appear to be mitigated through financial product innovation. These managed funds are subsequently sold to various investors, including mutual funds, pension funds and insurance. What remains unknown, however, is that the risk of default is increasing, interconnectedness through the financial network is more complex and systemic risk is building.

The five characteristics are clearly apparent in the case of the massive US housing credit crisis along with the proliferation of structured products. In general, such developments created interconnectedness and complex financial networks with all five of the characteristics. During an economic upswing, a financial network flourishes due to the attractive returns offered, which are imitated and traded extensively on the financial markets, and appear robust yet are actually vulnerable to shocks. Therefore, a small shock of bad financial products triggers a chain reaction on the financial markets and subsequently feeds through to systemic failure due to complex interconnectedness in the financial network. Spreading, the markets hoard liquidity and initiate fire sales of their financial assets, thus exacerbating the decline and propagation into a large financial crisis. Such characteristics of interconnectedness and financial networks demand vigilance as an integral element of maintaining FSS.

14.5. Macroprudential Policy Theory and Practices

As explained previously, the central bank is the most appropriate institution to conduct macroprudential (regulation and supervision) policy. The central bank's mandate to maintain FSS is primarily achieved through such macroprudential policy. In addition, the central bank supports FSS through monetary policy,

payment system policy and foreign capital flow management, including jurisdiction to regulate and supervise the money markets and foreign exchange market as an integral part of monetary policy.

14.5.1. Principles, Targets, and Instruments

As mentioned at the beginning of this chapter, macroprudential policy encompasses the macroregulation and supervision of financial services institutions, with a focus on systemic risk to foster FSS. The primary target is to mitigate the risks that emerge from procyclical macrofinancial linkages and the accumulation of systemic risk that appears from interconnectedness and the networks between and in financial institutions, markets and infrastructures, including the payment system. The primary target of macroprudential policy is to prevent the build-up of risk from boom-bust financial cycles because of factors in the financial system itself as well as interaction with the domestic and international economies (time series dimension). The second target is directed toward reinforcing financial system resilience and mitigating contagion risk from interconnectedness and networks in the financial system (cross-section dimension).³

The two main targets of macroprudential policy, from a macroperspective and focusing on systemic risk in terms of FSS, differ from the targets of microprudential regulation and supervision, which are to maintain the soundness of individual financial institutions, including banks and nonbanks. The sound of individual financial institutions is necessary but insufficient to maintain FSS. Default at an individual financial institution, if not considered systemic, must be viewed as an insolvency problem due to mismanagement, weak risk management or lack of competitiveness. The closure of an individual financial institution due to default can be seen as a natural process of banking system restructuring and, simultaneously, as a test of overall financial system resilience.

It is different for systemically important banks (SIB). In accordance with the four criteria stipulated by BIS (2012), namely size, interconnectedness, complexity, and substitutability, the PPKSK Act of April 2016 defines systemic banks as

banks that, because of the size of their assets, capital and liabilities; broad network or transaction complexity of banking services; as well as interconnectedness with other financial sectors, could precipitate the failure of some or all other banks or financial sectors, operationally and financially, in the event of distress or default.

In that context, failure of a domestic systemically important bank (D-SIB) would likely trigger systemic risk and failure of the financial system as a whole. Per the

³In addition to the two primary targets, another target of macroprudential policy, as conducted by the European Central Bank (ECB), is to develop prudential regulations that, from a general system perspective, can facilitate market incentives and disincentives (structural dimension).

Table 14.3: Macroprudential Instruments Applied in Asian Countries.

	SIN	HKG	INA	MAL	KOR	IND	THA	PHI	PRC	Total	%
<i>Credit related</i>	13	5	11	6	23	6	15	1	9	92	49.2
<i>Liquidity related</i>	0	0	7	3	3	18	1	10	31	79	42.2
<i>Capital related</i>	1	1	1	0	2	4	0	6	1	16	8.6
Total	14	6	19	9	28	28	16	17	41	187	100

Source: Lee, Asuncion, and Kim (2015).

PPKSK act, systemic banks are required to: (a) meet specific capital and liquidity adequacy ratios; and (b) prepare an action plan that at least contains the obligations of the controlling shareholders and/or other parties to inject capital and convert certain types of debt into capital.

Which macroprudential policy instruments can be used? Lim et al. (2011) proposed ten instruments, as listed in Table 14.3, to control procyclicality and systemic risk in relation to credit exposure, foreign exchange, liquidity and capital. In terms of credit procyclicality, the most common instruments are the LTV, debt-to-income ratio (DTI) and restrictions on credit growth to certain sectors. Concerning foreign exchange exposure, the instruments include the net open position (NOP), restrictions on foreign currency loans as well as mandatory hedging and regulating the maturity of external debt. Regarding liquidity instruments, the reserve requirement is generally used, which can be adjusted in accordance with prevailing liquidity conditions. Meanwhile, capital can be strengthened to mitigate procyclicality and systemic risk through a countercyclical capital buffer (CCyB), loan loss provisions in line with procyclical credit risk and profit-sharing regulations.

Galati and Moesner (2014) categorized macroprudential policy instruments according to risk, namely the risks associated with leverage/credit boom/asset price bubbles, liquidity/market risks as well as interconnectedness/market structure risks in line with the risk dimension, namely dynamically over time (time series) or statically across sectors (cross section). In general, LTV policy and the CCyB are used to mitigate procyclicality in the time series dimension for the first type of risk. Time series instruments are also used to overcome liquidity/market risks, such as the loan-to-deposit ratio (LDR) in accordance with prevailing conditions as well as additional liquidity requirements for SIB, in addition to static instruments, for example, additional capital requirements for derivative debt and levies on non-core liabilities. In terms of interconnectedness and market structure risks, cross-sectional instruments are typically employed, including more stringent liquidity and capital requirements for SIB or additional deposit insurance premiums to mitigate systemic risk.

Macroprudential instruments can be loosened and tightened as required in accordance with the level of risk anticipated from procyclicality and systemic

risk. Claessens, Gosh, and Mihet (2014), categorized macroprudential policy instruments that can be applied during periods of economic expansion and contraction or to mitigate contagion risk from interconnectedness and financial networks. The instruments can take the form of lending restrictions, limitations on instruments and activities or restrictions on financial sector balance sheets and buffer-based policies. For example, during a period of economic expansion, the LTV, RR and CCyB ratios could be raised but then subsequently relaxed during a period of economic contraction. Adjustments can also be made to regulations concerning provisions, margins and haircuts. Furthermore, contagion could be mitigated through more stringent regulations on banks or systemically important financial institutions.

Several macroprudential and microprudential regulation and supervision instruments may, in fact, be the same because they are both based on assessments of liquidity, market and credit risks. Nonetheless, the targets and perspectives are different. Macroprudential instruments aim to maintain FSS from a macrofinancial (or procyclicality risk) perspective and focus on systemic risk. Conversely, microprudential instruments target the soundness of individual financial institutions with a focus on consumer protection, which include liquidity ratios, minimum capital requirements per the risk profile and prudent risk management. For the few instruments that have both a macroprudential and microprudential dimension, the following three regulatory aspects must be considered: individual soundness, systemic risk and procyclicality. Therefore, the instruments can be set incrementally in line with the three regulatory dimensions based on assessments of prevailing financial system and macroeconomic conditions. This kind of approach solves many of the issues associated with instruments that have macroprudential and microprudential targets.

Capital regulations, as stipulated by the FSB in November 2015, require a minimum level of Total Loss Absorbency Capacity (TLAC) for Globally Systemically Important Banks (G-SIBs), as an example of this kind of approach. The magnitude of TLAC ensures that G-SIBs maintain an adequate level of capital to absorb losses and recapitalization capacity in the event of business default or resolution, thus the critical bank functions are maintained without incurring losses to the state or threatening FSS. Consequently, TLAC is typically set at between 16% and 20% for capital requirements based on risk-weighted assets (RWA) and 6%–6.75% based on total bank exposure. The regulations are applicable to G-SIBs specified prior to 2015.

First, Basel III stipulates a capital requirement of 8% in accordance with the risk profile, which must be met by all banks, systemic or otherwise. The 8% threshold aims purely to ensure the soundness of individual banks. *Second*, various capital buffers, under Basel III, are also required, including: the capital conservation buffer for systemic risk, CCyB to mitigate procyclicality, and a capital surcharge for G-SIBs. As a macroprudential instrument, the rate of the CCyB can be raised or lowered in accordance with assessments of procyclicality in the current period. *Third*, the additional capital required and subordinated debt instruments with a maturity of more than one year may be included in the calculation of TLAC. A structured regulatory approach based on individual

soundness targets, systemic risk and procyclicality, such as TLAC, can also be applied to other prudential banking instruments that have a micro and macro-prudential perspective.

14.5.2. Application in Various Jurisdictions

After the GFC of 2008/2009, central banks in various jurisdictions applied different macroprudential policies to mitigate macrofinancial risks (procyclicality) and systemic risk that could undermine FSS, in addition to monetary policy and foreign capital flow management to control inflation, stabilize exchange rates and maintain a healthy BOP. The same phenomenon occurred at central banks in Asia, which proved that the application of macroprudential instruments supports FSS using different types of instrument to mitigate macrofinancial and systemic risks in line with prevailing conditions in each country (Lee, Asuncion, & Kim, 2015). Table 14.3 presents the macroprudential instruments applied in 10 Asian countries to control credit, including LTV and DTI, which were most commonly applied in Indonesia, South Korea, Singapore and Thailand. Meanwhile, liquidity-related macroprudential instruments, such as RR and NOP, were applied in China, India and the Philippines. In contrast, capital-related macroprudential instruments, such as the CCyB, at that time, were not commonplace, except in India. Of the three instrument types, credit-related macroprudential instruments were the most popular among central banks in Asia at that time.

Of the 10 countries, China applied the most macroprudential policy instruments from 2000 to 2013 to regulate balance sheet exposure at financial institutions, rein in the credit boom and escalating house prices, and simultaneously avoid systemic risk. From 2006 to 2008, the People's Bank of China (PBoC) and China Banking Regulatory Commission simultaneously implemented macroprudential instruments to reinforce FSS. Macroprudential policy was tightened as many as 40 times to offset heightened financial system instability risk at that time, including 31 times using liquidity-related instruments, specifically the reserve requirement. The experience demonstrated that tightening credit-related macroprudential policy instruments successfully dampened the credit boom and rising house prices but failed to reduce debt accumulation (leverage). Meanwhile, liquidity-related macroprudential instruments initially succeeded in reducing leverage, but the impact on rising new house prices only become apparent several periods thereafter.

South Korea systematically applied a number of macroprudential instruments prior to the GFC of 2008/2009 and even before the Asian financial crisis of 1997/1998, including several liquidity ratios to overcome fragilities in the banking system and in terms of foreign currency transactions. In November 2009, for example, the Bank of Korea promulgated regulations requiring the banks to increase their share of long-term foreign loans from 80% to 90% and then subsequently to 100%. In addition, the Bank of Korea also introduced restrictions on foreign currency derivatives at banks and reserves for foreign currency loans, as well as tighter liquidity ratios for domestic banks. Furthermore, the Bank of Korea tightened macroprudential policy in 2006 and 2007 to overcome

spiraling house prices, which were exacerbated by the US crisis in 2007–2008. In fact, from 2000 to 2013, the Bank of Korea issued macroprudential instruments approximately 28 times, most of which were credit related. The impact of tighter macroprudential policy quickly became apparent in terms of slowing credit expansion and leverage, but only subsequently did the impact on rising house prices become evident.

In India, bank credit growth has become an important consideration underlying monetary and macroprudential policymaking at the RBI. Consequently, macroprudential policy has been instituted since 2004 through several instruments, including liquidity and capital-related instruments, to maintain FSS. The primary focus is concentrated on excessive credit growth in the housing sector. To that end, capital-related macroprudential policy, as applied by the RBI, focuses on the banking industry through application of the CCyB. In addition, tighter regulations on risk weights and asset reserves have also been enforced for housing loans. RBI also applied liquidity-related macroprudential instruments, primarily by raising the reserve requirement. Although credit growth persisted, the range of macroprudential policy instruments controlled excessive leverage and rising house prices and, therefore, helped to prevent an accumulation of systemic risk and asset price bubbles. The challenge was, however, to control financing to the housing sector from the shadow banking system, including from developers, as well as rising house prices (Table 14.4).

In Singapore, although the Monetary Authority of Singapore (MAS) maintains stringent regulation and supervision over the financial system, financial system instability risk has increased since 2010 due to rising house prices, which have already exceeded the peak recorded in 2008. Higher property prices have stoked concerns of rising inflation expectations and systemic risk because most property is purchased using comparatively cheap bank loans. Housing loans have continued to accelerate and household debt has also escalated over the past three years to 76% of GDP at the end of 2012, with large exposure at domestic banks to the housing sector. In response, MAS quickly strengthened surveillance and assessments of FSS as well as the policies required, one of which was to tighten regulations in the property sector by raising the loan installment-to-income ratio (IIR), lowering the threshold of the LTV, introducing an additional stamp duty levied on property purchases and increasing the minimum cash downpayment for property purchases. The regulations specifically targeted the speculative segment of the property market. In general, tighter macroprudential policy successfully reduced house prices and housing loans and, therefore, become an integral element of maintaining FSS in Singapore.

In addition to the four countries mentioned, numerous other Asian countries have also applied macroprudential policy, including Hong Kong, Malaysia, Philippines, Thailand and Indonesia. In general, macroprudential policy is used to mitigate systemic risk and, thus, maintain FSS during periods of excessive credit growth and rising property prices, in addition to other sectors, such as the automotive sector in Indonesia. Such problems were also linked to economic booms and/or the influx of foreign capital flows to the Asian region from 2010 to 2013 due to loose monetary policy in various jurisdictions after the GFC in 2008/2009,

Table 14.4: Macroprudential Instruments: Types and Risk Dimensions.

Risk Types	Risk Dimension: Time Series or Cross-sections
Leverage/credit/ asset price booms	<i>Time Series:</i> • Countercyclical capital buffers • Through-the-cycle valuation of margins or haircuts for collateral used in securitized funding markets (like repos) • Countercyclical change in risk weights for exposure to certain sectors • Time-varying LTV, Debt-To-Income (DTI) and Loan-To-Income (LTI) caps • Time-varying caps and limits on credit or credit growth • Dynamic provisioning • Rescaling risk-weights by incorporating recessionary conditions in the probability of default assumptions (PDs)
Liquidity/market risk	<i>Time Series:</i> • Time-varying systemic liquidity surcharges • Levy on non-core liabilities • Time-varying limits in currency mismatch or exposure (e.g., real estate) • Time-varying limits on loan-to-deposit ratio • Stressed VaR to build additional capital buffer against market risk during a boom <i>Cross-sections:</i> • Capital charge on derivative payables • Levy on non-core liabilities
Interconnectedness/ market structure/ financial infrastructure	<i>Cross-sections:</i> • Higher capital charges for trades not cleared through CCPs • Systemic capital surcharges • Systemic liquidity surcharges • Powers to break up financial firms on systemic risk concerns • Deposit insurance risk premia sensitive to systemic risks • Restrictions on permissible activities (e.g., ban on proprietary trading for SIBs)

Source: Galati and Moessner (2014).

as explained in Chapter 13. Macroprudential instruments vary from one country to another, including credit, liquidity and capital-related instruments, depending upon the specific problems faced and prevailing financial system conditions. The most common instruments include the LTV and DTI ratios, reserve requirement and CCyB. The application of macroprudential policy has proven effective in terms of controlling excessive credit growth and rising property prices and, therefore, is an integral element of preventing build-ups of systemic risk, while also helping to maintain FSS.

14.6. Concluding Remarks

This chapter has elucidated an invaluable lesson from the GFC of 2008/2009, namely the need to maintain FSS as well as macroeconomic stability in order to support robust and sustainable economic growth. The chapter focused on interconnectedness between the financial system and economy, or macrofinancial linkages, as reflected in financial procyclicality and systemic risk. The GFC of 2008/2009 and previous crisis episodes were all preceded and caused by an accumulation of systemic risk that surfaced due to excessive credit growth, asset price bubbles, particularly property prices, uncontrolled external debt, and sudden capital reversals. It is impossible, however, to prevent the various crisis triggers through microprudential regulation and supervision, macroeconomic policy (fiscal and monetary policy) and foreign capital flows management.

Macroprudential policy encompasses the regulation and supervision of financial institutions from a macroperspective, namely macrofinancial linkages, with a focus on systemic risk to maintain FSS. There are two overarching targets: preventing financial procyclicality (time series dimension) and mitigating systemic risk (cross-section dimension) that appear from macrofinancial linkages. The instruments used to control procyclicality and mitigate systemic risk are credit-related, foreign currency-related, liquidity-related, and capital-related. Macroprudential instruments can also be adjusted per the types of risk, namely leverage/credit boom/asset price bubble risks; liquidity/markets risks; and interconnectedness/market structure risks in line with the risk dimension, either dynamically over time (time series) or statically across sectors (cross section). Furthermore, macroprudential policy can be loosened or tightened according to the level of risk anticipated from potential procyclicality and systemic risk. The central bank is the most appropriate institution to implement macroprudential policy.

In general, macroprudential policy instruments have become commonplace at central banks in most jurisdictions, including Asia, in addition to monetary and exchange rate policy as well as foreign capital flow management. The GFC of 2008/2009 ultimately confirmed the importance of deepening and implementing policy to control macrofinancial linkages and to avoid a build-up of systemic risk that not only endangers FSS but also threatens macroeconomic stability and sustainable economic growth. The problems are becoming increasingly complex as foreign capital flows gain mobility through stronger global financial integration. The challenge is how central banks integrate monetary policy, macroprudential policy, and foreign capital flow management into an optimal policy mix to achieve price and exchange rate stability on one hand, and also support sustainable economic growth on the other. This issue will be explored and addressed in Chapter 15 concerning the central bank's policy mix.

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Chapter 15

Central Bank Policy Mix

15.1. Introduction

The global financial crisis of 2008/09 had fundamental implications on the central bank's mandate, policy and institutional arrangements. In terms of the mandate, central banks can no longer merely focus on price stability alone but also on maintaining financial system stability. To achieve that dual mandate, central banks institute monetary policy, foreign capital flow management, and macroprudential policy within an optimal policy mix, in addition to payment system policy. Institutionally, the central bank strengthens internal capacity through assessments, policy scenarios, and decision-making that underlie the policy mix. Furthermore, the central bank also coordinates with the government in terms of macroeconomic policy as well as with other relevant authorities to maintain financial system stability.

During the two decades preceding the global financial crisis of 2008/09, central banks in various jurisdictions focused on price (inflation) stability to support economic growth. This was driven by academic consensus at the time regarding the trade-off between inflation and economic growth in the short term and monetary policy neutrality in the long term, per a synthesis of Keynesian and Neoclassical economics, as elaborated in Chapter 3. From a political economic perspective, that focus represented a reformulation of central banking policy and institutional reforms through crisis resolution in the wake of the Asian financial crisis of 1997/98. In fact, the bank supervision function was transferred away from the central bank in several countries, including South Korea and Indonesia. In practice, a monetary policy framework that targets price stability, otherwise known as the inflation targeting framework (ITF), became popular and was embraced in various advanced and emerging market economies (EMEs) alike, as elucidated in Chapters 8 and 9.

The global financial crisis of 2008/09 fundamentally changed the perspective of the central bank's mandate and policy practices but not because ITF-based monetary operations had failed. In contrast, ITF had successfully achieved low inflation, stimulated economic growth, and reduced interest rates in many jurisdictions. In fact, monetary stability had spurred economic booms in many countries, including the United States, during an era that became known as the

Great Moderation. The problem was that a focus on price stability alone, made central banks less responsive to crisis risks contained in macrofinancial linkages. On the other hand, microprudential regulation and supervision focuses on the soundness of individual financial institutions, especially the banks. The US crisis is a concrete example of where macrofinancial linkages during an economic boom in a superpower heightened systemic risk that culminated in a devastating financial crisis and impacted all corners of the globe. In addition, the global financial crisis of 2008/09 also had other implications for central bank policy, namely the influx of volatile foreign capital flows to EMEs as a consequence of monetary expansion in advanced countries to stimulate the economic recovery process. Such conditions further complicated the central banks' policy response to achieve price (and exchange rate) stability and to maintain financial system stability.

In brief, to support sustainable economic growth, the central bank cannot merely rely on monetary policy to achieve price (and exchange rate) stability alone. Indeed, the monetary policy strategy of the central bank, as described in Chapters 6–9, as well as institutional strengthening, as explained in Chapters 10–12, are no longer considered sufficient. Central banks are now required to support financial system stability through macroprudential regulation and supervision from a macrofinancial perspective and with a focus on systemic risk as elucidated in Chapter 14. Furthermore, the central bank is also required to apply foreign capital flow management in order to provide the maximum gains in terms of sustainable economic growth, without triggering the risk of macroeconomic and financial system instability due to a balance of payments crisis, external debt crisis, and/or sudden-stop capital reversal crisis, as explored in Chapter 13. The question, therefore, is how the central bank can integrate and combine those three policies, namely monetary policy, macroprudential policy, and foreign capital flow management, into an optimal policy mix?

This chapter discusses the latest ideas and practices regarding the importance of the central bank's policy mix to achieve price stability and support financial system stability. Various ideas that originated at international forums and the central banking community will be proposed, along with several theories currently being developed in the academic community. Following on from this introduction, the discussion is broken down into four main topics. The first part deals with the conception of the central bank policy mix in terms of integrating the dual targets of price stability and financial system stability, formulation of the monetary policy mix, macroprudential policy, and foreign capital flow management, as well as the transmission mechanisms to attain the dual targets. The second and third sections introduce and examine two common modeling approaches used to formulate the policy mix, namely the structural macroeconomic model and the Dynamic Stochastic General Equilibrium (DSGE) model. The fourth section explains how the policy mix at Bank Indonesia has been formulated and applied during three distinct episodes since 2010. In general, the policy mix applied by Bank Indonesia plays an important role in terms of strengthening national economic resilience to global economic and financial shocks and uncertainty since the global financial crisis of 2008/09.

15.2. Conceptual Dimension of the Central Bank Policy Mix

Fundamentally, the central bank policy mix represents the optimal integration of monetary policy, macroprudential policy, and foreign capital flow management, as applied by the central bank to achieve price stability and maintain financial system stability. The terminology has a broader scope than the Flexible Inflation Targeting Framework (FITF) proposed in several studies (Agénor & da Silva, 2013; Svensson, 2009; Ito, 2010; Juhro, 2015; Warjiyo, 2013a). *First*, and departing from FITF, the ultimate target is not only price stability but also maintaining financial system stability. *Second*, and this is similar to FITF, the instruments used are monetary policy, macroprudential policy, and foreign capital flow management in one optimal mix. And *third*, which is also like FITF, policy mix formulation requires an analysis framework and macroeconomic projections that take into consideration macrofinancial linkages. Formulation of the central bank policy mix also requires assessments of the more complex transmission mechanisms, not only of the three component policies on price stability but also on support for maintaining financial system stability. Three aspects that are the fundamental concepts of the central bank policy mix framework are discussed in the following subsections.

15.2.1. Integration of the Price Stability and Financial System Stability Targets

Chapter 14 discussed the need for central banks to maintain financial system stability. Therefore, support has recently gained momentum for central banks to have a dual mandate, namely price stability and supporting financial system stability (Acharya, 2015; Bank for International Settlements (BIS), 2012; Kawai & Morgan, 2012; Sinclair, 2000). How can a central bank integrate both mandates? Several ideas have been developed in the context of the close correlation between monetary stability and financial system stability. Regarding the first mandate, namely to achieve price stability, consensus has been reached among EMEs that central banks should not only focus on inflation but also maintain exchange rate stability, as explained in Chapter 13. The nascent issue is, therefore, how a central bank can address other price dimensions, such as financial asset prices (stocks and bonds) as well as property prices. Concerning the second mandate, namely to support financial system stability, as explained in Chapter 14, central banks institute macroprudential policy through the regulation and supervision of financial institutions from a macrofinancial linkages perspective and with a focus on systemic risk, in addition to monetary policy (including foreign capital flow management) and payment system policy.

The importance of price stability, as an integral part of macroeconomic stability and financial system stability to support sustainable economic growth, is inevitable. One dimension of price stability is goods and services, commonly referred to as inflation, which is measured using the price change above the base

Consumer Price Index (CPI).¹ Another dimension of price stability is with respect to other currencies, commonly referred to as exchange rates. Unlike conditions in advanced countries, the role of exchange rate stability is crucial in EMEs due to a number of salient considerations. In general, foreign exchange markets in EMEs remain inefficient, thus often leading to volatile fluctuations because of domestic economic shocks and/or foreign capital flows. The exchange rate impacts inflation directly through import prices and indirectly through domestic demand. Exchange rate fluctuations also influence financial system stability through their impact on the balance sheets of financial institutions as well as stock prices and bond yields. In fact, exchange rate (currency) shocks often stoke concerns in the business community and among the public and, therefore, are a good barometer of political stability in many EMEs. The various considerations outlined ensure that exchange rate stability remains an important part of the price stability mandate, in addition to inflation, as is the case in Indonesia. This is achieved through monetary policy, using interest rate instruments, foreign exchange intervention, and foreign capital flow management, as discussed in Chapters 4 and 13.

A new dimension has become the focus of post-global crisis discussions between central bankers and academia, namely how the central bank should deal with other asset prices, such as stock and bond prices as well as property prices? Several economists suggested the central bank should maintain focus on price stability in support of maintaining financial system stability (Bordo, 2007; Schwartz, 1995). Nevertheless, the global financial crisis of 2008/09 proved that price stability was insufficient and, therefore, the central bank should also support financial system stability (Borio & Lowe, 2002; Borio et al., 2003; Shirakawa, 2012). In fact, the global financial crisis demonstrated that too great of a central bank focus on short-term price stability to maintain economic stability actually created instability due to asset price bubbles, debt accumulation, excessive credit growth and other macrofinancial imbalances that culminated in a crisis. This is the essence of the financial instability hypothesis in a capitalist economy, as coined by Minsky (1982).

Concerning the linkages between monetary policy and property prices, an interesting debate continues to rage regarding whether the success of the Federal Reserve in terms of achieving low and stable inflation and interest rates for two decades during the Great Moderation was a contributing factor to the financial crisis that befell the United States. Taylor (2010), for example, stated that the Fed's policy during that time was excessively loose and the Federal Funds Rate (FFR) was too low for a prolonged period, which precipitated a housing sector boom and, therefore, macrofinancial imbalances that culminated in a crisis. Bernanke (2010), on the other hand, disagrees with that point of view. He argued that

¹CPI measures prices at the consumer level as a weighted average of each good or service. Inflation can also be measured at the producer level using the Producer Price Index (PPI). Another measure of inflation is the GDP deflator, which measures the level of prices of all components in an economy in the current year against a base year when calculating nominal and real GDP.

the main cause of excessive housing bubbles was the various types of cheap and attractive credit and mortgage-backed securities against a backdrop of weak lending and underwriting standards. Therefore, a more appropriate solution would be prudential regulation rather than monetary policy. Meanwhile, Filardo (2001) found that the monetary authority must respond to rising asset prices while it has access to trusted information on inflation and output, although it is difficult to differentiate between the cause of rising asset prices, namely due to fundamental factors or bubbles.

The latest empirical evidence after the global financial crisis of 2008/09 has shown the importance of central bank policy in response to asset price bubbles, particularly property prices. Jorda, Schularick, and Taylor (2014), for instance, proved the importance of linkages between monetary conditions, house prices and credit growth from data stretching over 140 years from 14 advanced countries. Specifically, the empirical evidence showed that a 1% decline in the policy rate would raise house prices (in the ratio to income) and housing loans (in the ratio to GDP) by more than 4% and 3%, respectively, over a horizon of 4 years. History has shown that potential instability from loose monetary policy must be considered seriously in the narrow terms of price stability on CPI inflation compared to the adverse impact that may result on the economy. Meanwhile, Williams (2015) suggested that there is an expensive trade-off using monetary policy to mitigate house prices when there is a conflict between the targets of macroeconomic stability and financial system stability. Using the same data, a 1% increase in the monetary policy rate was shown to depress real house prices (after considering inflation) by more than 6% over 2 years, while real GDP per capita fell by nearly 2%. Such empirical evidence confirms that central banks must equally consider price stability and financial system stability when formulating monetary policy.

The discussion then leans toward the issue of how monetary policy responds to macrofinancial imbalances and a build-up of systemic risk, which manifest in procyclical housing bubbles, credit booms, external debt, and volatile foreign capital flows. This issue is also linked to another debate, *lean vs clean*: namely whether it would be better for a central bank to lean against the wind in order to prevent bubbles from bursting or better to wait for the bubble to burst before cleaning up the mess through aggressively loose monetary policy. The “clean” view was embraced by the Federal Reserve under Alan Greenspan. Some of the basic considerations include that an investment boom is driven by increased productivity, bubbles are due to a lower risk premium and irrational exuberance, and raising interest rates might be ineffective in mitigating the bubbles and actually burst them, thus adversely impacting the economy (Greenspan, 2002).

Nonetheless, the global financial crisis of 2008/09 dispelled that view. The Crisis clearly showed potential risk from excessive credit growth and leverage, driven by bubbles. Therefore, it is considered more appropriate for the central bank to lean against the wind to prevent bubbles rather than cleaning up the devastating crisis impact. Experience from the Australian case, for instance, showed that leaning monetary policy can be successful. In response to excessive credit growth in the housing sector during 2002 and 2003, the Reserve Bank of Australia gradually

raised its policy rate despite inflation remaining under control and on target (Bloxham, Kent, & Robson, 2011). The justification for a tighter monetary policy stance in terms of the ITF was that the Reserve Bank of Australia was increasingly concerned about excessive credit growth. Consensus now strongly supports leaning against the wind and for central banks to seriously consider aspects of financial system stability, even though financial system stability is not part of the mandate. Bernanke (2009), for instance, stated that the Federal Reserve played a significant role in terms of mitigating the crisis in the United States and, therefore, must maintain its institutional capacity to support financial system stability and foster economic recovery with low inflation.

The second central bank mandate, namely to maintain financial system stability, also contains several applicable aspects and policy instruments. Chapter 14 has already explained that maintaining financial system stability is determined by several salient factors, namely the soundness of individual financial institutions, controlling procyclical lending and housing bubbles, domestic macroeconomic stability (inflation and fiscal deficit) and external stability (exchange rate, current account deficit, external debt, and risk of sudden capital reversal), interconnectedness and financial networks, as well as controlling the risk of information contagion. Those aspects, in general, form part of the FSS indicators, such as the Financial Soundness Indicator (FSI), as well as Early Warning Indicators (EWI).² Furthermore, those aspects also show where a central bank supports financial system stability through policy coordination with the government and other relevant authorities.

The central bank's mandate to support financial system stability is primarily achieved through macroprudential policy, namely the regulation and supervision of financial institutions from a macrofinancial perspective with a focus on systemic risk. As discussed in Chapter 14, macroprudential policy is principally directed toward two main targets to maintain financial system stability. The first target is to prevent an accumulation of risk from a boom-bust financial cycle,

²Several studies have developed measures for the key aspects of FSS. In general, however, simple FSS indicators, covering default risk and asset prices, are inadequate. Concerning the soundness of individual financial institutions, the IMF developed the FSI, which encompasses indicators of capital adequacy, liquidity, credit quality, income and profitability, as well as sensitivity to market risk (IMF, 2001). The IMF and FSB also collaborated to develop and analyze Early Warning Exercises (EWE), which are typically presented each year in April and October. This assessment covers: (1) sectoral and market fragilities in the external, fiscal and corporate sectors, asset prices, market valuation, and bubble detection, as well as risk-taking behavior on the financial markets; (2) modeling sovereign risks, especially systemic risks; and (3) assessing the systemic implications of spillover and contagion analysis, analysis of Large Complex Financial Institutions (LCFI) as well as global scenarios (IMF, 2010). Another ambitious IMF project is to develop a more comprehensive and integrative measure of systemic risk, known as "SysMo" for six key risks, namely financial institutions, asset prices, sovereign risks, the broader economy, cross-border linkages, and crisis risks (Blancher et al., 2013).

specifically credit booms and housing bubbles, either due to internal factors of the financial system itself or interaction between the domestic and international economies. The main macroprudential policy instruments available to control financial procyclicality include: the loan-to-value (LTV) ratio or debt-to-income ratio (DTI), countercyclical capital buffer (CCB), and loan-to-deposit ratio (LDR) as well as additional liquidity requirements for systemic banks. Meanwhile, the second target of macroprudential policy is to strengthen resilience and mitigate contagion risk from interconnectedness and financial networks in the financial system. The instruments available include mapping, regulation and interbank money market restructuring to control systemic risk from the perspective of interconnected exposure in terms of liquidity, credit, counterparty transactions, and the operational failure of financial infrastructure, including the payment system. In addition, more stringent liquidity and capital requirements can be applied to systemic banks or additional deposit insurance premiums to mitigate systemic risk.

In addition to macroprudential policy, central banks also support financial system stability through monetary and payment system policy and foreign capital flow management. Monetary policy supports financial system stability by preventing economic shocks that could threaten financial system stability through controlled inflation, exchange rate stability, and a managed current account deficit. Monetary policy instituted through interest rate instruments and the statutory reserve requirement can be used to lean against risk accumulation due to financial procyclicality, particularly excessive credit growth, and housing bubbles. In addition, monetary policy also covers the regulation and supervision of money markets and the foreign exchange market, which also supports financial system stability from the perspective of mitigating systemic risk that can emerge from interconnectedness and financial networks, especially on the interbank money market and foreign exchange market. The same considerations are at play concerning central bank jurisdiction in the payment system, which supports financial system stability in terms of interconnectedness and networks in the case of uninterrupted transactions, instrument security, and payment system infrastructure resilience. Meanwhile, the central bank's authority regarding foreign capital flow management also supports financial system stability by preventing economic shocks from currency instability risk, sudden-stop, and balance of payments crises as well as excessive external debt accumulation.

15.2.2. Basic Concepts of the Central Bank Policy Mix

The previous section explained the close linkages between price stability and financial system stability as the dual mandate of the central bank after the global financial crisis of 2008/09, and how monetary policy, macroprudential policy, and foreign capital flow management are mutually complementary in pursuance of that dual mandate. The question, therefore, is how the central bank integrates those instruments into an optimal policy mix to achieve the dual mandate? As described in Chapters 6–9, the monetary policy framework established at various central banks has been proven to achieve price stability but with a trade-off in

terms of economic growth. The most common instruments are interest rates and liquidity on the interbank money market to influence domestic demand through the interest rate, exchange rate, money supply, credit, risk-taking behavior, and market expectations channels. That monetary policy framework is supported by an analysis system and macroeconomic projections, accompanied by simulations of a consistent policy response, scheduled policymaking process, as well as transparent communications, as explained in Chapters 10–12. Furthermore, as elucidated in Chapter 13, the impact of the global economy through exchange rates, foreign capital flows, and the balance of payments has also already been included in the macroeconomic projections. In fact, a policy mix response between interest rate instruments, intervention on the foreign exchange market and foreign capital flow management is practiced by various central banks in EMEs to overcome the monetary policy trilemma in an open economy.

The established monetary policy framework could easily accommodate macroprudential policy to manage macrofinancial imbalances and systemic risk in order to support financial system stability as mentioned in Chapter 14. The existing framework must be expanded to cover macrofinancial linkages in the financial system. In addition to the existing analysis as well as macroeconomic and external projections, analyses, and projections of macrofinancial imbalances are required along with interconnectedness in the macroeconomic assessment. To that end, macrofinancial assessments should focus on four types of procyclicality and systemic risk that are often cited as the causes of a crisis, namely assets (financial and property), price bubbles, credit booms, external debt accumulation, and volatile foreign capital flows. Macroeconomic and macrofinancial analyses and projections should underlie formulation of the optimal policy mix to be instituted by the central bank.

Fig. 15.1 illustrates how the macroprudential policy framework is integrated into the monetary policy framework already established at the central bank.

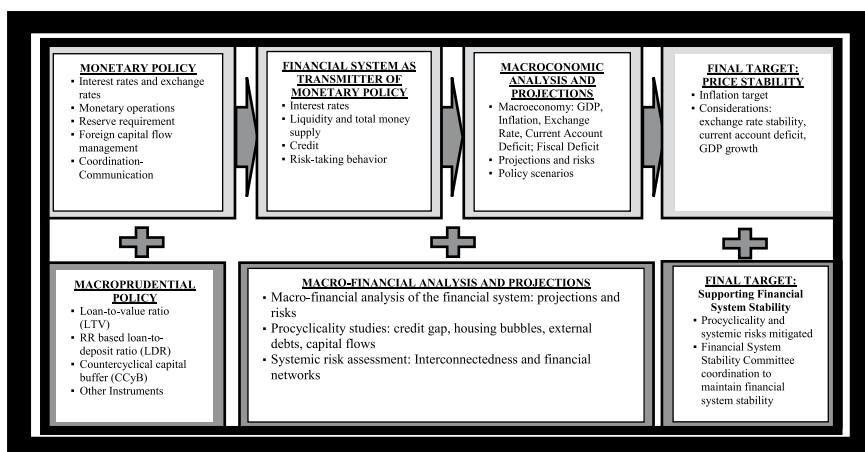


Fig. 15.1: Integration of the Monetary Policy and Macroprudential Policy Framework.

The upper panel illustrates the commonly understood monetary policy framework to achieve price (and exchange rate) stability, including: inflation as the final target, macroeconomic analysis, and projections, assessment of monetary policy transmission and the monetary policy instruments available. The external dimension of the monetary policy framework not only covers assessments of exchange rates and BOP conditions in the macroeconomic analysis and projections, but also the impact of global spillovers and the foreign capital flow management required. Such a policy framework refers to FITF, which has been applied by numerous central banks, including Bank Indonesia. The integration of macroprudential policy into that policy framework is presented in the lower panel, covering: the complementary targets to support financial system stability, analysis, and projections of macrofinancial linkages from the financial system in the economy, assessments of procyclicality and systemic risk, as well as the macroprudential policy instruments available.

In implementation, three salient aspects demand attention, namely: (1) how the dual targets of price stability and financial system stability can be integrated; (2) the mix of policy instruments to use; and (3) the effectiveness of the transmission mechanism. The description in the upper part explains how price stability and financial system stability can be integrated into the dual mandate of the central bank. In terms of the instruments, integration between monetary policy, macroprudential policy, and foreign capital flow management is required to ensure mutual complementariness when pursuing the dual mandate of the central bank. Accordingly, three fundamental policymaking frameworks underlie formulation of the central bank policy mix as follows.

First, monetary policy is still directed toward achieving price stability, while also simultaneously paying more attention to (financial and property) asset prices in order to maintain financial system stability. Regarding the exchange rate, as elucidated in Chapter 13 and reiterated above, interest rate policymaking already takes into consideration the impact of the exchange rate on achieving the inflation target through an extension of the Taylor rule. In fact, many EMEs, including Indonesia, adhere to two instruments-two targets, namely the interest rates and foreign exchange market intervention to achieve the inflation target and exchange rate stability. Meanwhile, the interest rate policy response to other asset prices, particularly property prices, is achieved by leaning against asset price movements that are shown to deviate from fundamentals. The monetary policy response to asset prices, including exchange rates, stock prices, and bond yields, as well as property prices, is formulated to achieve price stability and support financial system stability.

Second, macroprudential policy covers the regulation and supervision of financial services institutions from a macrofinancial perspective, with a focus on systemic risk to nurture financial system stability. As noted in Chapter 14 and reiterated above, macroprudential policy to control financial procyclicality, specifically credit booms and housing bubbles, is achieved by applying LTV, DTI, CCB as well as other instruments. Meanwhile, systemic risk is mitigated using macroprudential instruments in the form of interbank money market regulations and financial infrastructure, as well as enforcing liquidity and capital requirements on

financial institutions shown to have a high level of interconnectedness and intricate financial networks. The range of macroprudential policy instruments used to maintain financial system stability also reinforce the effectiveness of monetary policy transmission to achieve price stability. During an economic upswing, for example, interest rate policy effectiveness to control excessive credit growth could be strengthened through more stringent LTV or DTI enforcement for sectors less sensitive to interest rates, such as the property sector.

Third, foreign capital flow management could further strengthen monetary policy effectiveness to achieve price stability, or macroprudential policy effectiveness to support financial system stability from global spillovers. As mentioned in Chapter 13, foreign capital flows exert influence on exchange rates, asset prices, liquidity, and risk-taking behavior at home. Similarly, external risks to domestic macroeconomic conditions can also originate from external debt, especially short-term external debt and that used to finance the domestic-oriented economy. The risks associated with foreign capital flow volatility and external debt, in general, heighten during periods of economic expansion and also go hand-in-hand with financial procyclicality and systemic risk that emerge in the domestic financial system. Therefore, foreign capital flow management is crucial to support exchange rate stability and also represents an important element of BOP and sudden-stop crisis prevention efforts, which can often culminate in a financial crisis.

In addition to integration of the three policies mentioned, financial market deepening is another vital aspect of supporting the efficacy of the central bank's policy mix. Mature and efficient financial markets enhance the effectiveness of monetary policy transmission through the interest rate channel, exchange rate channel, liquidity channel and bank credit channel, as well as in terms of risk-taking behavior. Financial market deepening also supports intermediation of foreign capital flows for economic financing, as well as provides exchange rate flexibility and mitigates the risks to financial system stability that appear. Furthermore, financial product innovation and investment portfolio diversification through financial market deepening supports monetary and financial system stability. Nonetheless, more stringent prudential regulations and market conduct are required to accompany financial market development. As will be explained in subsequent sections, financial product innovation and portfolio diversification also increase interconnectedness and financial networks and, therefore, exacerbate systemic risk. Similar considerations are also at play concerning central bank jurisdiction over the payment system. Increased interconnectedness and instrument innovation in the payment system would enhance the reliability and efficiency of various financial and economic transactions, in addition to improving monetary policy effectiveness. Therefore, more stringent regulation and supervision of the payment system would be required to ensure such positive developments do not trigger systemic risk to financial system stability.

15.2.3. Central Bank Policy Mix Transmission Mechanism

Effective transmission of the central bank policy mix is crucial to achieve price stability and maintain financial system stability. Chapter 5 explained how the

monetary policy transmission mechanism influences domestic demand and the attainment of price stability through the interest rate, exchange rate, asset price, liquidity, and money supply as well as credit channels along with risk-taking behavior and market expectations. How does the central bank policy mix transmission mechanism support financial system stability? In this context, two policy dimensions are discernible in terms of sources of financial system instability, namely short-term cyclical factors and long-term structural factors. The former policy dimension can manifest as monetary policy, capital flow management and/or macroprudential policy, congruent with the current financial cycle, and can target specific sources of risk to financial system stability. Meanwhile, the latter policy dimension can be in the form of microprudential policy and/or macroprudential policy, directed toward strengthening financial system resilience to possible shocks through regulations pertaining to liquidity, credit risk exposure, market risk exposure, including currency risk, as well as capital requirements. Furthermore, stronger structural policy reduces the expense of applying cyclical policy.

Policy to control cyclical risk from financial system stability is formulated in stages, as illustrated in Fig. 15.2 (IMF, 2015). The first phase determines whether FSS risk is excessive. Several financial variables are available, including rapidly rising asset prices during an economic boom. Other variables, such as credit growth and high leverage or corporate debt can also be regarded as potential sources of economic disruptions. In general, rapidly rising asset prices also drive excessive credit growth, thus precipitating asset price corrections and default risk. How far FSS risk is considered excessive will depend on the respective conditions in each jurisdiction, as well as the resilience of the corresponding financial system. Generally, such conditions occur when several of the financial variables simultaneously indicate excessive or rapid escalation.

The second stage is to determine the appropriate policy response to mitigate excessive FSS risk. Typically, the first option focuses on macroprudential policy, in particular because such policy can be effectively directed toward certain sources of financial system fragilities (Blanchard, Dell’Ariccia, & Mauro, 2010;

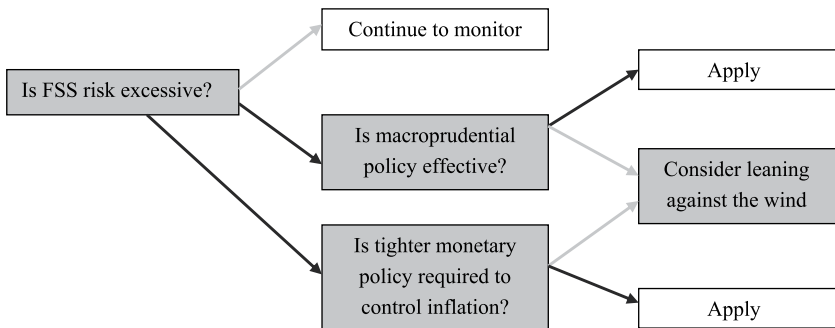


Fig. 15.2: Macroprudential and Monetary Policy: Maintaining FSS in the Cyclical Dimension. *Source:* Adapted from IMF (2015).

IMF, 2013b). For example, FSS risk often surfaces due to rising house prices and excessive credit growth in the property sector, therefore, LTV and DTI can effectively target this sector specifically. Other policies can also be considered to reinforce macroprudential policy, such as tighter fiscal policy. Tighter monetary policy to achieve price stability would also help maintain financial system stability, for instance during an economic boom, when inflation generally accelerates and/or the current account deficit widens, which is often accompanied by high credit growth and external debt accumulation. When FSS risk builds up but inflation remains under control, monetary policy can be considered to lean against the wind in terms of maintaining financial system stability and macroeconomic stability if indications of economic overheating are present.

Transmission between monetary policy and financial system stability is facilitated through two links, namely between the interest rate and financial variables, and between the financial variables and the probability of a shock to macroeconomic conditions (IMF, 2003). As presented in Table 15.1, a change to the monetary policy stance would influence financial system stability through the balance sheet channel due to tighter loan constraints and the probability of losses that could arise. For example, loose monetary policy removes guarantee constraints because asset prices increase and borrower credit worthiness improves, while the cost of external finance decreases and, therefore, credit conditions ease in general (Gertler & Gilchrist, 1994; Jiménez, Ongena, Peydró, & Saurina, 2009). Conversely, raising interest rates could threaten loan instalments and, therefore, exacerbate default risk and financial instability.

A change of monetary policy stance could also affect risk-taking behavior at financial institutions (Jiménez, 2009; Merrouche & Nier, 2010). With asymmetric information, low interest rates provide an incentive for the banks to more aggressively extend loans by lowering lending standards. Furthermore, low interest rates could also shift risk to other economic players more willing to take risks in pursuit of higher gains (Landier, Sraer, & Thesmar, 2011). The adverse influence

Table 15.1: Effect of Monetary Policy of FSS.

Sources of Financial Instability	Channel	Predicted Impacts (Improves Stability)	
		↓ r	↑ r
Borrowing Constraint	Balance sheet (default) channel	↑	↓
Risky Behavior of Financial Institutions	Risk taking channel	↓	↑
	Risk shifting channel	↑	↓
Externalities through Aggregate Prices	Asset price channel	↓	↑
	Exchange rate channel	↑	↓

Source: Adapted from IMF (2003).

tends to be stronger if monetary policy is too accommodative during a boom period for a prolonged period, as more and more investors favor speculative and Ponzi actions rather than hedging activity, as foretold by Minsky (1982). On the other hand, aggressively easing monetary policy during a recession to stimulate the real economy and financial system tends to have a stronger impact due to the incentives thus available to financial institutions and economic players to take on more risk.

Monetary policy can exacerbate asset price and exchange rate fluctuations and, therefore, the balance sheets and net worth of financial institutions and corporations. Low interest rates can raise asset prices as collateral, thus increasing leverage and asset price booms, which would compound the financial cycle (IMF, 2009). Oppositely, a tight monetary policy stance could constrain collateral due to a loss of corporate worth and/or net worth, which could prompt fire sales by the borrowers to meet repayments and by financial institutions to meet the requirement for liquidity. This phenomenon often occurs when economic conditions and the financial system are under distress.

In addition, for EMEs and developing countries, higher interest rates can attract a deluge of foreign capital inflows and spur exchange rate appreciation. From there, the liquidity from non-resident capital tends to accelerate credit growth and foreign debt accumulation and, thus, heightens risk to financial system stability from currency exposure, maturity, and default (Hahm, Mishkin, Shin, & Shin, 2012; Merrouche & Nier, 2010). This phenomenon was found in eastern European countries prior to the crisis with European banks exploiting low rates in the United States. After the global financial crisis and before the US Taper Tantrum, the influx of foreign capital to EMEs such as Brazil, Turkey, South Africa, and Indonesia also drove credit growth, external debt accumulation, and rising house prices. Vigilance and an appropriate response are required in terms of monetary policy, macroprudential policy, and foreign capital flow management in order to pre-emptively analyze, project, and control the domestic and external economic risks as well as the risks contained in the financial system to achieve price stability and maintain financial system stability.

15.3. Structural Macroeconomic Modeling: Integrated Inflation Targeting

Several studies have already developed models to support formulation of the central bank's policy mix. The models include macrofinancial linkages and the influence of foreign capital flows in the macroeconomic analysis and projections. Several models were even developed prior to the global financial crisis under the auspices of the FITF, using a structural macroeconomic economic model approach based on the New-Keynesian Philips Curve (NKPC). In several EMEs, model development was initially motivated to accommodate the impact of foreign capital flows and exchange rates when formulating monetary policy to achieve price stability. As explained in Chapter 13, the model accommodated the external sector, including the influence of global risks, and was extended to determine the interest rate by including the real exchange rate in the Taylor rule.

After the global financial crisis of 2008/09, the financial sector was included in several models through measures of the credit gap and credit risk to support the integration of macroprudential policy in the formulation of monetary policy and foreign capital flow management in the FITF.

15.3.1. From Flexible to Integrated Inflation Targeting

Flexible inflation targeting is already practiced by many central banks in EMEs to accommodate the effect of foreign capital flows and exchange rates in monetary policymaking to achieve price stability. The global crisis, however, forced central bankers to discuss how to best include financial system stability with the goal of price stability. Consequently, a consensus shift occurred away from Jackson Hole in 2009 to leaning monetary policy in order to support financial system stability and/or even to place the goal of financial system stability on equal standing to price stability, as presented in [Table 15.2](#) (Smets, 2014).

The first approach was based on a viewpoint that emerged in advanced countries prior to the crisis, as a modification of Jackson Hole Consensus. The approach asserts that monetary policy should remain focused on price stability, while FSS should be achieved using effective and credible macroprudential policy. FSS aspects are considered indirectly in terms of monetary policymaking if the assessment of macrofinancial linkages and systemic risk demonstrates an impact on price stability and real economic activities. This view is based on the conviction that the targets, instruments and transmission mechanisms between monetary policy and macroprudential policy can easily be separated. The considerations are that: interaction between the two policies is limited; monetary policy does not contribute significantly to macrofinancial imbalances that cause crises; and short-term interest rate policy is not an effective instrument to overcome those imbalances. Nonetheless, this approach recognizes the importance of exchanging information and assessments between monetary and macroprudential policymakers, irrespective of whether the two policies fall under different committees at the central bank or different authorities altogether (Adrian, Covitz, & Liang, 2013).

The global financial crisis of 2008/09 changed the prevailing Jackson Hole Consensus view to the second or third approaches, especially in EMEs and several advanced countries. The central bank must take into consideration price stability and financial system stability when formulating monetary and macroprudential policy. In general, various studies have shown that: (1) monetary and macroprudential policy mutually reinforce one another when leaning against the financial cycle, driven by excessively optimistic expectations and control of risk-taking behavior; (2) there are potential coordination problems because of “push-me, pull-you” factors between the two policy instruments if separated; and (3) including macroprudential policy does not change but actually enriches the central bank’s monetary policy framework (Smets, 2014). In their study, Cecchetti and Kohler (2012) showed that interest rate policy and macroprudential policy (such as LTV or bank capital requirements) mutually interact and reinforce one another in terms of their influence on bank credit on the supply and demand

Table 15.2: Integration of Price Stability and FSS: Three Modeling Approaches.

	Modification of Jackson Hole Consensus	Leaning against the Wind Approach	Price Stability and FSS as Targets
Monetary Policy	• Framework largely unchanged; • Limited impact on credit and risk-taking behavior • Interest rate is not an instrument for macrofinancial imbalances	• FSS as the second target; policy horizon extended • Influences credit and risk-taking behavior • Gets in all of the cracks	• Twin targets with the same considerations • Prevents macrofinancial imbalances during economic upswing and deteriorating balance sheet
Macroprudential Policy	Granular and effective	Does not fully control the financial cycle; arbitration	Indistinguishable from monetary policy; coordinated
Interaction	Limited interaction and easily separable targets, instruments and so on	Financial fragilities influence monetary policy transmission and price stability	Price stability and FSS are inextricably linked
Issues	• Coordination? • Lender of last resort?	• Coordination? • Encumber monetary policy?	Problem of time inconsistency?
Model	Svensson (2012, 2013); Collard et al. (2013)	Borio and White (2003); Cecchetti, Genberg, Lipsky, and Wadhwani (2000); Cecchetti, Genberg, and Wadhwani (2002)	Woodford (2011, 2012)

Source: Adapted from Smets (2014).

sides. Consequently, this study emphasizes coordination between the two policies to be more optimal and effective.

Various studies have shown the need for central banks to lean against the wind in terms of asset prices in order to maintain financial system stability and achieve price stability (Borio & Lowe, 2002). Rajan (2005) found that price stability was insufficient to maintain financial system stability and recommended that central banks lean against financial imbalances through tighter monetary policy even when there is no risk to price stability. This view is consistent with the European Central Bank (ECB), as found by Trichet (2004) and Issing (2011). The financial cycle interacts closely with the economic cycle and, therefore, cannot be overcome using macroprudential policy alone. Monetary policy affects risk-taking behavior in the financial sector and, in contrast, financial sector fragilities influence the effectiveness of policy transmission and attainment of price stability. Therefore, financial system stability must be included as the second target of central bank policy. Woodford (2011, 2012) developed a model to analyze the implications of financial imbalances on monetary policy, revealing that an optimal response would take into consideration financial system stability, such as the marginal risk of financial crisis or credit gap, in addition to the inflation gap and output gap. This model recommends extending the policymaking horizon at the central bank to beyond 2 years in order to capture the financial imbalances. Fahr, Motto, Rostagno, Smets, and Tristani (2013) modeled macroeconomic simulations using financial frictions in the euro area and showed how monetary policy that leans against the credit gap could reduce the trade-off between price stability and the output gap, thus improving macroeconomic performance in general.

The third view recognizes that price and financial system stability are interconnected and difficult to separate. Fundamentally, monetary policy strives to stabilize the financial system and deepen the financial markets, hence strengthening the effectiveness of monetary policy transmission. An inefficient financial system clearly influences monetary policy and, therefore, coordination with macroprudential policy is a must. Brunnermeier and Sannikov (2012) developed a model to test the close correlation between monetary policy and financial system stability based on financial frictions in the monetary policy transmission mechanism. The model showed that financial frictions create close linkages between price stability and financial system stability. When an economy moderates or enters a recession, the optimal monetary policy response must identify and overcome balance sheet declines in the financial system that clog up the flow of funds to productive sectors in the economy. Less effective accommodative monetary policy in Europe and Japan, with interest rates approaching zero, is an example of such balance sheet issues. In contrast, during an economic boom, vigilance is crucial to identify macroeconomic and macrofinancial imbalances in order to institute monetary and macroprudential policy that prevents a crisis from unfolding.

All three views clearly stress the importance of linkages between price stability and financial system stability but differ in terms of which monetary policy response would be best to reinforce macroprudential policy and vice versa. It all depends on conditions in each respective jurisdiction. *First*, if the interaction

between price stability and financial system stability increases, monetary and macroprudential policy coordination will be more effective under one roof, namely at a central bank with the dual mandate of price stability and financial system stability. *Second*, if the macroprudential policy instruments are not fully effective in controlling the financial cycle, monetary policy could reinforce such instruments in pursuance of the central bank mandate to support financial system stability. And *third*, if the expense of monetary policy to overcome a crisis becomes too large, there is a strong argument to give the central bank the dual mandate of price stability and financial system stability. In practice, the various monetary policy instruments could bolster financial system stability, for instance through the minimum reserve requirement and intervention on the foreign exchange market.

Based on various studies and the previously mentioned ideas, Agénor and da Silva (2013) recommended that central banks in EMEs apply a new policy framework, known as the Integrated Inflation Targeting Framework (IITF). IITF is defined as

a flexible IT regime in which the central bank's mandate is explicitly extended to include a financial stability objective, the policy interest rate is set to respond directly to a (well-defined) measure of excessively rapid credit expansion and monetary and macroprudential policies are calibrated jointly to achieve macroeconomic (price) and financial stability.

Table 15.3 presents a comparison between IITF and FITF before and after the global financial crisis of 2008/09. Calibration must be based on macroeconomic models that calculate macrofinancial linkages with the monetary and macroprudential policy response, which can strengthen the transmission mechanism to the financial system and economy.

IITF application requires the central bank to understand the macroeconomic and macrofinancial interactions in the financial system and economy. Modeling is required, including analysis of the transmission mechanism. In general, commercial banks tend to dominate the financial systems in EMEs, therefore, analysis of credit and risk-taking behavior is also crucial, including the impact on the financial system and economy as well as the resilience to domestic shocks and global spillovers. The monetary and macroprudential policy response will depend on the macroeconomic and macrofinancial analyses as well as the transmission mechanisms. Departing from standard ITF and FITF, the policy response under the new framework is not merely determined by the inflation gap, output gap and exchange rate volatility, but also by the credit gap as a proxy of macrofinancial imbalances. Measurement of the credit gap is based on a comparison between actual credit growth and a reference of optimal credit growth to achieve sustainable economic growth. It is important for the central bank to analyze, predict and detect excessive credit expansion, or credit booms, as a key indicator of macrofinancial imbalances, which generally trigger financial system instability and can culminate in crisis. Furthermore, as discussed in Chapter 13, central banks in

Table 15.3: Central Bank Framework: Integrated Inflation Targeting.

Macroeconomic Target	Macroeconomic Indicator	FITF (Pre-crisis)	FITF (Post-crisis)	IITF (Post-crisis)
Price stability	Inflation gap (I) Output gap (O)	Taylor-rule with I and O	Taylor-rule with I, O and FX factors	Central bank rule with I, O, C, and FX factors
Financial stability	Credit gap (C)	Microprudential instruments (MiP)	Micro-Macroprudential Instruments (MiP and MaP)	Monetary and macroprudential instruments calibrated together (Mon+MaP) and Coordinated Micro-Macroprudential instruments (MiP+MaP)
External stability	Exchange rate (FX) volatility	Free-floating exchange rate regime	Floating exchange rate regime with FX intervention	Floating Exchange Rate regime with FX intervention and foreign capital flow management

Source: Adapted from Agénor and da Silva (2013).

EMEs must also consider exchange rate volatility through inclusion of the real exchange rate gap when determining the appropriate policy response.

Central banks in EMEs must also take into consideration aspects of credibility and expectations that could be influenced through application of the new policy framework. Specifically, the considerations underlying the policy response formulation process must also be communicated clearly to the public, including stressing the importance of controlling excessive bank credit expansion to maintain financial system stability and sustainable economic growth, without neglecting price stability, which has remained an integral part of existing monetary policy frameworks. The issue of credibility is important because of possible inconsistencies between monetary policy and the dual mandate of price and financial system stability. Smets (2014), for instance, stated two challenges that must be confronted by the central bank. *First*, central bank policy cannot fully prevent a financial crisis, which could undermine the central bank's reputation and, therefore, affect monetary policy independence and credibility. *Second*, when both objectives are merited with the same degree of importance, inconsistencies could appear where macroprudential policy does not lean adequately against credit booms and housing bubbles because of political pressures and the reliance on monetary policy to overcome some of the macrofinancial imbalances.

15.3.2. Model Structure and Analysis Framework

Several studies have addressed how the central bank formulates its policy mix using structural macroeconomic modeling and expanding the Taylor-rule policy response. Accordingly, there are three dimensions that must be considered when modeling. *First*, the central bank policy response in terms of exchange rate stability to achieve price stability. This was discussed in Chapter 13, which recommended that the central bank utilize two instruments, namely the interest rate and foreign exchange market intervention, to achieve two objectives, namely the inflation target and exchange rate stability (Aizenman, Hutchison, & Noy, 2011; Blanchard, Adler, & Filho, 2015; Edwards, 2006; Mohanty & Klau, 2004; Ostry, Ghosh, & Chamon, 2012a). *Second*, the central bank policy response in terms of asset prices. To that end, a monetary policy response taken in accordance with the Taylor rule, extended to address asset price stability, such as stock prices and property prices, would increase macroeconomic stability (Cecchetti et al., 2000, 2002). *Third*, the central bank policy response in terms of financial system stability. In this regard, several studies recommend that the central bank should consider credit growth or financial system instability risk in the monetary policy response (Curdia & Woodford, 2009; Woodford, 2012).

The central bank's policy mix response has also been modeled according to a structural macroeconomic analysis framework, using the NKPC approach (Cecchetti et al., 2000; Woodford, 2012) evolved from the forward-looking structural model developed by Batini and Nelson (2000). This approach is also practiced at various central banks, including México (Penaloza, 2011) and Indonesia (Harmanta, Tarsidin, Bathaluddin, & Idham, 2012). The structure and analysis framework of the structural macroeconomic model are described as follows.

The IS equation was expanded to capture the impact of the exchange rate, asset prices, and credit gap as follows:

$$y_t = \beta_1 y_{t-1} + \beta_2 y_{t+1} - \beta_3 r_t + \beta_4 y_t^* + \beta_5 z_t + \beta_6 cr_t + \varepsilon_t^y \quad (1)$$

where y = output gap, r = real interest rate gap, y^* = foreign demand gap, z = real exchange rate gap, and c = credit gap. Gap measurement is based on the potential level, long-term value or fundamental value of each respective variable.³ Meanwhile, the Philips-Curve equation is expanded to include the real exchange rate as follows:

$$\pi_t = \lambda_1 \pi_{t-1} + (1 - \lambda_1) E_t \pi_{t+1} + \lambda_3 y_t + \lambda_4 z_t + \varepsilon_t^\pi \quad (2)$$

where π = inflation gap, measured from the target set, which is influenced by backward-looking and forward-looking expectations, output gap, and real exchange rate gap. In this context, the real exchange rate gap influences nominal exchange rate fluctuations as follows:

$$E_t \Delta S_t = \sigma \Delta S_{t+1} + (1 - \sigma) z_t \quad (3)$$

where ΔS = change in nominal exchange rate.

Macro-financial imbalances are modeled in the structural macroeconomic model framework through inclusion of the credit gap, in addition to the output gap and real exchange rate gap, which is also influenced by interest rate spread as well as two policy variables, namely LTV ratio and reserve requirement (RR) as follows:

$$cr_t = \delta_1 cr_{t-1} + (1 - \delta_1)(\delta_2 r_t - \delta_3 spread_t + \delta_4 y_t + \delta_5 LTV_t - \delta_6 RR_t) + \varepsilon_t^{cr} \quad (4)$$

In this case, interest rate spread is influenced by loss loans or default risk in the following equation:

$$spread_t = \nu_1 spread_{t-1} + (1 - \nu_1) v_2 def_t + \varepsilon_t^{spread} \quad (5)$$

where default credit risk is determined by:

$$def_t = \theta_1 def_{t-1} + (1 - \theta_1)(\theta_2 spread_t + \theta_3 cr_{t-1} - \theta_4 y_t) + \varepsilon_t^{def} \quad (6)$$

Fundamentally, this equation shows that default credit risk will escalate as spread and the credit gap increase, and the output gap declines.

³Output gap is measured according to potential output value, the real interest rate gap according to the long-term value or real rate of interest and the real exchange rate gap according to the fundamental value, and the credit gap is measured according to the optimal level of credit growth.

The central bank formulates its policy mix through monetary policy with interest rate instruments using the Taylor rule equation, expanded with the real exchange rate gap, as well as macroprudential policy instruments, including LTV and RR. The Taylor rule equation to determine the (nominal) monetary policy interest rate, expanded with the real exchange rate gap can be expressed as follows:

$$i_t = \rho_1 i_{t-1} + (1 - \rho_1)(\rho_2 \pi_t + \rho_3 y_t + \rho_4 z_t) + \varepsilon_t^i \quad (7)$$

While LTV and RR are set to control the credit gap as follows:

$$LTV_t = \mu_1 LTV_{t-1} + (1 - \mu_1)(-\mu_2 cr_t) + \varepsilon_t^{LTV} \quad (8)$$

$$RR_t = \kappa_1 RR_{t-1} + (1 - \kappa_1)(\kappa_2 cr_t) + \varepsilon_t^{GWM} \quad (9)$$

It is important to note that macroprudential policy formulation per equations (15.8) and (15.9) must be supported by bank credit procyclicality analysis in order to determine the timing and magnitude required for pre-emptive and forward-looking control of excessive credit growth.

15.3.3. Central Bank Policy Mix Formulation

The aforementioned structural macroeconomic model can be used as a macroeconomic and macrofinancial projection and analysis framework as a solid basis to formulate the optimal mix of monetary policy, exchange rate stabilization and macroprudential policy by the central bank. In general, this approach is applied by central banks to bridge more complex DSGE modeling approaches. Several salient aspects must be stressed when formulating the policy mix as follows. *First*, to ensure a pre-emptive and forward-looking policy mix, the macroeconomic and macrofinancial projection horizon must be extended from the typical 2–3 years or more in order to: (1) detect indications of macrofinancial imbalances in the financial cycle that generally have a longer lag than the macroeconomic cycle; and (2) provide a sufficient lead time for interest rate and macroprudential policies to unwind the macroeconomic and macrofinancial imbalances. In general, the lag for inflation and economic growth to react to interest rate policy is usually six to eight quarters.

Second, in-depth research and modeling are required to broaden understanding of macrofinancial linkages in order to strengthen policy mix formulation at the central bank based on the structural macroeconomic modeling framework mentioned previously. As elucidated in Chapter 14, financial procyclicality often precipitates macroeconomic and financial system instability risk that generally originates from: bank credit expansion, asset and housing bubbles, external debt accumulation, and sudden capital reversal. In the structural macroeconomic model above, the influence of macrofinancial linkages from bank credit expansion and foreign capital flows are shown by adding the variables: credit gap and

real exchange rate gap. Risks in the form of asset and housing bubbles as well as external debt accumulation can be detected through both variables. Nevertheless, a separate model is required to investigate the behavior of the financial cycle in terms of the four macrofinancial linkages mentioned in order to detect indications of risk in the form of macroeconomic, macroexternal and macrofinancial imbalances. This will facilitate the pre-emptive and forward-looking policy mix required.

Moreover, research such as this is also crucial to determine the appropriate timing and magnitude required of the policy mix. Fig. 15.3 shows how the monetary and macroprudential policy cycle must be directed to control the financial cycle in line with the economic cycle, thus avoiding macroeconomic and macrofinancial imbalances. To that end, the policy mix must be tightened at the appropriate time, when there are indications and projections that the financial cycle could spur risks in the form of macroeconomic and macrofinancial imbalances during an economic upswing or boom period. The opposite is also true, namely that the policy mix must be loosened to avoid the adverse effects of financial procyclicality during an economic downswing or bust period, or when the bubbles burst. To that end, further research into the behavior of the financial and economic cycles, as described in Chapter 14, is critical. Such research, into the four aspects of financial procyclicality that oftentimes trigger macroeconomic and macrofinancial imbalances, should be conducted as an aggregate and by sector. Furthermore, research into corporate and household behavior, with a more micro approach, would also benefit the understanding of macroeconomic and macrofinancial imbalances.

Third, consistency and coordination when formulating the central bank policy mix are crucial to ensure that monetary policy credibility, which has thus far remained strong, can be maintained (Angelini et al., 2011; Cecchetti & Kohler, 2012; Curdia & Woodford, 2010; Smets, 2014). Similar to the findings above, Smets (2014) emphasized the importance of monetary and macroprudential policy

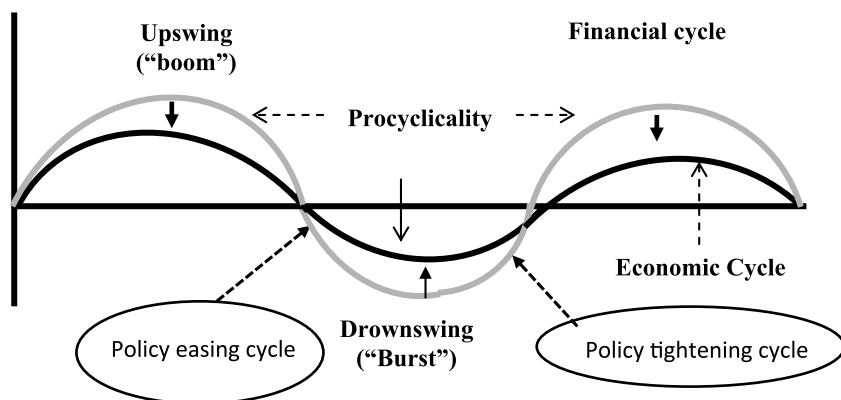


Fig. 15.3: Monetary and Macroprudential Policy Mix Cycle Maintains Macroeconomic and Financial System Stability in a Cyclical Dimension.

consistency, especially when divergence between the price stability and financial system stability targets arises, in order to maintain central bank credibility. In this regard, several models have been developed that include macrofinancial linkages in the formulation of the monetary and macroprudential policy mix. Angelini et al. (2011) modeled the FSS target in the form of a policy mix response to stabilize the credit-to-GDP ratio, like the credit gap in the structural macroeconomic model above. Curdia and Woodford (2010) favored credit spread when modeling the impact of financial frictions on public prosperity. Credit spread shows the financial frictions in bank intermediation due to the costs incurred by the banks when selecting credit demand and credit default because a portion of borrowers do not repay their loans.

Meanwhile, Cecchetti and Kohler (2012) analyzed under which conditions monetary policy, through the interest rate, and macroprudential policy, through the CCB, would be mutually complementary or substitutes to achieve price stability and support maintained financial system stability. In general, their research demonstrated that tight coordination between monetary policy and macroprudential policy is crucial to optimally achieve both targets without reducing monetary policy credibility to attain price stability. In more detail, the analysis framework of Cecchetti and Kohler (2012) was based on a structural macroeconomic model with the inclusion of macroeconomic and macrofinancial linkages and, thus, could better capture the transmission mechanism of the monetary and macroprudential policy mix through the interest rate, exchange rate, asset price, bank credit, corporate balance sheet, and household worth channels.

In this case, aggregate demand was expressed in the following equation:

$$y^d = -\alpha(\rho - \pi^e) - \beta(i - \pi^e) - \delta\pi + \eta \quad (10)$$

where y^d = aggregate demand, i = short-term nominal interest rate, ρ = nominal lending rate, π^e = inflation expectations, π = inflation, and η = random demand shocks. The short-term nominal interest rate (1) is determined by the central bank, while the lending rate (ρ) is determined by credit market equilibrium.

For simplification purposes, the magnitude of credit is assumed to be determined by the amount of bank capital, which can be expressed as follows:

$$L^S = -\kappa.k + \tau.B \quad (11)$$

where L^S = bank credit supply, k = capital requirements, and B = bank capital. Moreover, the real value of bank capital is assumed to rise as the real value of output increases as follows:

$$B = by \quad (12)$$

Meanwhile, credit demand depends on the real lending rate and real output as flows:

$$L^D = -\phi(\rho - \pi^e) + \omega y \quad (13)$$

On the other hand, aggregate supply shows output determined by the difference between inflation and inflation expectations pursuant to the Philips Curve as follows:

$$y^s = \gamma(\pi - \pi^e) + \varepsilon \quad (14)$$

where y^s = aggregate output supply and ε = random supply shock that does not correlate with the random demand shock in equation (15.10) above. Ultimately, the model shows a balance between the real sector and financial sector as follows:

$$y^S = y^d = y \quad \text{and} \quad L^S = L^D = L \quad (15)$$

When analyzing the monetary and macroprudential policy mix, Cecchetti and Kohler (2012) also considered interest rate spread in addition to achieving the inflation target and considered economic growth in the objectives of the central bank as follows:

$$\min_{(i,k)} W_{baucan} = \pi^2 + \lambda y^2 + \zeta(\rho - i)^2 \quad (16)$$

where $(\rho - i)$ = spread between lending rate and the monetary policy rate, and ζ = FSS weight in the objective function of the central bank. In this regard, the central bank minimizes losses to public welfare by selecting an optimal policy mix between interest rate instruments on the monetary policy side and capital requirements on the macroprudential policy side.

Through follow-up analysis, Cecchetti and Kohler (2012) compared the optimal solutions that had the least adverse impact on public welfare if monetary policy and macroprudential policy were independent, coordinated or partially coordinated. In general, the results showed that tight coordination between monetary and macroprudential policy provides the optimal solution. In other words, achievement of price stability and financial system stability requires close coordination between both policies to provide the optimal solution toward public betterment.

In more detail, when no coordination exists between monetary and macroprudential policy, determining the policy rate to minimize the adverse impact on the public (15.16) can be expressed as follows:

$$i_{MPindependen}^* = \left[-\frac{\alpha(\omega - b\tau) + \phi(1 - \gamma\delta\lambda)}{\beta\phi(1 + \gamma^2\lambda)} \right] \cdot \varepsilon + \frac{1}{\beta} \cdot \eta \quad (17)$$

Meanwhile, the capital requirements determined in the optimal solution are as follows:

$$k_{FSindependen}^* = \frac{(\delta\varepsilon + \gamma\eta)(\omega - b\tau)}{(\gamma + \delta)\kappa} \quad (18)$$

In this regard, because monetary policy does not take into consideration the impact of changes to capital requirements, the impact on public welfare is more

significant. Increasing demand, for instance, tends to exaggerate the interest rate response because tightening the capital requirements by raising the CCB rate is not considered. The opposite is also true, a negative supply-side shock also tends to amplify the reduction to the policy rate because loosening the capital requirements is not considered. In fact, excessive loosening of both instruments would be a risk to maintained financial system stability. Similarly, the target of maintaining financial system stability would also be undermined if both policy instruments were independent or not coordinated.

The analysis above demonstrates the importance of close coordination between monetary and macroprudential policy to improve public welfare. If both policies are coordinated, the optimal solution when determining the monetary policy rate can be expressed in the following equation:

$$i_{MPkoordinasi}^* = \left[-\frac{(1-\gamma\delta\lambda)}{(\alpha+\beta)(1+\gamma^2\lambda)} \right] \cdot \varepsilon + \frac{1}{(\alpha+\beta)} \cdot \eta \quad (19)$$

While the optimal solution to determine the capital requirements is as follows:

$$k_{FSkoordinasi}^* = \left[-\frac{(\alpha+\beta)(\omega-b\tau)+\varphi(1-\gamma\delta\lambda)}{(\alpha+\beta)(1+\gamma^2\lambda)\kappa} \right] \cdot \varepsilon + \frac{\varphi}{(\alpha+\beta)\kappa} \cdot \eta \quad (20)$$

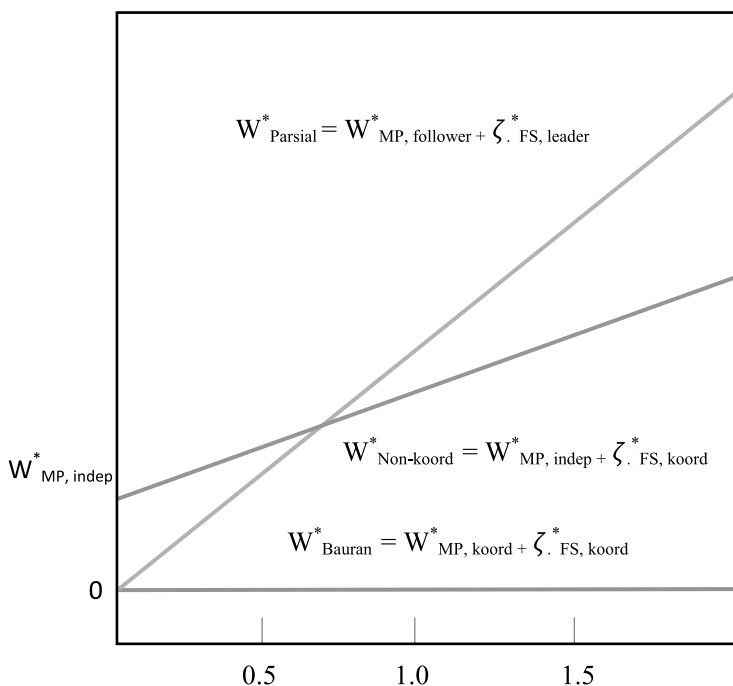
The optimal solution precipitates the same social losses as the optimal monetary policy solution under the ITF as discussed in Chapters 6–9, namely:

$$W_{Bauran}^* = \left[\frac{\lambda\varepsilon^2}{(1+\gamma^2\lambda)} \right] \quad \text{for } i = i_{MPkoordinasi}^* \quad \text{and } k = k_{FSkoordinasi}^* \quad (21)$$

From the optimal solution, macroeconomic imbalances (inflation gap and output gap) and macrofinancial imbalances (interest rate spread) can be overcome through close coordination between determining the policy rate and determining the capital requirements. The interest rate will not be too tight (loose) if macroeconomic imbalances occur during an economic boom (bust) period. Likewise, the capital requirements will not be too tight (loose) if macrofinancial imbalances occur. In other words, coordination between the two policies would produce an optimal solution to achieve price stability and support financial system stability.

A similar analysis can be conducted for partially coordinated monetary and macroprudential policy. In this regard, as with game theory, the monetary policy response can be determined after the macroprudential policy response to overcome macrofinancial imbalances. Oppositely, the macroprudential policy response can be determined after the monetary policy response to overcome macroeconomic imbalances. The two alternative policy responses do not produce the optimal solution, however, because interaction between the macroeconomic and macrofinancial imbalances is not overcome through an optimal policy mix.

Visually, Figs. 15.4 and 15.5 present a comparison of the influence of monetary policy and macroprudential policy formulation, which is independent, coordinated, or partially coordinated, on public losses.



Public Losses (W); Financial Stability Weight (ζ)

Fig. 15.4: Coordination and Losses after a Demand Shock.

The two graphs show that the optimal solution of the monetary and macroprudential policy mix produce the lowest public loss compared to policies that are independent or partially coordinated. This is applicable to a demand-side shock, as shown in Fig. 15.4, as well as a supply-side shock as shown in Fig. 15.5. In fact, public losses from that policy mix are lower than if monetary policy is formulated independently. The two graphs also show that the solution from partially coordinated policies is lower than policies that are independent to a certain point but thereafter the impact tends to amplify. Such conditions are inextricably linked to the behavior of the financial cycle, which tends to exacerbate the economic cycle during an economic bust period, and vice versa.

15.4. DSGE Modeling with Macrofinancial Linkages

The main criticism of the NKPC approach when modeling FITF is the assumed homogeneity of financial institutions and economic players, thereby leading to aggregate analyses. The global crisis demonstrated the importance of heterogeneity among financial institutions, especially banks, similar to liquidity, capital, and risk-taking behavior, as well as among economic players, similar to project feasibility, demand for external financing and risk-taking behavior. Heterogeneity

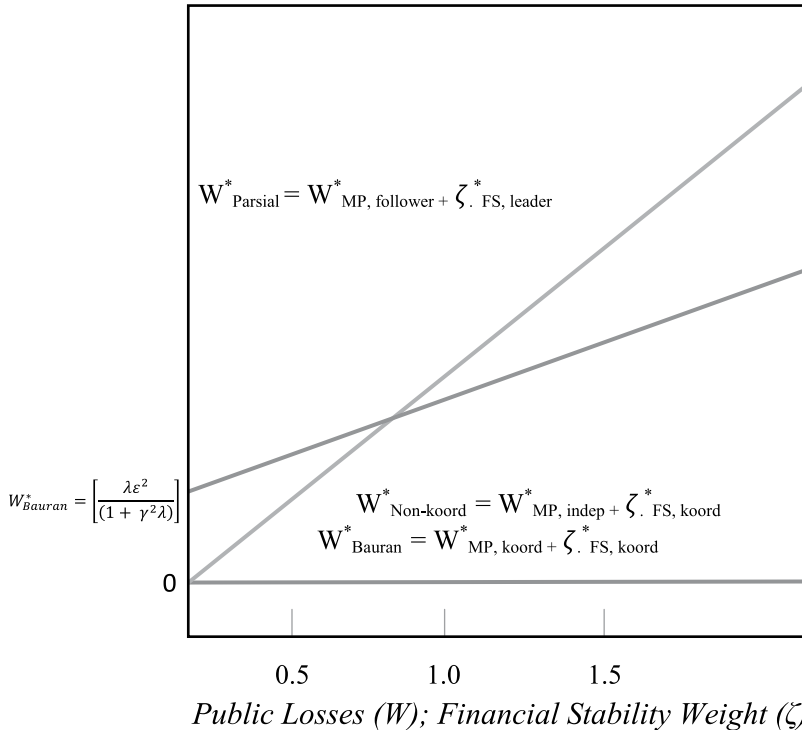


Fig. 15.5: Coordination and Losses after a Supply Shock.

among banks and economic players creates financial procyclicality and systemic risk when analyzing financial system stability. Similarly, a number of studies have developed macroeconomic models with macrofinancial and external sector linkages using the DSGE approach.

Several central banks have also developed the DSGE approach to support formulation of the mix between monetary policy, macroprudential policy, and foreign capital flow management (Brázdik, Hlaváček, & Maršál, 2011; Roger & Vleck, 2012). In advanced countries, the DSGE approach is used for policy formulation at the ECB using the NAWM model (Christoffel, Coenen, & Warne, 2008) and CMR model (Christiano, Motto, & Rostagno, 2010); at the US Federal Reserve using the EDO model (Chung, Kiley, & Laforge, 2010); at the Reserve Bank of New Zealand using the KITT model (Lees, 2009); at the Bank of Canada using the ToTEM model (Murchison & Rennison, 2006); at the Bank of England using the BEQM model (Harrison et al., 2005); at the Bank of Japan (BOJ) using the JEM model (Fujiwara, Teranishi, & Hara, 2004); at the Sveriges Riksbank using the SIGMA model (Erceg, Guerrieri, & Gust, 2006) and RAMSES model (Adolfson, Laseen, Linde, & Villani, 2007); and at the Banca d'Italia (Angelini et al., 2011; Gerali, Neri, Sessa, & Signoretti, 2010). Meanwhile, in EMEs, DSGE development was found in Brazil (de Carvalho & de Castro, 2015; Ferreira &

Nakane, 2015); in India (Anand, Peiris, & Saxegaard, 2010; Banerjee & Basu, 2015); and Bank Indonesia (Harmanta, Purwanto, Rachmanto, & Oktiyanto, 2013).

15.4.1. Macrofinancial Modeling Approaches in DSGE

There are three basic approaches commonly applied in studies that include financial frictions in DSGE models, namely financial accelerator, collateral constraints, and through explicit modeling of financial intermediation, as presented in Table 15.4 (Roger & Vleck, 2012). Two changes to New Keynesian economics are required to include financial frictions in DSGE models. The first is to change economic players from homogenous (representative) to heterogeneous. This provides a basis for the analysis of borrowing between economic players due to differences in terms of consumption, productivity, and financial constraints. Second is to introduce asymmetric information between economic players that creates financial frictions due to borrower selection and the costs incurred to assess business feasibility and loan monitoring, which manifests as the external financing premium.

The financial accelerator approach departs from the assumption of asymmetric information regarding project feasibility, thus creating the external financing premium above the cost of internal capital incurred by a firm seeking a loan, congruent with the mindset of Bernanke, Gertler, and Gilchrist (1999). This is modeled using DSGE with heterogeneous economic players according to the value to net worth and, therefore, the external financing premium will correlate inversely with the net worth of the borrower and limit the size of the loan. During an economic upswing, a borrower has a higher net worth, thus improving business feasibility and reducing the cost of external funding. Conversely, during an economic downswing, the loss of net worth reduces business feasibility and raises the cost of borrowing. Therefore, the external financing premium causes countercyclical behavior, creates an amplification mechanism and accelerates the investment and real output response to the shocks that emerge. For example, higher output due to technological advancement is amplified by rising asset prices and corporate net worth. Thereafter, such improvements reduce the external financing premium and amplify the output and asset price gains.

The collateral constraints approach includes financial frictions in the DSGE model through net worth, which becomes collateral when receiving a new loan per the model developed by Kiyotaki and Moore (1997). Departing from the financial accelerator approach, net worth of the borrower has a direct influence on the size of the loan received and an indirect impact through the external financing premium. This approach assumes that the legal enforcement of loan agreements in the event of default is somewhat weak, thus necessitating the banks to request physical collateral on the disbursed loans, including land, property, or capital. In the DSGE model, economic borrowers (impatient agents) have a lower discount factor than other economic agents, and prefer to borrow at current interest rates to finance consumption using existing assets as collateral. Congruent with the financial accelerator approach above, the investment and output amplification

Table 15.4: Three Approaches to Model Financial Frictions in DSGE.

	Financial Accelerator	Collateral Constraints	Bank Intermediation
Financial Intermediation	No explicit financial intermediation.	No explicit financial intermediation.	Explicit financial intermediation through the banking sector.
Sources and characteristics of financial frictions	External financing premium sensitive to net worth of economic players due to <i>ex post</i> asymmetric information.	Loans determined by collateral value of economic players due to weak loan agreements.	Cost of financial intermediation increases with volume of financial services because of loan selection and/or credit default.
Motivation to borrow and lend	Heterogeneous economic players: households as savers and corporations as borrowers.	Heterogeneous economic players in the form of different discount factors because of consumption preferences.	Advance production costs required or assume a fixed portion of the loan is used for investment.
Key Variables	External financing premium	Credit limits and spreads	Interest rate spreads
Transmission Mechanism	Borrower net worth fluctuations determine the cost of external financing.	Asset price fluctuations influence collateral value and, therefore, credit constraints.	Interest rate spread determines size of loan.
Implications of financial frictions	Amplification of asset price shocks. Moderate amplification of non-financial shocks. Strong amplification is very sensitive to model assumptions.	Amplification of asset price shocks. Moderate amplification of non-financial shocks. Strong amplification is very sensitive to model assumptions.	Shock transmission is weak because of procyclical interest rate spreads, except if combined with rigid interest rates or another financial friction approach.

Source: Adapted from Roger and Vleek (2012).

mechanism of bank credit manifests through fluctuating asset prices as collateral. For example, lower asset prices would translate to a lower value of collateral and, therefore, the bank would reduce the credit ceiling offered to the borrower. Consequently, spending by the borrower would also subsequently decrease and this would exacerbate the decline of asset prices, investment, and output.

Initially, DSGE modeling per the financial accelerator and collateral constraints approaches assumed that the borrower could access direct funding from the lender without an intermediary. Nonetheless, inclusion of the banking sector enriched the DSGE model analysis, despite the inherent complexities and difficulties. Furthermore, financial frictions could be enriched by impact analysis of the intermediation costs, balance sheet conditions, and risk-management behavior. Modeling typically differentiates wholesale banking from retail banking. In general, DSGE modeling focuses on: (1) endogenously determining the interest rate spread per the business cycle in line with the risk-taking behavior of economic players, be they bankers, corporations or households; (2) modeling bank capital channels in the financial and economic cycles, thus facilitating analysis of the transmission mechanism and impact of capital requirements on the banks' balance sheets; and (3) modeling bank risk-management.

Several academic and central bank studies combine two or more of the approaches depending on the objectives of the analysis and desired policymaking requirements. The KITT model at the Reserve Bank of New Zealand, for instance, applies the collateral constraints approach, where households borrow using the house as collateral and there are collateral constraints to using offshore loans (Lees, 2009). At the Bank of England, the DSGE model applies the financial frictions approach with an external financing premium that is enhanced by the banks incurring costs on the capital accumulated (Markovic, 2006). The Banca d'Italia applies the collateral constraints approach, with the cost of capital accumulation leading to interest rate spread and the banks also face capital requirements (Gerali et al., 2010). At the ECB, Angeloni and Faia (2009) modeled the risk of bank runs, the probability of which increased as bank capital decreased, while Christiano et al. (2010) used the financial accelerator approach, augmented by the banking sector, which provided liquidity services. Meanwhile, the Bank of Canada developed a multi-country DSGE model with international financial flows, where each country had a heterogeneous banking sector and credit faced capital requirements (Beaton, Lalonde, & Snudden, 2010).

15.4.2. Monetary Policy and Macprudential Policy in DSGE Modeling

Monetary and macroprudential policy analysis using DSGE modeling is not a simple process. A balance is required between the structural complexity of the DSGE models themselves and the importance of analyzing both policies. In this regard, modeling in macroprudential policy studies using DSGE models has been developed by Gerali et al. (2010). In that model, the economy was characterized by a number of enterprises, heterogeneous households, and monopolistic banks. Patient households (savers) placed their savings at the banks, while impatient

households (borrowers) and enterprises borrowed from the banks despite collateral constraints. Firms produced consumer goods and investment using labor from households and capital. Bank assets included corporate and household loans, while the liabilities were savings deposits and capital. Banks strived to maintain capital in pursuance of the specific capital requirements, v_p , determined exogenously by the authorities, such as the Basel requirements. Box 15.1 explores the DSGE model developed by Gerali et al. (2010) in more detail.

Box 15.1: DSGE Models with Financial Frictions

DSGE model with financial frictions was developed by Gerali et al. (2010). They introduced two types of financial instrument in the banking system, namely savings (term deposits) and loans (credit). When seeking a loan from a bank, economic players face credit constraints in relation to the value of collateral held, which could be in the form of a house for individuals/households or physical capital (net value of fixed assets) for entrepreneurs/businesses. The heterogeneity of the economic players determines the sustainability of intermediation in the economy. The banking sector operates under monopolistic competition, meaning that the banking industry sets deposit and lending rates to maximise profits. In summary, this model is able to demonstrate that macro-financial linkages, together with the boom and bust economic and financial cycles, are exacerbated by collateral constraints when extending credit as well as differentiation in deposit and lending rates due to monopolistic competition in the banking industry.

A. Households and Entrepreneurs/Businesses

In the model setting of Gerali et al. (2010), there are two types of household, namely patient households as savers (P) and impatient households as borrowers (I), as well as one group of entrepreneurs as businesses (E). Patient households have a higher discount factor concerning their proclivity to consume in a future period (β_p) than impatient households (β_I), which is assumed to be the same as entrepreneurs (β_E).

Patient household, i , maximises the intertemporal utility expectation function as in Equation (15.1.1) depending on the current level of individual consumption, c^p , and aggregate consumption in the previous period scaled by group-specific consumption habits, α^p , household services, h^p , and working hours, l^p . Preferences are influenced by two shocks, consumption, ε^c , and household demand, ε^h . The real budget constraints in Equation (15.1.2) encompasses current consumption, accumulated housing services at real prices, q^h , and the current position of term deposits, d_t . Meanwhile, income is derived from salaries and wages, $w^p l^p$, interest receipts in the previous period, $(1 - d_{t-1})d_{t-1}/\pi_t$, and a lump-sum transfer in the form of

dividend payments from a business or the bank (with the patient households as the owners) after deducting labour costs and insurance.

$$\max E_0 \sum_{t=0}^{\infty} \beta_t^P \left[(1 - \alpha^P) \varepsilon_t^c \log(c_{t,i}^P - \alpha^P c_{t-1}^P) + \varepsilon_t^h \log h_{t,i}^P - \frac{l_{t,i}^{P(1+\phi)}}{1+\phi} \right] \quad 15.1.1$$

$$\text{s.t. } c_{t,i}^P + q_t^h \Delta h_{t,i}^P + d_{t,i} = w_t^P l_{t,i}^P + (1 + r_{t-1}^d) d_{t-1,i} / \pi_t + s_{t,i}^P \quad 15.1.2$$

Impatient household maximizes the intertemporal utility expectation function as in Equation (15.1.3) depending on consumption, c^I , household services, h^I , and working hours, l^I . The parameter, α^I , measures the effect of habits specific to this group, while ε^c and ε^h are consumption and housing shocks. The real budget constraints in Eq. 15.1.4 consists of current consumption, accumulated housing services as well as interest and principal payments on loans in the previous period, b_{t-1}^{bH} , with lending rates for households, r^{bH} , determined by the bank. In addition to salaries and lump-sum transfers, income is also derived from loan withdrawals, b^I .

$$\max E_0 \sum_{t=0}^{\infty} \beta_t^I \left[(1 - \alpha^I) \varepsilon_t^c \log(c_{t,i}^I - \alpha^I c_{t-1}^I) + \varepsilon_t^h \log h_{t,i}^I - \frac{l_{t,i}^{I(1+\phi)}}{1+\phi} \right] \quad 15.1.3$$

$$\text{s.t. } c_{t,i}^I + q_t^h \Delta h_{t,i}^I + (1 + r_{t-1}^{bH}) b_{t-1,i}^I / \pi_t = w_t^I l_{t,i}^I + b_{t,i}^I + s_{t,i}^I \quad 15.1.4$$

$$(1 + r_t^{bH}) b_{t,i}^I \leq m_t^I E_t [q_{t+1}^h h_{t+1,i}^I \pi_{t+1}] \quad 15.1.5$$

Furthermore, impatient households face credit constraints as in Equation (15.1.5), which stated that the expected value of the property used as collateral in period t must be sufficient as a guarantee for the principal and interest payments, using m^I as the LTV ratio for housing loans. From a macro perspective, the magnitude of the LTV ratio determines the value of the bank loan extended to a household based on the value of the house used as collateral. From a micro perspective $(1 - m^I)$ could be interpreted as the proportional cost of the guarantee held by the bank in the case of loan delinquency or default. The central bank is assumed to set the LTV ratio in order to influence the availability of credit in the economy and is, therefore, exogenous to the household.

Entrepreneur maximizes the intertemporal utility expectation function as in Equation (15.1.6). The real budget constraint equates the total spending and total income, with consumption, c^E , physical capital, k^E , bank loans, b^E , capacity utilisation, u_p , labor input, l^E , depreciation, δ , the price

of each physical capital unit measured from consumption, q^k , real utility cost, $\psi(u_p)k_{t-1}$, and production function, y^E . Moreover, similar to impatient households, the size of the loan a bank can guarantee to an entrepreneur is limited by the value of the collateral in the form of physical capital as in Equation (15.1.8), where m^E = the LTV ratio for loans extended to the corporate sector.

$$\max E_0 \sum_{t=0}^{\infty} \beta_E^t \log(c_{t,i}^E - \alpha^E c_{t-1}^E) \quad 15.1.6$$

$$\begin{aligned} \text{s.t } c_{t,i}^E + w_t^P l_{t,i}^{E,P} + (1+r_t^{bE})b_{t-1,i}^E / \pi_t + q_t^k k_{t,i}^E + \psi(u_{t,i})k_{t-1,i}^E \\ = \frac{y_{t,i}^E}{x_t} + b_{t,i}^E + q_t^k (1-\delta)k_{t-1,i}^E \end{aligned} \quad 15.1.7$$

$$(1+r_t^{bE})b_{t,i}^E \leq m_t^E E_t [q_{t+1}^k \pi_{t+1} (1-\delta)k_{t,i}^E] \quad 15.1.8$$

Demand for deposits and loans were modeled in Gerali et al. (2010) using the Dixit–Stiglitz monopolistic competition framework to describe the market power of the banking industry. Hence, corporate and household deposit and credit contracts were differentiated slightly by each bank branch, j , with a constant elasticity of substitution (CES) of ε^d , ε^{bH} , and ε^{bE} , respectively, for term deposits, household loans and corporate loans. Therefore, not only could each bank offer different deposit and lending rates for each customer but also apply a narrower (wider) spread from the monetary policy rate during an economic upswing (downswing). The credit demand functions for impatient households and entrepreneurs in Eq. 15.1.9 as well as the deposit demand function in Eq. 15.1.10 illustrate that the size of the loan/deposit would depend on the overall volume and interest rates borne/paid by bank, j , relative to the average in the economy.

$$b_{t,j}^I = \left(\frac{r_{t,j}^{bH}}{r_t^{bH}} \right)^{-\varepsilon_t^{bH}} b_t^I \quad b_{t,j}^E = \left(\frac{r_{t,j}^{bE}}{r_t^{bE}} \right)^{-\varepsilon_t^{bE}} b_t^E \quad 15.1.9$$

$$d_{t,j} = \left(\frac{r_{t,j}^d}{r_t^d} \right)^{-\varepsilon_t^d} d_t \quad 15.1.10$$

B. Banking Industry

Bank intermediation is modeled in Gerali et al. (2010) by using deposits as the sole financial asset for saving and credit as the only financial asset for borrowing, to reflect emerging market economies (EMEs) and developing

countries which are still dominated by the banking industry. There are two important characteristics relating to financial frictions in the model that produce financial procyclicality: (1) the monopolistic competition; and (2) capital requirement constraint.

Monopolistic competition allows the banks to differentiate deposit rates for each patient household and lending rates for each impatient household and entrepreneur, in addition to credit volume based on the value of collateral. In addition, industry dynamics also allow the banks to set deposit and lending rates in response to shocks and other cyclical factors in the economy. Consequently, this model facilitates impact analysis of financial frictions on bank behavior in terms of the interest rate channel and credit channel of the monetary policy transmission mechanism.

The balance sheet constraint faced by the banks stated that each bank could fund its loan disbursements by mobilising funds and capital, which can only be accumulated from retained earnings. The capital requirements constraint stated that the minimum capital-to-asset ratio must equal the requirements set by the central bank. Hence, bank capital acts as a channel that influences reciprocity between the financial system and real economy. When macroeconomic conditions improve, bank profit and capital increase along with the banks' ability to disburse loans. In contrast, during an economic recession, bank losses and decreasing capital tend to limit bank lending.

The structure of the banking industry consists of two types of *retail bank* (*loan* and *deposit*) and one type of *wholesale bank*. Loan retail bank disburses loans to impatient households and businesses, while deposit retail bank mobilizes the funds deposited by patient households. Both types of retail bank operate under monopolistic competition and, thus, can differentiate deposit and lending rates with different spreads for each customer. Meanwhile, wholesales banks operate under perfect competition, maintaining the group's capital position through financial intermediation and preventing violations of the capital requirements set by the central bank.

Wholesale bank branch operates under perfect competition by collecting bank capital, K_b , and wholesale term deposit mobilisation, D_r , and disbursing wholesale credit, B_r , from and to retail bank branches. The objective is to choose the amount of loans and deposits to maximise the current value of all future cash flows with balance sheet constraints as in Equation (15.1.11), subject to the balance sheet constraint as in Equation (15.1.12). The bank faces an adjustment cost, A_{Kb} , if the capital-to-assets ratio, K^b/B_r , deviates from the capital requirements set by the central bank, v^b . Bank capital is accumulated from retained earnings in each period as in Equation (15.1.13), where Π^b = overall profit of the three bank branches, $(1 - \omega^b)$ = the bank's dividend policy and δ^b = funds used to manage bank capital. Dividend policy is assumed to be fixed and, therefore, not a choice for the bank. Wholesale lending and deposit rates, R^b and R^d , are determined exogenously.

$$\underset{\{B_t, D_t\}}{\text{Max}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P \left[(1+R_t^b)B_t - B_{t+1} + D_{t+1} - (1+R_t^d)D_t + \Delta K_{t+1}^b - \frac{A_{Kb}}{2} \left(\frac{K_t^b}{B_t} - v^b \right)^2 K_t^b \right] \quad 15.1.11$$

$$\text{s.t. } B_t = D_t + K_t^b \quad 15.1.12$$

$$K_t^b = (1-\delta^b)K_{t-1}^b + \omega^b \Pi_{t-1}^b \quad 15.1.13$$

Solving the problem using the budget constraint and assuming banks have access to funds from the central bank's lending facilities at the monetary policy interest rate so that the deposit rate will be the same as the interbank rate, $R^d=r_t$, yields Equation (15.1.14) which links wholesale lending rates on the interbank money market to the monetary policy rate and capital-to-assets ratio. The equation determines bank loan availability, depending on capital conditions. Furthermore, rearranging the equation, the spread between the lending rate and monetary policy rate in Equation (15.1.15) correlates inversely with the capital-to-assets ratio, implying that if the bank's capital is limited, the spread would be wider and, thus, the ability to lend would become more limited.

$$R_t^b = r_t - A_{Kb} \left(\frac{K_t^b}{B_t} - v^b \right) \left(\frac{K_t^b}{B_t} \right)^2 \quad 15.1.14$$

$$S_t^W \equiv R_t^b - r_t = -A_{Kb} \left(\frac{K_t^b}{B_t} - v^b \right) \left(\frac{K_t^b}{B_t} \right)^2 \quad 15.1.15$$

Retail loan bank branch obtains a wholesale loan, $B_{t,j}$, at a wholesale lending rate, R^b , and subsequently differentiates the loan into credit for impatient households or businesses with different spreads. The objective is to maximize profit as in Eq. 15.1.16 subject to the demand for credit in Eq. 15.1.9, with $B_{t,j} = b_{t,j}^I + b_{t,j}^E$. Retail loan bank incurs a cost when adjusting the lending rate per the parameters, A_{bH} and A_{bE} , which is proportional to the aggregate lending rate.

$$\underset{\{r_{t,j}^{bH}, r_{t,j}^{bE}\}}{\text{Max}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P \left[r_{t,j}^{bH} b_{t,j}^I + r_{t,j}^{bE} b_{t,j}^E - R_t^b B_{t,j} - \frac{A_{bH}}{2} \left(\frac{r_{t,j}^{bH}}{r_{t-1,j}^{bH}} - 1 \right)^2 r_t^{bH} b_t^I - \frac{A_{bE}}{2} \left(\frac{r_{t,j}^{bE}}{r_{t-1,j}^{bE}} - 1 \right)^2 r_t^{bE} b_t^E \right] \quad 15.1.16$$

For each respective lending rate offered to households and businesses (indexed using $s = \{I, E\}$), log-linearizing the first order condition produces Equation (15.1.17). The equation shows that the lending rate set by the retail bank incorporates the current value of the wholesale interest rate. Adjustments to changes in the wholesale interest rate are inversely proportional with the intensity of the adjustment cost (measured using A_{bs}) and positively proportional with the level of competition in the bank credit sector (measured using the inverse of ε^{bs}).

$$\hat{r}_t^{bs} = \frac{A_{bs}}{\varepsilon^{bs} - 1 + (1 + \beta_p) A_{bs}} \hat{r}_{t-1}^{bs} + \frac{\beta_p A_{bs}}{\varepsilon^{bs} - 1 + (1 + \beta_p) A_{bs}} E_t \hat{r}_{t+1}^{bs} + \frac{\varepsilon^{bs} - 1}{\varepsilon^{bs} - 1 + (1 + \beta_p) A_{bs}} \hat{R}_t^b \quad 15.1.17$$

In the case of flexible interest rates, interest rate determination yields Equation (15.1.18), which basically shows that retail loan interest rates are set as a mark-up from the wholesale loan interest rate. Furthermore, analysing the spread between the lending rate and monetary policy rate as in Equation (15.1.19), The retail lending rate spread increases if the policy rate increases and is proportional to the wholesale spread, S^W . Furthermore, a larger market share (namely a lower substitution elasticity, ε^{bs}) would determine a wider absolute spread, reflecting the bank's ability to innovate in pursuit of interest rate margin.

$$r_t^{bs} = \left(\frac{\varepsilon_t^{bs}}{\varepsilon_t^{bs} - 1} \right) R_t^b \quad 15.1.18$$

$$S_t^{bs} \equiv r_t^{bs} - r_t = \frac{\varepsilon_t^{bs}}{\varepsilon_t^{bs} - 1} S_t^W + \frac{1}{\varepsilon_t^{bs} - 1} r_t \quad 15.1.19$$

Retail deposit bank branch mobilises the deposits, $d_{t,j}$, of patient households and subsequently deposits the funds at a wholesale bank branch at an interest rate, r_t . The objective is to maximize profit as in Equation (15.1.20) subject to the demand for deposit in Equation (15.1.10), with $D_{t,j} = d_{t,j}$. Retail deposit bank incurs a cost when adjusting the deposit rate per the parameter, A_d , and is proportional to the aggregate deposit rate.

$$\text{Max}_{\{r_{t,j}^d\}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P \left[r_t D_{t,j} - r_{t,j}^d d_{t,j} - \frac{A_d}{2} \left(\frac{r_{t,j}^d}{r_{t-1,j}^d} - 1 \right)^2 r_t^d d_t \right] \quad 15.1.20$$

Log-linearizing the first-order condition produces Equation 15.1.21. The equation shows that the deposit rate set by the retail bank incorporates the current value of the monetary policy rate. Adjustments to changes in the wholesale interest rate are inversely proportional with the intensity of the adjustment cost (measured using A_d) and positively proportional with the level of competition in terms of mobilising bank funds (measured using the inverse of ε^d).

$$\hat{r}_t^d = \frac{A_d}{1 + \varepsilon^d + (1 + \beta_p) A_d} \hat{r}_{t-1}^d + \frac{\beta_p A_d}{1 + \varepsilon^d + (1 + \beta_p) A_d} E_t \hat{r}_{t+1}^d + \frac{1 + \varepsilon^d}{1 + \varepsilon^d + (1 + \beta_p) A_d} \hat{r}_t \quad 15.1.21$$

In flexible interest rates, interest rate determination yields Equation (15.1.22), which basically shows that retail deposit interest rates are set as a mark-down from the monetary policy rate.

$$r_t^d = \left(\frac{\varepsilon_t^d}{\varepsilon_t^d - 1} \right) r_t \quad 15.1.22$$

In general, the real profit of a bank is the total net revenue (intermediation margin minus the other costs) of the wholesale bank and retail bank branch. After omitting the inter-branch transactions, bank profitability is determined as in Equation (15.1.23), with A^B the adjustment cost of changes in the deposit and lending rates.

$$\Pi_t^b = r_t^{bH} b_t^I + r_t^{bE} b_t^E - r_t^d d_t - \frac{A_{Kb}}{2} \left(\frac{K_t^b}{B_t} - v^b \right)^2 K_t^b - A_t^B \quad 15.1.23$$

Model Estimation

Estimation can be performed using a linearization model around the steady-state and proceed with Bayesian approach. Gerali et al. (2010) uses 12 observational variables: real consumption, real investment, real house prices, real deposits, real household and corporate credit, overnight rate, deposit rate, as well as the lending rates for households and businesses. For further clarification and estimation results, refer to Gerali et al. (2010).

When analyzing the interaction between monetary policy and macroprudential policy, Angelini et al. (2011) used two macroprudential instruments, namely the LTV ratio and the CCB, namely v_t . In this context, banks accumulate capital naturally by acquiring retained earnings as follows:

$$K_{b,t} = (1 - \delta_b) K_{b,t-1} + \Pi_{b,t-1} \quad (22)$$

where $K_{b,t}$ = bank capital, Π = retained earnings, and δ = capital depreciation. The model assumes that banks only accumulate capital from retained earnings.

The lending rates offered to households (H) and enterprises (E) are determined by the first form equation of bank profit maximization as follows:

$$R_t^{B,i} = \mu_t^{B,i} \left[R_t + \phi \left(\frac{K_{b,t}}{L_t} - \nu_t \right) \right] + A_t^{B,i} \quad i = H, E \quad (23)$$

where μ_t = time-varying mark-up applied by banks due to their monopolistic nature, R_t = monetary policy rate, ϕ = the cost incurred by the bank if the ratio of capital to credit (L) is less than the capital requirement, ν_t , as determined by the central bank as the macroprudential authority, and A_t = cost of adjusting the bank interest rates due to rigidity. Equation (15.23) can be interpreted as a function of bank credit supply. If credit increases, the ratio of capital to assets decreases below ν_t , thus compelling the bank to raise lending rates, which, in turn, reduces demand for new loans.

The time deposit rate, R^D , as the interest rate for savers to choose between consumption and saving is determined as follows:

$$R_t^D = \mu_t^D R_t + A_t^{B,D} \quad (24)$$

where μ_t = time-varying mark-down below the monetary policy rate that can be determined by the banks due to their monopolistic nature, and A_t = cost of adjusting the term deposit rate.

Equations (15.23) and (15.24) show that in this model, monetary, and macroprudential policy play independent roles. The monetary policy rate, R_t , has an immediate impact on lending and deposit rates, while the macroprudential instrument, ν_t , only affects lending rates. The independent formulation of both instruments creates increasingly significant misalignment between deposit rates and lending rates, which will ultimately have a larger impact on consumption, saving and investment.

The second macroprudential instrument is the LTV ratio, which also has an impact on the economy. In this case, the optimal solution regarding the level of consumption/saving by saver households (s) and by borrower households (b) is determined using the Euler equation to maximize the utility function of each respective household type.

$$\frac{1}{c_t^s} = \beta^s E_t \left(\frac{1}{c_{t+1}^s} \frac{(1 + R_t^D)}{\pi_{t+1}} \right) \quad (25)$$

$$\frac{1}{c_t^b} = \beta^b E_t \left(\frac{1}{c_{t+1}^b} \frac{(1 + R_t^{B,H})}{\pi_{t+1}} \right) + (1 + R_t^{B,H}) \lambda_t \quad (26)$$

where β^i is the discount factor with $\beta^s > \beta^b$, c = consumption, and μ = inflation. For savers (Equation 15.25), the optimal level of consumption depends on the current real deposit rate and future expectations. On the other hand, for

borrowers (Equation 15.26), the optimal level of consumption depends on the current real lending rate and future expectations, in addition to the Lagrange multiplier, λ_r , of the loan constraints as follows:

$$(1 + R_t^{B,H})b_t^H \leq m^H E_t[q_{t+1}^h h_t \pi_{t+1}] \quad (27)$$

where q_t = house prices, h_t = current stock of houses, b_t = value of bank loan, and m_t = LTV ratio for housing loans.

Equations (15.25)–(15.27) show how macroprudential policy can influence the economy indirectly of monetary policy. The monetary policy rate, R_t , influences deposit and lending rates through Equations (15.23)–(15.24) and, thus, the choice to consume or invest by savers through Equation (15.25.) and the decision to consume or borrow by borrowers through Equation (15.26). By stipulating the LTV ratio, m_r , macroprudential policy influences loan constraints, λ_r , and, hence, the level of consumption and borrowing by borrowers through Equation (15.26) but not the level of consumption and investment by savers. When collateral constraints tighten and λ_r increases, the ability of borrowers to cover household consumption and investment decreases and, therefore, the corresponding level of spending on both items also declines. It is important to note that the role of independent macroprudential policy is due to collateral and borrowing constraints. When such constraints are not present, monetary and macroprudential policies are mutually influential.

15.4.3. Central Bank Policy Mix Formulation

Using the structural DSGE model above, Angelini et al. (2011) analyzed monetary and macroprudential policy formulation with the following considerations: (1) sources of economic shock, whether technological or demand-side; and (2) the formulation process of both policies, whether independent or coordinated into a mix. In harmony with the analysis of Cecchetti and Kohler (2012), using the structural DSGE model above, Anglenia et al. (2011) also showed that the coordinated formulation of a monetary and macroprudential policy mix provides the most optimal solution with a DSGE model.

In this regard, the central bank formulates monetary policy by determining the policy rate according to the Taylor rule as follows:

$$R_t = (1 - \rho_R)R^* + (1 - \rho_R)[\omega_\pi(\pi_t - \pi^*) + \omega_y(y_t - y^*)] + \rho_R R_{t-1} \quad (28)$$

where ρ_R = monetary policy rate adjustment inertia, and ω_i = weight of the central bank's preferred inflation and economic growth targets. The monetary policy response is directed toward minimizing the loss function on public welfare as follows:

$$W^{MP} = \sigma_\pi^2 + k_{y,MP}\sigma_y^2 + k_R\sigma_{\Delta R}^2 \quad (29)$$

where σ_2 = variance of inflation, output and changes to the monetary policy rate. A positive k_R weight indicates a cost of adjusting the monetary policy rate, meaning that the central bank will avoid an excessive change to the policy stance in order to prevent an excessive impact on volatility.

Meanwhile, macroprudential policy is formulated by setting the CCB rate or LTV ratio. Each instrument is analyzed for simplicity. In this regard, the capital requirements are determined as follows:

$$v_t = (1 - \rho_v)v^* + (1 - \rho_v)\omega_v X_t + \rho_v v_{t-1} \quad (30)$$

where the parameter v^* measures the capital requirement v_t at a steady state. The level of v^* can be interpreted as the capital requirements necessary to ensure the soundness of individual banks, with adjustments to the capital surcharge for systemically important banks (SIB), as discussed in Chapter 14. The CCB, v_t , is adjusted with the dynamics of the macroeconomic variable X_t , which is measured by the level of output. Therefore, the CCB rate can be raised when output is projected to climb and, thus, macroprudential policy is directed toward controlling procyclical bank lending by requiring additional capital. The reverse is also true, namely that if a recession is expected, the CCB rate can be eased to stimulate bank lending and support stronger economic growth.

With a similar mechanism, macroprudential policy using the LTV ratio can be determined as follows:

$$m_t = (1 - \rho_m)m^* + (1 - \rho_m)\omega_m X_t + \rho_m m_{t-1} \quad (31)$$

where parameter m^* measures the capital requirement m_t at a steady state. Like the capital requirements, the level of m^* can be interpreted as the LTV ratio required to ensure the soundness of individual banks, while the LTV ratio, m_t , is adjusted over time with the dynamics of the macroeconomic variable X_t , which is measured with house prices or housing loans. Therefore, the LTV ratio can be raised (lowered) to control (stimulate) bank lending if rising (falling) house prices or housing loans are projected to accelerate during a period of economic overheating (recession).

When formulating macroprudential policy, the central bank minimizes the loss function on public welfare as follows:

$$W^{FS} = \sigma_{ly}^2 + k_{y,FS}\sigma_y^2 + k_m\sigma_{\Delta m}^2 \quad (32)$$

where σ^2 = variance of the credit to output ratio, output and changes to macroprudential instruments. In the equation, macroprudential instruments are represented by the LTV ratio. The same analysis can be conducted for the CCB instrument. Equation (15.32) shows that the desired target of the central bank to maintain financial system stability is controlling bank credit expansion, indicated by variability in the credit to output ratio. A positive weight, k_m , indicates a cost

of adjusting the macroprudential instrument, meaning that the central bank will avoid an excessive change to the policy stance in order to prevent an excessive impact on volatility.

Formulating a coordinated monetary and macroprudential policy mix is analyzed through joint calibration of both instruments to minimize the total loss function on public welfare as follows:

$$W^{Mix} = W^{MP} + W^{FS} = \sigma_{\pi}^2 + \sigma_{lly}^2 + (k_{y,MP} + k_{y,FS})\sigma_y^2 + k_R\sigma_{\Delta R}^2 + k_m\sigma_{\Delta m}^2 \quad (33)$$

Several observations can be found from the central bank's objective function above. *First*, the target of the central bank policy mix is to achieve price stability (in this case controlling inflation within the target corridor) and financial system stability (in this case a stable credit to output ratio). *Second*, when formulating the policy mix, the central bank also nurtures sustainable economic growth toward the potential output. *Third*, the central bank must also consider the adjustment cost of changing policy stance to avoid excessive volatility. The optimal solution of the central bank policy mix produces a set of parameters (ω_{π} , ω_y , ω_m , ρ_R , ρ_m) to minimize the objective function (15.33) above.

When the two policies are formulated independently, however, the monetary policy response is determined by minimizing the function (15.29), assuming the macroprudential policy instrument is congruent with Equations (15.30) and (15.31). And vice versa, the macroprudential policy response in Equation (15.32) is determined assuming monetary policy is consistent with Equation (15.28). There is no joint calibration between the two policy instruments. This lack of coordination can occur at a central bank if the committee to formulate monetary policy is separate from the committee to formulate macroprudential policy. Conditions will become even less coordinated, or more independent, if the responsibility for macroprudential policy is not under the auspices of the central bank. Independent policies cause economic players, households and firms, to be passive, meaning that their behavior and interactions are not considered when formulating policy and merely accept the policy response set by the central bank as a given.

A comparison between the alternative policymaking approaches by Angelini et al. (2011) revealed several interesting findings. Under normal conditions, meaning that economic dynamics are influenced more by technology shocks and supply-side shocks, actively applying macroprudential policy instruments, such as LTV or the CCB, would not have a significant impact on macroeconomic stability. In fact, if both policies are formulated independently, the central bank could actually formulate conflicting policies and, therefore, trigger excessive interest rate and LTV/capital volatility without significantly strengthening macroeconomic stability (inflation and output) and financial system stability (credit to output variability). This kind of conflict demonstrates the mutual influence of both policies on interest rates and bank lending, but the impact on inflation, output, and credit to output can vary.

The results are very different if the macroeconomic fluctuations are more attributable to financial or demand-side shocks. Under such conditions,

macroeconomic stability is strengthened significantly by macroprudential policy derived from monetary policy. Monetary and macroprudential policy coordination produces broader benefits in terms of macroeconomic stability (especially output) and financial system stability (credit to output variability). The LTV macroprudential instrument is significantly effective if the financial shocks occur in a specific sector, although this may be accompanied by slightly higher inflation and interest rate volatility because monetary policy supports macroprudential policy to stabilize credit and output. The benefits of stronger macroeconomic and financial system stability will not be realized if the two policies are formulated independently (not coordinated).

The results explain why many countries apply macroprudential policy, particularly in the wake of the global financial crisis, namely due to economic shocks from the financial side (advanced countries) or demand side. It is important to note that macroprudential policy is not a substitute for monetary policy, nor can both policies be considered a panacea to overcome all problems. A mix of the two should be applied and formulated in a coordinated way in accordance with the sources of shocks in the economy, particularly financial shocks and shocks in specific sectors. Moreover, independent policy formulation under normal conditions may not safeguard macroeconomic or financial system stability, and could even exacerbate conditions.

In reality, technology shocks and financial shocks can occur, which both influence the economy. During an economic upswing due to productivity factors, for instance, higher corporate net worth could lower the external financing premium or higher collateral asset prices could raise the bank credit ceiling. Consequently, technology shocks to increase output could be accompanied by financial shocks, thus exacerbating the amplification of the financial cycle by the economic cycle, as found by Bernanke et al. (1999), Kiyotaki and Moore (1997), and Minsky (1982). The analysis above shows that a monetary and macroprudential policy mix calibrated through close coordination could control the financial cycle to remain synchronized with the economic cycle and, thus, not spur macroeconomic and financial system instability risk. This is the essence and advantages of a central bank policy mix rather than traditional monetary policy.

15.5. Bank Indonesia Policy Mix

In terms of monetary policy, Bank Indonesia applies the ITF introduced in the early 2000s, which has been officially implemented since the middle of 2005. The final goal of monetary policy is price stability (inflation). To that end, exchange rate stability is consistent with attainment of the inflation target and supports general macroeconomic stability, including management of the current account deficit and non-resident capital flows. The primary policy instrument is the interest rate, which is supported by exchange rate policy instruments, foreign capital flow management, statutory reserve requirements, and other prudential regulations. Although directed toward price stability, monetary policymaking pays due consideration to macroeconomic and financial system conditions in general, particularly economic growth, the balance of payments and monetary policy transmission effectiveness.

ITF-based monetary policy has facilitated a downward inflation trend within the $4\pm 1\%$ target corridor since 2015. The Government introduced energy reforms at the end of 2014 – the main driver of higher inflation – but inflation remained under control and continues to decline toward the medium-term target of $3\pm 1\%$. Institutionally, ITF has reinforced monetary policy credibility at Bank Indonesia. Disciplined, regular and formal economic assessments underlying policy formulation at the monthly BI Board of Governors' Meeting has placed Bank Indonesia at the forefront of macroeconomic conditions and the national economic outlook, including the necessary policy response. The regular economic evaluations are also very helpful to Bank Indonesia when coordinating with the government (central government and regional administrations) in terms of monetary and fiscal policy as well as structural reforms to underpin sustainable economic growth and macroeconomic stability. Furthermore, aggressive communication by Bank Indonesia has proven an effective instrument to anchor inflation expectations and relay the macroeconomic outlook.

In general, the established monetary policy framework represents a solid foundation for Bank Indonesia to institute macroprudential policy and, thus, support financial system stability through a policy mix. Several domestic and external challenges remain, however, that must be addressed to ensure a successful policy mix. *First*, Indonesia's economy is spread out across a vast archipelago and continues to rely on commodities, and hence, often faces inflation shocks and volatile food prices, along with balance of payments uncertainty. Mitigating the internal and external balance requires a monetary policy that is closely coordinated with the government's fiscal policy and structural reforms. *Second*, the financial system remains dominated by the banking sector with an immature financial market and, therefore, the economic cycle is replete with financial cycle accelerators in the banking system. Mitigating procyclicality and systemic risk in the banking system through macroprudential (regulation and supervision) policy is critical not only to support financial system stability but also in terms of monetary policy effectiveness. And *third*, Indonesia's economy is comparatively small with a very open capital account system, thus foreign capital flow management is a crucial element of ensuring that the benefits of non-resident capital reach the economy, while simultaneously avoiding economic instability.

The three challenges must be considered in the Bank Indonesia policy mix, moreover because all three are interconnected and mutually reinforcing. During an economic upswing, for instance due to financial deregulation, a commodity price boom or propitious global economic conditions, expanding domestic demand accelerates credit growth, creates property bubbles, intensifies inflationary pressures, widens the current account deficit and increases government external debt. We have experienced boom and bust periods in the economic and financial cycles during mini crisis and large crisis episodes throughout Indonesia's economic history. The economic bonanza after financial sector liberalization from 1983 to 1988 subsequently culminated in the Asian financial crisis of 1997/98, which raised the curtain on various systemic risks and macrofinancial imbalances that had accumulated in previous periods. Developments during the decade before and then after the global financial crisis of 2008/09 also revealed a

similar pattern, albeit at a comparatively lower level. The mini crisis of 2005, for instance, was distinguishable for rapid growth of domestic demand, bank credit and foreign capital inflows from the economic boom due to high international commodity prices. Similar conditions occurred again in 2010 to 2013 because global commodity prices stoked the current account deficit when they fell. Macroeconomic and macrofinancial imbalances such as these cannot be overcome through monetary policy, foreign capital flow management or even microprudential regulation and supervision. Thus, macroprudential policy is an additional instrument of critical importance, which must be integrated, along with monetary policy and foreign capital flow management, into the Bank Indonesia policy mix.

15.5.1. Policy Mix Targets and Instruments

As explained in the previous section, two important aspects must be considered when including the FSS dimension and macroprudential policy into the central bank's policy mix, namely: (1) the need to broaden price stability beyond inflation to also encompass (financial and property) asset prices; and (2) the necessary response to macrofinancial linkages in terms of financial system stability, specifically in relation to procyclicality and systemic risk accumulation in the financial system. Concerning the first aspect, in addition to CPI inflation, Bank Indonesia also focuses on assessments of exchange rates, bond yields and stock prices, as well as property prices. Bank Indonesia considers the exchange rate for macroeconomic analysis and projections as well as the required policy response. Consistent with ITF, the interest rate is the primary instrument used to achieve the inflation target. Nonetheless, Bank Indonesia is not averse to intervention on the foreign exchange market in the case of exchange rate volatility or misalignment from the currency's fundamental value. Regarding other asset prices, Bank Indonesia uses separate models to enhance the macroeconomic and macrofinancial assessments and projections, as well as monetary and/or macroprudential policy instruments considered more appropriate to tackle the risks faced.

Referring to the second aspect, to deepen understanding of macrofinancial linkages, Bank Indonesia has expanded its macroeconomic projection model to include the external sector, incorporating external default risk as a proxy of sudden-stop risk, as well as the banking sector, containing the credit gap as a measure of procyclicality (Harmanta et al., 2012, 2013). The model presents various policy scenarios using interest rates and reserve requirements on the monetary policy side and/or LTV on the macroprudential policy side. The model is forward-looking with a horizon of 2 years, therefore, the projections provide valuable input when determining the timing and magnitude of the appropriate policy response to lean against price stability risk, capital reversal risk as well as FSS procyclicality and systemic risk using monetary instruments, foreign capital flow management or macroprudential policy instruments or, indeed, a combination of the three. To hone the understanding of procyclical behaviors and other macrofinancial linkages, particularly during credit booms and property bubbles, models and studies have been developed to assess cyclical behaviors and systemic risk

accumulation, as an aggregate and by sector (Alamsyah, Adamanti, Yumanita, & Astuti, 2014; Harun, Taruna, Nattan, & Surjaningsih, 2014).

From the perspective of financial system stability, several approaches and technical systemic risk assessments have been developed as recommended in the literature (Bisias, 2012) or as practiced by central banks (such as by the European Banking Authority (EBA), 2015). The focus is on systemic risk assessment (not individual soundness) of SIB, namely

banks that, due to the size of assets, capital and liabilities; scope of network or transaction complexity of banking services; as well as interconnectedness with other financial sectors, could trigger default in some or all other banks or the financial sector as a whole, operationally or financially, if the bank in question were to experience disruptions or default. (PPKSK Act, 2016)

Interconnectedness in the interbank money market and payment system is also evaluated. Credit risk is analyzed using several methods, including the “probability of default” and “transition matrix.” The banking system is also regularly stress tested to analyze resilience to macroeconomic shocks through an integrated and/or balanced approach based on the results of risk surveys conducted. Corporate sector and household risks are also assessed in terms of financial performance and the impact on banking system performance. To assess macrofinancial linkages between the financial system, from a procyclicality and systemic risk standpoint, balance sheet statistics were developed that highlight (asset and liability) linkages between economic players and the financial system, public and private, nationally, and domestically.

Based on a holistic assessment of the macroeconomic analysis and projections, macrofinancial linkages as well as time series and cross-section dimensions of systemic risk, Bank Indonesia’s policy mix is formulated using the following four instruments (Warjiyo, 2014a, 2015b). *First*, like ITF, interest rate policy is directed to ensure forward-looking inflation projections are in the $4\pm 1\%$ target corridor for 2016 and 2017. *Second*, exchange rate policy is directed toward maintaining exchange rate stability on the market in line with the currency’s fundamental value to remain consistent with achieving the inflation target and to guard against excessive volatility that could trigger financial system and macroeconomic instability. *Third*, foreign capital flow management is used to support exchange rate policy as well as macroeconomic and financial system stability, especially during an influx of foreign capital or a sudden capital reversal. *Fourth*, macroprudential policy is directed toward supporting financial system stability and the effectiveness of monetary policy transmission. Bank Indonesia also maintains close policy coordination with the central government and regional administrations in terms of macroeconomic management. A coordination mechanism also exists between Bank Indonesia, the Financial Services Authority (OJK) and Deposit Insurance Corporation (LPS) for financial system stability matters. Clear communication is crucial for the success of the policy mix.

How does the policy mix respond to the various risks associated with attaining the inflation target and supporting financial system stability in the future? This issue has been a hot topic of debate due to possible conflicts between monetary policy to achieve price stability and macroprudential policy to support financial system stability. The policy mix is also said to violate the Tinbergen rule, namely that there should be at least the same number of instruments as there are targets. Nevertheless, recent understanding is that in many cases the two policy instruments are mutually complementary to achieve the two targets (Yellen, 2014). [Table 15.5](#) presents four cases of price stability risk and FSS risk that could emerge from the assessment of macroeconomic and macrofinancial projections over a 2-year horizon as well as the corresponding monetary and macroprudential policy mix stances available. In the first quadrant, where low price stability and FSS risks are projected, the monetary policy and macroprudential policy stance is neutral or loose. Conversely, in the fourth quadrant, where high price stability and FSS risks are projected, a tight monetary and macroprudential policy stance should have been adopted.

Potential conflicts can occur in the second and third quadrants. In the second quadrant, where price stability risk is projected low but FSS risk is projected high, a tight macroprudential policy stance is clearly required. Under such conditions, monetary policy can support the effectiveness of macroprudential policy by leaning against the projected future FSS risk. This case is similar to conditions in the United States prior to the global crisis, which has been debated by Taylor (2010) and Bernanke (2010) as explained above. In the third quadrant, where price stability risk is projected high but FSS risk is projected low, a tight monetary policy stance is required. Under such conditions, macroprudential policy

Table 15.5: Four Cases of Price Stability and Financial System Stability.

		Projected Price Stability Risk	
		Low	High
Projected Financial System Stability (FSS) Risk	HIGH	<u>Quadrant II</u> • NEUTRAL/LEANING monetary policy • TIGHT macroprudential policy	<u>Quadrant IV</u> • TIGHT monetary policy • TIGHT macroprudential policy
	LOW	<u>Quadrant I</u> • NEUTRAL/LOOSE monetary policy • NEUTRAL/LOOSE macroprudential policy	<u>Quadrant III</u> • TIGHT monetary policy • NEUTRAL/LEANING macroprudential policy

can support the effectiveness of monetary policy by leaning against the projected future price stability risk. How far and the macroprudential policy instruments applied depend on the factors causing price stability risk. A reasonable choice would be macroprudential policy that can strengthen monetary policy transmission to achieve price stability. For example, when price stability risk stems from strong domestic demand, driven by rapid bank credit growth in the property sector, adjusting the LTV ratio would be the most appropriate policy action.

The factual issues that occur in the real world, however, are not always as simple as described in [Table 15.5](#). Nonetheless, the four cases presented in the table can be referred to as guiding principles that help assess the existing problems and select a policy mix that addresses the potential conflicts that arise in the dual mandate of price stability and maintaining financial system stability. It is important to reiterate that the choice of monetary and macroprudential instruments depends on an assessment of the factors causing the projected high risks to price stability and financial system stability. This kind of approach is also used to address potential conflicts between price stability and autonomous monetary policy, exchange rate stability and free foreign capital flows, namely the policy trilemma that plagues an open economy and is regularly explained in international finance (Obstfeld, 2015). The following sections will explore how the policy mix of Bank Indonesia is applied to reinforce monetary stability and financial system stability during periods of economic turbulence and in terms of international finance since the global financial crisis of 2008/09.

15.5.2. Interest Rate and Exchange Rate Policies

In pursuance of the monetary policy strategy based on the ITF, interest rate policymaking is directed toward ensuring inflation projections remain within the target corridor. The issue, therefore, is how to address exchange rate fluctuations on the market that can edge inflation projections outside of the target range. Such conditions have been faced by EMEs when confronting highly volatile foreign capital flows since the global financial crisis of 2008/09. Differing from advanced countries, exchange rate stability is crucial for EMEs due to various factors, including underdeveloped financial markets, the monumental impact on banking conditions and financial system stability, as well as rigidity in the economy. Under such conditions, targeting the exchange rate as an element of achieving the inflation target will strengthen monetary policy credibility based on ITF (Ostry et al., 2012b). Specifically, targeting the exchange rate can help mitigate the undesirable impact of foreign capital flows on achieving the inflation target, either directly through exchange rate pass-through or indirectly through domestic demand. Many EMEs already consider the exchange rate when formulating interest rate policy using the Taylor rule (Aizenman et al., 2011; Mohanty & Klau, 2004). Foreign exchange intervention represents another option. If foreign capital flows are creating a misalignment between the exchange rate and its fundamental value and inflation has moved outside of the target range, an interest rate response combined with foreign exchange intervention would be more effective and, thus, reinforce monetary policy credibility.

Bank Indonesia introduced such an approach in 2010, which has proven superior to the standard ITF application that is merely based on interest rate instrument. Three episodes since the global financial crisis of 2008/09 have backed up such claims, including the period from 2010 until the Fed's Taper Tantrum in May 2013, the period after the Taper Tantrum until the middle of 2015 and then the period thereafter. During the first period, Indonesia enjoyed the benefits of global spillovers, in particular high commodity prices and an influx of foreign capital flows (Warjiyo, 2013b). Economic growth peaked at 6.5% in 2011 and remained relatively robust at 6.3% in 2012. Furthermore, inflation was successfully brought down to its historically lowest level of 3.8% in 2011, even dipping below the target corridor of $5 \pm 1\%$, but accelerating thereafter to 4.3% in 2012. Indonesia received a deluge of foreign capital flows during that period, driven by excess global liquidity seeking higher yields and a promising domestic economic outlook. The exchange rate appreciated due to the foreign capital flows, coupled with a current account surplus due to high commodity prices. The challenge was how to manage the foreign capital flows in order to prevent and mitigate systemic risk to financial system stability, similar to conditions when bank credit growth soared to 23% per annum from 2010 to 2012. Such conditions mirrored the second quadrant, where price stability risk was low but FSS risk was relatively high, as discussed above.

Consistent with ITF, Bank Indonesia lowered its policy rate by 75bps from 6.5% in 2010 to 5.75% in 2012. Further reductions would have been inconsistent with ITF because inflation had already hit its lowest level in history, while being ineffectual in terms of overcoming the torrent of foreign capital inflows that were driven more by push factors than pull factors (Indawan, Fitriani, S., Permata, & Karlina, 2013). In addition, further reductions to the policy rate would have also been inconsistent with the goal of supporting financial system stability due to excessive credit growth. Therefore, Bank Indonesia intervened (purchased) on the foreign exchange market to stem the flow of foreign capital and strengthen the exchange rate. In addition, to more effectively sterilize the impact on liquidity expansion in the domestic banking industry, the statutory reserve requirement was raised from 5% to 8% in November 2011. The position of official reserve assets increased significantly from just USD 66.2 billion at the beginning of 2010 to peak at USD 112.8 billion in 2012, which provided a larger buffer to overcome the foreign capital outflows after the Fed's Taper Tantrum in the middle of 2013.

The situation reversed in the second period. Foreign capital flowed out of Indonesia with rapidity after the Federal Reserve announced plans to normalize monetary policy, which stoked concerns and rattled the markets from May to August 2013. The sudden and large capital reversal from government bonds and stocks triggered herding behavior that exacerbated the risks to monetary and financial system stability (Warjiyo, 2014b). The problems were compounded by a current account deficit that swelled to 4.4% of GDP as well as declining export performance as commodity prices tumbled, while imports remained high on the back of strong domestic demand. Inflation spiked to 8.4% in 2013, as the government raised fuel prices in July, and was subsequently recorded at 8.3% in 2014 as the government reformed its energy policy and scrapped fuel subsidies, while

limiting diesel subsidies in October. In terms of financial system stability, bank credit growth remained high at 21.4% in 2013. Such conditions reflect the fourth quadrant, where the risks to price stability and FSS are high, as described above.

Bank Indonesia immediately responded to stabilize the situation by raising the policy rate and tightening macroprudential policy. Bank Indonesia was one of the first authorities to raise its policy rate after the Fed's Taper Tantrum in May 2013. Bank Indonesia first raised its policy rate in June 2013, followed by further hikes in the following months, totaling 175 bps to 7.50%, over a 6-month period through to November 2013. The main goal of aggressively raising the policy rate was to pre-empt intense inflationary pressures when the government hiked fuel prices in July 2013. Furthermore, the move was part of coordinated macroeconomic and fiscal policy to reduce the excessive current account deficit. The timing of the increase was critical because of the need to simultaneously control the foreign capital reversal after the Fed's Taper Tantrum. The aggressive interest rate response provided a clear signal to the markets, thereby reinforcing monetary policy credibility.

Bank Indonesia also intervened (sold) on the foreign exchange market to overcome the capital reversal and stabilize the exchange rate after the Taper Tantrum. The exchange rate stabilized thereafter in September 2013. Consequently, the position of foreign exchange reserves decreased sharply to USD 92 billion in September 2013 before rebounding to USD 99 billion at the end of the year. Foreign exchange intervention was also accompanied by the purchase of government bonds on the secondary market, a tactic known as dual intervention (Warjiyo, 2013c). Fundamentally, the tactic makes sterilization more effective because the purchase of bonds on the secondary market prevents liquidity from evaporating as it is absorbed due to the intervention (selling) of foreign exchange. Dual intervention, as a tactic, also strengthens the effectiveness of foreign exchange intervention in terms of stabilizing the exchange rate. Through this tactic, Bank Indonesia provided a clear signal that it would not hesitate to supply foreign exchange and was prepared to buy the bonds sold by non-resident investors, thus preventing herding behavior and contagion from the capital reversal. Moreover, dual intervention to safeguard monetary stability is more consistent with the need to maintain financial system stability. By stabilizing the foreign exchange market and government bond market, dual intervention helped stabilize the financial markets in general.

The aggressive and timely monetary policy response bore fruit and reinforced credibility. Market confidence quickly recovered and foreign capital inflows returned at the end of 2013 and throughout 2014. Furthermore, macroeconomic and financial system stability were maintained. In fact, inflation decelerated from 8.3% after the energy reforms in 2014 to 3.3% at the end of 2015, while the current account deficit reduced from 3.3% to 2.0% of GDP over the same period. Such conditions are like those found in the first quadrant of the table, where the risks to price stability and financial system stability are low. Nevertheless, economic growth slumped from 5.2% in 2014 to 4.8% in 2015 and credit growth was squeezed to around 10%. Therefore, with stability maintained and clearer communication from the Fed that the normalization process would be gradual, Bank

Indonesia eased its monetary policy stance to reinforce looser macroprudential policy taken the year earlier. Bank Indonesia reduced its policy rate three times in the first 3 months of 2016, totaling 75 bps, to 6.75%. The reserve requirement was also lowered by 50 bps in November 2015 and again by 100 bps to 6.5% in February 2016. Bank Indonesia was confident that easing monetary and macroprudential policy would strengthen the fiscal stimuli introduced to catalyze economic growth with inflation controlled in the target range for 2016 and 2017 at $4\pm 1\%$. Through coordinated macroeconomic and FSS policies, combined with accelerating the government's structural reforms, economic growth is expected to improve on that posted in 2015, and then continue to accelerate in 2016 and 2017.

15.5.3. Foreign Capital Flow Management

Foreign capital flow management in Indonesia aims to complement, not substitute, sound macroeconomic policy. The best defense to reap the benefits and mitigate the adverse effects of foreign capital flows is to strengthen economic fundamentals, institute prudent FSS, and macroeconomic policies as well as accelerate structural reforms in the real sector. Congruently, foreign capital flow management in Indonesia is based on three guiding principles. *First*, the goal is to mitigate the adverse impact of foreign capital flows in terms of exchange rate, monetary, financial system and macroeconomic instability in general. *Second*, selective foreign capital flow management is applied to specific types of foreign capital, namely short-term and speculative, while long-term and productive foreign capital is promoted to finance the national economy. And *third*, foreign capital flow management instruments and regulations remain consistent with the open capital account regime. As far as possible, the regulations do not differentiate between residents and non-residents. Enforcement is temporary when required, namely by tightening regulations when foreign capital flows are too large and vice versa.

The following section outlines several foreign capital flow management regulations that have been implemented. When foreign capital flows inundated the domestic financial system after the global financial crisis through to the Taper Tantrum, Bank Indonesia managed foreign capital flows in the form of a 6-month holding period for Bank Indonesia Certificates (SBI) and capped short-term offshore loans at 30% of capital. This meant that investors had to wait 6 months before releasing their SBI, thereby limiting short-term foreign capital flows and encouraging longer terms. Similarly, restrictions on short-term foreign loans prevented the banks from using short-term funds to extend long-term loans, while simultaneously mitigating currency risk. During the subsequent period, as foreign capital inflows subsided and outflows were even recorded in the wake of the Fed's Taper Tantrum in 2013, foreign capital flow management was eased by relaxing the holding period to just 1 month for SBI, while a number of transactions were excluded from the restrictions on short-term foreign loans by banks. The measures helped to manage short-term and speculative foreign capital flows and, hence, were consistent with and supported the achievement of price (and exchange rate) stability as well as financial system stability.

Bank Indonesia also introduced regulations at the end of 2014 to strengthen risk management for non-bank corporate external debt. In Indonesia, government external debt has been restricted pursuant to prevailing regulations that limit the fiscal deficit to 3% of GDP. For the banking industry, in addition to the restrictions on short-term offshore loans mentioned above, each foreign loan withdrawn by a bank would have to be preapproved by Bank Indonesia in order to remain consistent with the goals of maintaining macroeconomic and financial system stability. Applicable to non-banks, the new regulations promulgated by Bank Indonesia at the end of 2014 strengthened risk management for external debt by requiring: (1) a minimum currency hedging ratio of 25% of net foreign currency liabilities maturing in the upcoming three and 6 months; (2) a minimum liquidity ratio of 50% (after calculating foreign currency assets in the hedging ratio) of net foreign currency liabilities maturing in the upcoming three and 6 months; and (3) a minimum credit rating of one level below investment grade. Monitoring of more than 2,000 firms that submitted financial statements to Bank Indonesia showed that 88% had met the requirements. In addition, the new regulations also encouraged corporations to hedge against currency risk, evidenced by the rapid increase of swap and forward transactions on the foreign exchange market.

15.5.4. Macprudential Policy

As described above, the application of macroprudential policy in Indonesia can and must strengthen the effectiveness of monetary policy to achieve price (and exchange rate) stability as well as financial system stability. Consequently, monetary policy transmission can be more effective, especially through the banking system, and financial system stability can also be better maintained. If monetary and macroprudential policies were not mutually reinforcing, the interest rate response on the monetary policy side to achieve price (and exchange rate) stability and support financial system stability would have to be notably larger. For example, to control excessive bank credit expansion, a much larger hike to interest rates would be required if not accompanied by tighter macroprudential policy. The impact would also not only trigger economic losses but also financial system losses.

At Bank Indonesia, in addition to expanding macroeconomic analyses and projections by including macrofinancial linkages when formulating the policy mix, several approaches and models have also been developed to assess the optimal level of bank credit growth (Utari, Arimurti, & Kurniati, 2012). Models such as these are applied as an aggregate and by loan type (consumer loans, working capital loans, and investment loans) as well as by economic sector. Comparisons can be drawn between actual credit growth and the optimal level of credit growth, which reveal whether bank credit growth as an aggregate, by loan type and by economic sector, is excessive. Such assessments are critical to determine when and which macroprudential policy instruments should be applied to reinforce monetary policy and foreign capital flow management to achieve price (and exchange rate) stability as well as support financial system stability.

Bank Indonesia followed such an approach by enforcing an LTV ratio averaging 70% for the automotive and property sectors in June 2012 (Warjiyo, 2015a). As explained above, while inflation was kept under control, FSS risk began to build on strong credit growth and an influx of foreign capital flows. Raising the policy rate was not an option because the move would attract even more foreign capital and drive credit growth further. Therefore, to control foreign capital inflows, Bank Indonesia intervened (purchased foreign exchange), raised the statutory reserve requirement and enforced several foreign capital flow management regulations. Macroprudential regulations, namely the LTV ratio, strengthened the Bank Indonesia policy mix. Furthermore, reinforcing policy to maintain macroeconomic and financial system stability in the wake of the Fed's Taper Tantrum, Bank Indonesia tightened the LTV ratio in August 2013 on purchases of a second and additional property, applicable to specific house and apartment types. This policy measure was part of monetary and fiscal policy tightening to control high inflation and the current account deficit. Tighter macroprudential policy enhanced the effectiveness of raising the policy rate, which obviated the need for further interest rate hikes. Furthermore, the policy move also paid due consideration to lending rate rigidity in response to changes in the monetary policy rate with a wide spread between deposit and lending rates. The tightening of macroprudential policy was also accompanied by tighter supervisory actions imposed on certain banks, primarily those with excessively high property credit growth.

Thus far, experience has shown that macroprudential regulation and supervisory actions have strengthened the effectiveness of monetary policy transmission and supported financial system stability (Purnawan & Nasir, 2015; Wimanda, Permata, Bathaludin, & Wibowo, 2012; Wimanda, Maryaningsih, Nurliana, & Satyanugroho, 2014a). In general, loans extended to the property sector declined from around 25% in 2012 to 15% in 2015 and to 11% more recently. Growth of housing loans also fell from around 45% in 2012 to 15% in 2015 and to 7% recently. Meanwhile, real estate loans began to decelerate after the LTV was tightened in August 2013, falling from 35% at that time to 15% in 2015 but then accelerating again to around 20% recently. The recent surge in real estate loans since 2015 has been attributed to increased capital spending as an integral part of the Government's fiscal stimulus package, which also fed through to growth of construction loans at around 28%. It is important to note that the automotive and property sectors contain a large import component and, hence, controlling credit to both sectors helped narrow the current account deficit, which was a macroeconomic sticking point in 2013.

During the third period, as part of the policy to stimulate economic growth, while maintaining macroeconomic stability and financial system stability, Bank Indonesia relaxed macroprudential regulations by raising the LTV ratio (or, reducing downpayments on loans) to an average of around 10% in June 2015. As described above, macroeconomic and macrofinancial projections at the time indicated that future inflation risk and FSS risk would be kept under control, similar to conditions in the first quadrant. Nonetheless, reductions to the monetary policy rate were constrained by ubiquitous uncertainty surrounding the proposed FFR hike through to December 2015. Therefore, Bank Indonesia began to ease

its policy mix by relaxing macroprudential policy instruments in June 2015 and subsequently lowering the reserve requirement in November 2015 before finally reducing the monetary policy rate at the beginning of 2016. Easing the policy mix by lowering the policy rate and reserve requirement on the monetary side and raising the LTV ratio (lowering loan downpayments) on the macroprudential side, coupled with fiscal stimuli and accelerated structural reforms by the government, catalyzed economic growth, while maintaining macroeconomic and financial system stability.

In terms of macroprudential policy, Bank Indonesia began to apply the CCB at the end of 2015. Congruent with the ongoing spirit of loosening the policy mix, the CCB rate was set at 0% and is reviewed every 6 months. CCB application by Bank Indonesia, including international macroprudential standards, is critically important.

15.5.5. Institutional Arrangements

Central bank policy mix effectiveness requires strong institutional arrangements at the central bank and in terms of coordination with the government and relevant authorities. To that end, Bank Indonesia has refined its macroeconomic projection and policy analysis framework by including macrofinancial linkages, specifically for the banking and external sectors. The integrated analysis and projection framework is used as the basis underlying policy mix formulation, including interest rate policy, exchange rate policy, the reserve requirement, and macroprudential policy (LTV). Research has also been conducted to investigate the behavior of foreign capital flows and financial procyclicality, particularly in terms of bank lending and the housing sector. More analysis and research is required, however, to broaden our understanding of macrofinancial linkages, procyclicality and systemic risk. The data and statistics have also been enhanced through inclusion of FSS indicators and an early warning system as well as cross-sector financial balance sheet statistics at the national and regional levels.

The decision-making process has also been strengthened. In practice at several central banks, there are disparate views about whether to maintain separate monetary policy and FSS policy committees or integrate them. Kohn (2015), for instance, recommended maintaining separate committees due to the different targets and focus, instruments, and accountabilities. Such conditions are found at the Bank of England, where three committees relating to FSS have been formed, separate to the monetary policy committee, since the global financial crisis, namely the Prudential Regulation Authority (PRA) dealing with microprudential policies, the Financial Conduct Authority (FCA) for the financial markets, and the Financial Policy Committee (FPC) to formulate macroprudential policies.

At Bank Indonesia, monetary, macroprudential, and payment system policy formulation and decision-making are the preserve of the monthly BI Board of Governors' Meeting. The process does involve three committees, however, namely the monetary policy committee, the FSS committee and the payment system committee, led by the respective deputy governor responsible for policy formulation. A joint committee was also established to strengthen the policy mix by integrating

the projections and analyses of the three policy aspects along with formulating the policy mix required. Strengthening the projection models and policy mix formulation process has enriched our understanding of macroeconomic conditions as well as the macrofinancial linkages in the financial system, along with formulating a more optimal policy mix to achieve price stability and support financial system stability. This is very different to the standard monetary policy framework that is merely based on the ITF.

Bank Indonesia also coordinates closely with the government and other relevant authorities. Monetary and fiscal policy coordination between Bank Indonesia and the Government is applied to state budget formulation as well as various other aspects of macroeconomic management, centrally and through regional administrations. Although Bank Indonesia was granted independence to execute its tasks and duties pursuant to prevailing regulations, the Bank Indonesia policy mix represents an integral part of the national policy mix, particularly in terms of macroeconomic policy, financial system stability and structural reforms (Warjiyo, 2013a). Concerning financial system stability, coordination is facilitated through the Financial Stability Policy Committee (KKSK), chaired by the Minister of Finance and with the Governor of Bank Indonesia, Chairman of the Financial Services Authority (OJK), and Chairman of the Deposit Insurance Corporation (LPS) as members. Furthermore, the recently enacted Financial System Crisis Prevention and Resolution Act (PPKSK Act) provides a legal foundation for FSS policy coordination, details the respective jurisdictions of each respective institution and contains procedures for the handling of systemic banks as well as the crisis prevention and resolution mechanism. Bilaterally, coordination is also implemented between the macroprudential regulation and supervision conducted by Bank Indonesia and the microprudential regulation and supervision by the Financial Services Authority (OJK).

15.6. Concluding Remarks

This chapter has explained the concepts, theories, and implementation of the central bank's policy mix in pursuance of the dual mandate of achieving price stability and maintaining financial system stability. The policy mix consists of four key elements, namely interest rate policy, exchange rate policy, foreign capital flow management, and macroprudential policy. The central bank's dual mandate of achieving price stability and financial system stability is mutually complementary. Four cases were presented to demonstrate the consistency between the targets of price and financial system stability, followed by the appropriate policy mix response. The ITF-based monetary policy framework represents a solid basis to support the policy mix of the central bank. The key, however, is to expand and extend the macroeconomic projection and policy analysis framework under ITF to include the macrofinancial linkages found in the financial system, particularly through assessments of procyclicality and systemic risk, as well as a policy mix that is consistent with the specific challenges faced. The structural macroeconomic theoretical foundations and modeling approaches per IITF and DSGE, as the basis of central bank policy mix formulation, were also described.

Bank Indonesia's experience of applying a policy mix since 2010 has evidenced its superiority over the standard ITF. Three episodes of macroeconomic and FSS issues in Indonesia have shown that application of the BI policy mix has supported economic resilience in Indonesia and successfully maintained macroeconomic and financial system stability. Underpinning the policy mix formulation process, Bank Indonesia has also expanded the existing macroeconomic projection models to include macrofinancial linkages, particularly in the banking system and external sector. Research and analysis has also been developed to broaden our understanding of macrofinancial linkages, specifically in terms of credit, property and procyclicality, external debt, foreign capital flow volatility as well as systemic risk assessment in the banking system, particularly in terms of interconnectedness and financial networks in the financial markets and payment system. The policymaking process was further reinforced by the establishment of a joint committee to oversee monetary, macroprudential, and payment system policy, which integrates all the three aspects and provides optimal policy mix response recommendations.

Close coordination is also maintained with the Government and other relevant authorities. Nationally, to achieve robust economic growth, while preserving macroeconomic and financial system stability, a national economic mix is required, consisting of: macroeconomic policy, FSS policy, and structural reforms. Accelerating structural reform policy aims to boost economic growth from a national production capacity (potential output) perspective. Furthermore, macroeconomic policy coordination intends to manage the economic cycle and prevent internal imbalances, namely high inflation, and external imbalances, such as an excessive current account deficit. Meanwhile, FSS policy coordination aims to manage the financial cycle to dampen boom-bust cycles and prevent a build-up of systemic risk that could culminate in a financial crisis.

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